

SURVEY

Advances in Spectral Imaging: A Review of Techniques and Technologies

SANI MUKHTAR¹, AMIR ARBABI², AND JAIME VIEGAS¹, (Member, IEEE)

¹Department of Electrical Engineering, Khalifa University of Science and Technology, Abu Dhabi, United Arab Emirates

²Department of Electrical and Computer Engineering, University of Massachusetts Amherst, Amherst, MA 01003, USA

Corresponding author: Jaime Viegas (jaime.viegas@ku.ac.ae)

ABSTRACT Spectral imaging systems enable a significant leap forward in identifying and characterizing materials and objects through their unique spectral signatures. This review explores the advances in spectral imaging systems (SIS) designs, analyzing both the techniques and technologies. It provides an overview of the fundamental principles of spectral imaging, including different panchromatic, multispectral, and hyperspectral imaging systems. Progress in the design of multispectral imaging (MSI) systems are comprehensively surveyed with focus on the optical technologies employed. Advances in hyperspectral imaging (HSI) system are discussed, covering various acquisition techniques such as staring, snapshot, whiskbroom, and pushbroom techniques. In the case of staring technique, an in-depth examination is provided on different filtering technologies, including dispersive and diffractive optics, acousto-optic tunable filters (AOTFs), Fabry-Perot interferometers (FPI), liquid crystal tunable filters (LCTFs), and linear variable filters (LVFs). A detailed overview and progress in snapshot techniques is provided, followed by a thorough examination of whiskbroom and pushbroom techniques. Finally, it offers insights into the potential future directions in SIS research.

INDEX TERMS Imaging, optical imaging, spectral imaging, sensors, hyperspectral imaging, panchromatic imaging, multispectral imaging, optics, optical filters, snapshot, staring, whiskbroom, pushbroom, review.

I. INTRODUCTION

Optical imaging has revolutionized various fields, ranging from medical diagnostics and biotechnology to environmental monitoring and industrial inspection. Among the various optical imaging techniques, spectral optical imaging systems have emerged as a powerful tool for acquiring rich information about the spectral characteristics of the target. By integrating spectroscopy with imaging, it generates spectral data cubes, offering richer insights than conventional imaging or spectroscopy alone [1]. In principle, the acquired spectral images are spectrally and spatially distributed and described as a function of position (r) and wavelength (λ) as $f(r, \lambda)$, which can be a discrete or continuous function of wavelengths [2], [3], [4]. Thus, as J. Qin put it “If conventional imaging tries to answer question *where* (spatial information) and conventional spectroscopy tries to answer the question *what* (spectral information), the spectral imaging

tries to answer the question *where is what* (spatial and spectral information)” [5]. The design of spectral imaging systems has witnessed significant developments over the past decades - from novel imaging modalities and sensor technologies to innovative materials and data processing techniques. These advancements have not only improved the performance and capabilities of spectral imaging systems but have also opened new possibilities and applications in various fields [6], [7], [8], [9], [10], [11]. The rapid expansion of spectral imaging research has led to a vast and growing body of literature, spanning specific applications to novel approaches. Keeping up with the latest developments in this evolving field has become increasingly challenging for researchers and practitioners. Numerous review articles synthesize and evaluate the available literature on this topic. Thus, placing this work within the context with those reviews emphasizes its contribution, providing a clear distinction from prior studies while highlighting its originality and significance.

Over the past two decades, numerous reviews have been published in academic literature, detailing the fundamental

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principles and methodologies of various spectral imaging techniques, while examining their wide-ranging applications across multiple fields [4], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28]. Between 2019 to 2022, various systematic review articles were published analyzing the implementation of different spectral imaging modalities across a diverse field of applications, encompassing but not limited to medical applications [29], [30], [31], [32], [33], [34], [35], biometric systems [36], food quality detection [37] agricultural domains [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], and studies of seafloor ecosystems [19], [47]. In 2023, an increasing number of review articles extensively surveyed the recent advancements in spectral imaging techniques across a broad spectrum of applications, including precision agriculture [48], [49], forensic sciences [50], quality estimation of tobacco [51], food and meat safety evaluation [52], [53], land cover and land use classification [54], [55], biomedical imaging [56], [57], [58], and plastic waste detection [59]. Notably, Zahra et al. [60] provided a comprehensive review highlighting the current states of the arts pre-processing methods, publicly accessible spectral imaging databases, and available software designed for the analysis of hyperspectral images. More recently, several reviews have emerged, further reflecting the growing interest and advancements in spectral imaging. These reviews encompass a wide range of topics, from technical innovations [61], [62] to diverse applications [63], [64], [65], showcasing the continuous evolution of spectral imaging technologies.

While existing review articles provide comprehensive analyses of specific spectral imaging techniques tailored to a specific application, they often lack a broader perspective on system design advancements. In contrast, this study focuses exclusively on recent advancements in spectral imaging system design techniques and technologies, along with their underlying principles. By examining the latest innovations in hardware and optical configurations, this review highlights key developments that have shaped the field in the past five years. Additionally, to provide the necessary historical and technical context, we briefly summarize earlier literature, ensuring a comprehensive understanding of the evolution of spectral imaging systems. This work contributes a unique perspective by emphasizing design methodologies and engineering innovations, offering insights into the future trajectory of spectral imaging technologies. The structure of this review article is organized as follows: Section II provides a general overview of spectral imaging, including its historical background and classifications. Section III discusses the acquisition techniques used in spectral imaging systems. Section IV details the advancements in multispectral imaging system design, while Section V focuses on the progress made in hyperspectral imaging techniques. Finally, Section VI offers conclusions and highlights potential avenues for future research.

II. GENERAL OVERVIEW

A. HISTORICAL BACKGROUND

George Ellery Hale, an eminent astronomer who played an important role in the design of several top-tier telescopes, is credited with inventing and practically utilizing an imaging spectrometer for scientific purposes. Even though he did not explicitly refer to it with such a name. In fact, the term “imaging spectrometer” did not come into common usage until several decades later. In the early 1900s, he constructed a spectrograph equipped with an extended entrance slit that could be continuously moved over the surface of the solar image at a uniform speed at the focal plane of a solar telescope. With a uniform translation of a photographic plate attached to the output of the device, he was able to capture an image of the sun at a single wavelength. This innovative device enabled Hale to conduct pioneering research on the sun, and decades later, revolutionized how astronomers study the sun [66], [67], [68]. Moreover, in the early 19th century, Janssen, an astronomer, employed a double-slit monochromator to observe the solar corona [69], [70]. This technique allowed Janssen to filter light from specific wavelengths to create a monochromatic image of the sun’s outermost atmosphere. This double-slit instrument was referred to as a spectroscope, Janssen was able to isolate specific spectral lines of the corona’s emission spectrum. Despite their limitations, the spectrograph and double-slit monochromator played a pivotal role in the development of spectral imaging and paved the way for modern techniques and equipment that are now utilized to gather detailed spectral data across a broad range of scientific fields.

The inception of the full utilization of spectral imaging technology and the beginning of the term imaging spectrometry began in the 1970s, when the idea for imaging spectrometry emerged from spectral measurements taken in the field to aid in analyzing Landsat-1 data. However, it was not until the advancements in electronics, computing, and software in the 1980s and 1990s that the Earth observation community started to adopt this technique on a wider scale. The outset of hardware development for imaging spectrometry can be attributed to NASA/Jet Propulsion Lab (JPL), which commenced the process with the construction of the Airborne Imaging Spectrometer (AIS) in 1983 [23], in which the application of two-dimensional integrated infrared area arrays in a scientific application was demonstrated [71]. The design of the AIS consists of fore-optics that collect and concentrate light that is directed onto a slit defining the cross-track line. The spectrometer disperses the light and captures each pixel in multiple spectral bands with a two-dimensional array of detectors on the focal plane. AIS captures images of 32 pixels across its width, all at once, and each pixel is imaged across 128 spectral bands within the wavelength range of 1.2 to 2.4 μm , and was utilized to survey various geological and vegetation sites from the air [72], [73].

In 1987, the JPL proposed the development of Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) to advance

the field of imaging spectroscopy by achieving high spectral resolution [74]. AVIRIS was engineered with six individual optical subsystems and five distinct electrical subsystems, enabling it to exploit the narrow yet distinctive diagnostic absorption features of surface materials within the $0.4\mu\text{m}$ to $2.45\mu\text{m}$ spectral range. This was made possible through its 224 contiguous spectral bands, each 10 nm wide, providing an unprecedented level of spectral resolution within aforementioned spectral range [75]. For decades, AVIRIS remained the leading source of high quality spectral data, which enables researchers to conduct highly precise and sophisticated imaging spectroscopy or spectral imaging analyses [76]. Over the last five decades, research in spectral imaging has expanded significantly, driven by its far-reaching applications in both military and civilian domains. This growth has been further accelerated by advancements in hardware technologies, optics, computational power, and image processing techniques, leading to enhanced precision, sensitivity, and acquisition speed. These technological advancements continue to encourage further exploration and innovation, solidifying spectral imaging as a critical tool across multiple scientific and industrial fields.

B. NOMENCLATURE AND CLASSIFICATIONS

Despite its vast potential, the field of spectral imaging faces a significant challenge in the inconsistent use of terminology, which starts with the name itself. The terms used to describe spectral imaging, such as imaging spectroscopy, imaging spectrometry, multispectral imaging, and hyperspectral imaging are frequently used reciprocally among authors in the scientific community [27]. Some authors differentiate these terms based on the number of spectral bands, referring to systems with fewer bands as spectral imaging and those with a larger number of bands as imaging spectrometry. Others define them by the spacing of spectral bands, where contiguous bands labeled as hyperspectral imaging, while non-contiguous bands fall under multispectral imaging [27].

For instance, in 1996, J. Breckinridge presented a comprehensive overview of the fundamental concepts that underlie the practice of imaging spectroscopy (spectral imaging) for the characterization of remote objects and gas vapors. In it, he defines and elucidates the technical nuances of the terms Multi-spectral, Hyperspectral, and Ultra-spectral, within the context of their applications and their specific instrument design approaches [66]. On the other front, Saleh, categorizes optical spectral imaging based on its spectral resolution, the number of spectral bands, their width, and their continuity, classifying them as panchromatic, multispectral, and hyperspectral imaging [2]. Tamim et al. [59] proposed a classification of spectral imaging based on the number of the spectral bands, defining Panchromatic imaging containing 1–2 bands, multispectral imaging as having 3–15 bands, super-spectral imaging as ranging from 16–99 bands, hyperspectral imaging covering 100–500 bands, and ultra-spectral imaging as consisting more than 501 bands. In contrast,

Polder and Gowen [77] recommended only employing terms like “imaging spectroscopy” or “spectral imaging” while avoiding descriptive adjectives such as multi, hyper, super, and ultra to specify the number of bands. This recommendation stems from the inconsistency in nomenclature, where the same number of wavebands might be labeled differently by various authors, leading to terms like “multispectral,” “hyperspectral,” or even “super-spectral.” This lack of a standardized classification system results in terminological ambiguity, potentially contributing to subjective interpretations and inconsistencies.

Consequently, a consistent nomenclature could help clarify the differences and similarities among the different types of spectral imaging systems and provide a more precise and accurate representation of the total number of spectral bands, their contiguousness, and width. Here, spectral imaging is conventionally categorized into panchromatic, multispectral, and hyperspectral imaging (see Figure 1(a)). Panchromatic imaging captures broad-spectrum data in a single band, while multispectral imaging divides the spectrum into discrete bands. Hyperspectral imaging goes further, offering continuous, narrow-band data for more detailed spectral analysis [54], [78] (see figure 2).

Over time, the scientific literature on spectral imaging has undergone a significant transformation. In its early stages, few studies existed, reflecting the emerging nature of the technology and its limited applications. However, as the field progressed, a steady increase in scientific publications emerged, reflecting advancements in terms of applications and system designs. This growth became particularly exponential in the late 1990s and early 2000s, as multispectral and hyperspectral imaging gained noticeability across various disciplines. The chart in Figure 1 (b) illustrates this upward trend, depicting the increase in scientific publications on different spectral imaging types from 1976 to 2024. The data used to generate the figure was sourced from [79], obtained through keyword-based search for “hyperspectral,” “multispectral,” or “panchromatic” solely within the title or abstract sections of the academic publications. This rapid expansion highlights the growing significance of spectral imaging across a wide range of scientific and industrial fields, reinforcing the need for a standardized terminology and classification. Establishing a universally accepted framework will not only enhance academic discourse but also facilitate cross-disciplinary collaboration, ultimately accelerating innovation.

1) PANCHROMATIC IMAGING SYSTEM

A Panchromatic Imaging System (PIS) is characterized by capturing a wide spectrum of light in a single channel, it is without any wavelength-specific details and is particularly valued for its high spatial resolution. The object classification is accomplished based on spatial characteristics [80], and it is unable to distinguish between different types of materials based on their spectral signatures. In addition, the

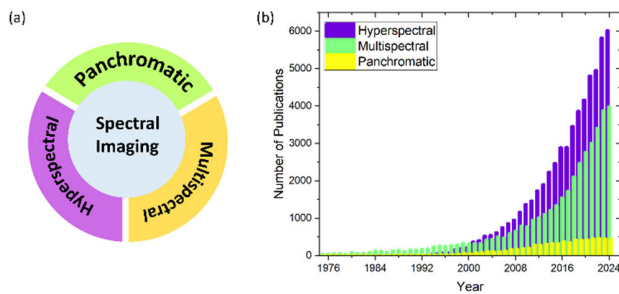


FIGURE 1. (a) Types of spectral optical imaging based on the number of spectral bands and contiguousness (b) Progression of Scientific Publications pertaining to the types of spectral imaging over the span of five decades.

broader range of wavelengths it captures can result in a lower signal-to-noise ratio and overall sensitivity, particularly, in the infrared and ultraviolet regions. Despite these, panchromatic imaging remains a valuable tool for many applications and is continually evolving with new technologies and techniques. For instance, high-resolution panchromatic sensors are now being integrated into both satellite and aerial imaging platforms, providing detail imagery for applications such as urban planning, agriculture, and environmental monitoring [81]. Moreover, Panchromatic imaging systems have been incorporated in several satellite imaging missions, such as the WorldView-3 mission by DigitalGlobe [82], the Panchromatic Imaging Fourier Transform Spectrometer (PanFTS) by NASA [83], and several other missions whose details can be found here [84], [85].

In recent years, there has been a significant increase in the use of panchromatic imaging technology in a variety of scientific applications. For instance, the detection of crop row orientations in agriculture [86], an airport management and automated military warfare and airplane detection system [87], the giant kelp detection in water [88], efficient hyperspectral spectral un-mixing [89], and so on. Although it lacks the spectral discrimination of multispectral and hyperspectral imaging, it offers very high spatial resolution and is often combined with other spectral imaging modalities to enhance the spatial details of multispectral or hyperspectral data through a process called pansharpening [90], [91]. This integration enhances the quality of the data, providing more detailed and informative imagery, particularly for remote sensing applications. However, keeping an excellent performance of the spectral and spatial resolution of the fused image is rather challenging due to the spectral distortion, making object classification even more demanding [92].

The development of sophisticated image processing algorithms has significantly improved the quality and utility of pansharpening technology. Algorithms such as Bayesian pansharpening, wavelet transform, deep learning techniques, and many others are being increasingly employed to enable more accurate pansharpening techniques, supporting a wide range of applications from environmental monitoring to defense and security. There are so many reports from the

literature on the development of different algorithms to achieve excellent visual quality of panchromatic sharpening. For comprehensive and thorough examination of PAN sharpening techniques, we cordially invite the reader to peruse these articles [93], [94], [95], [96], [97], [98], [99], [100], [101], [101], [102], [103], [104], [105], [106], [107], [108], [109]. They provide an in-depth analysis of the latest developments in PAN sharpening techniques and are equipped with detailed information for researchers and practitioners seeking to expand their understanding of this rapidly evolving field.

While panchromatic imaging excels in providing high spatial resolution, its primary limitation is that it combines all wavelengths into a single channel, producing a grayscale image. This means it lacks the spectral information available in multispectral or hyperspectral imaging, making it difficult to distinguish between different materials or surfaces based on their spectral signatures. Without spectral differentiation, panchromatic images cannot effectively distinguish between objects or surfaces that have similar reflectance but differ spectrally. This can limit their effectiveness in applications like vegetation analysis, mineral identification, or water quality assessment. Often, panchromatic imaging needs to be complemented with multispectral or hyperspectral data to provide a comprehensive analysis. This increases the complexity and cost of data acquisition and processing. Advanced image processing techniques, such as pansharpening or feature extraction, add to the computational complexity. Efficient processing pipelines are needed to handle the data effectively, which can be a challenge for resource-limited environments. Moreover, when used in conjunction with multispectral or hyperspectral data, precise geometric alignment (co-registration) is crucial. Misalignments can lead to inaccuracies in subsequent analysis and data fusion processes. Besides, the panchromatic images captured from aerial or satellite platforms can suffer from geometric distortions due to the angle of capture, altitude variations, and sensor movement. These distortions need to be corrected through geometric rectification processes, which can introduce errors if not done accurately. These limitations necessitate the integration of panchromatic images with other types of remote sensing data and advanced processing techniques to achieve accurate analyses in various applications.

2) MULTISPECTRAL IMAGING SYSTEM

Multispectral imaging (MSI) systems are designed to capture information from a predefined set of spectral bands, selected based on the specific requirements of an application. Unlike traditional panchromatic imaging, which records broad red, green, and blue (RGB) bands, MSI acquires data in multiple discrete spectral bands, enabling a more detailed analysis of an object's spectral properties [110], [111]. Typically, MSI utilizes fewer than twenty to twenty-five spectral bands, offering a balance between spectral resolution and data efficiency. This capability makes MSI a powerful tool in various scientific and industrial domains, including environmental

monitoring, remote sensing, agriculture, and biomedical imaging. MSI systems employ filters, sensors, or wavelength-selective devices to isolate different spectral bands, creating a stack of images where each layer corresponds to a specific wavelength. This method allows for the extraction of critical spectral features, facilitating material identification, classification, and scene analysis. Compared to hyperspectral imaging (HSI) systems, MSI offers higher temporal resolution and reduced data volume, making it more efficient in data acquisition and processing while still providing valuable spectral insights.

In an MSI system equipped with a set of spectral filters, the acquisition process involves selectively capturing specific wavelength bands from the scene. Each filter allows only a narrow spectral range to reach the sensor, ensuring that the recorded intensity corresponds to a particular portion of the spectrum. The captured intensity at each spectral band can be mathematically expressed as:

$$I_k(x, y) = \int_{\lambda} \mathcal{F}(x, y, \lambda) \cdot \Gamma_k(\lambda) \cdot \Upsilon(x, y, \lambda) d\lambda + T_k(x, y, \lambda) \quad (1)$$

where $I_k(x, y)$ represents the intensity recorded at pixel (x, y) for the k -th spectral band, $\mathcal{F}(x, y, \lambda)$ is the reflectance or transmittance of the object at wavelength λ , $\Gamma_k(\lambda)$ is the spectral response function of the k -th filter, $\Upsilon(x, y, \lambda)$ is the illumination spectrum, and $T_k(x, y, \lambda)$ accounts for the sensor noise and environmental interference. The formulation encapsulates the interaction between the incident light, the object's spectral properties, and the imaging system's response, providing a foundation for spectral feature extraction and material characterization.

The inception of MSI system trace back to the Earth Resource Technology Satellite (ERTS) program, initiated by NASA in partnership with the United States Geological Survey (USGS) in 1961. Later renamed the LandSat program, this initiative marked a pivotal moment in the deployment of MSI technology into space [112]. The LandSat program has contributed extensively to global land cover and land use studies by providing high-quality multispectral datasets used in analyzing vegetation dynamics, agricultural patterns, water resource distribution, and soil composition and more [113], [114]. In addition to the LandSat, other notable spaceborne MSI missions include: the SPOT (Satellite Pour l'Observation de la Terre) series by the French Space Agency, as well as commercial satellites such as QUICKBIRD by DigitalGlobe and IKONOS and ORBVIEW by GeoEye. These satellites provide commercially available high-resolution MSI data, which are widely used in scientific research, defense, and geospatial intelligence [114], [115]. Despite having a lower spectral resolution than hyperspectral imaging, MSI remains highly adaptable, with the number and range of spectral bands tailored to specific applications. The trade-off between spectral resolution and data efficiency makes MSI a preferred choice for applications requiring rapid, and cost-effective spectral analysis.

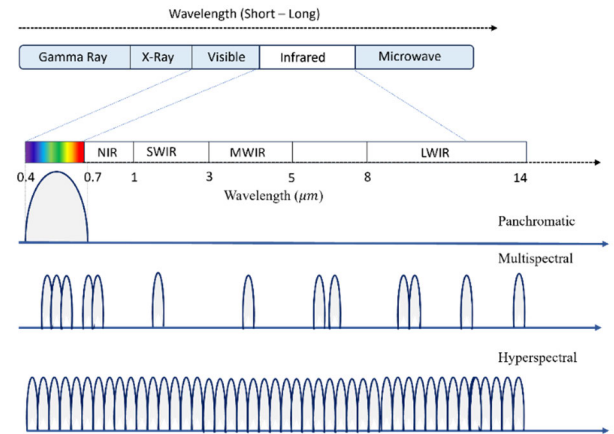


FIGURE 2. Classifications of spectral imaging: (i) Panchromatic imaging: employs a single wideband wavelength in the visible spectrum- this approach captures a broad range of wavelengths without differentiation, offering a comprehensive but undetailed view of the scene or object under observation. (ii) Multispectral imaging: employs many discretized spectrums at different intervals within a selected range. These imagers provide a more detailed and selective view, enabling analysis and identification of specific features or materials within the scene. (iii) Hyperspectral imaging: employs many narrowly spaced wavelength spectrums. This technique captures a high-resolution spectral signature for each pixel, allowing for precise identification and characterization of materials or substances based on their unique spectral fingerprints. The wealth of information obtained through hyperspectral imaging enables detailed analysis and discrimination of subtle variations within the observed scene or object.

3) HYPERSPECTRAL IMAGING SYSTEM

Advancements in imaging technologies have led to the development of multiband spectral imagers capable of simultaneously capturing hundreds of spectral bands per pixel. These hyperspectral imagers, can record continuous spectral data across hundreds of contiguous spectral bands, providing a powerful tool that revolutionizes data acquisition for numerous applications and research domains [116], [117], [118] (See figure 2). Unlike multispectral imaging, HSI system collects data across a continuous spectrum, providing a highly detailed “spectral signature” for each pixel. This allows for precise identification of materials based on their unique spectral properties. Initially, hyperspectral imaging was primarily developed for classified military surveillance operations. However, in recent years, significant progress has been made to reduce costs and manage the data processing demands associated with HSI. Hyperspectral imaging systems can now be deployed on various platforms, including ground-based (handheld or stationary), space-based, and airborne systems. Ground-based HSI systems are widely used in applications that require high spectral resolution, such as precision agriculture, biomedical imaging, food processing, and defense. In contrast, space-based and airborne systems capture spatial and spectral information remotely, either from satellites or onboard aircraft and drones. Several notable space missions are equipped with HS imaging capabilities. These include the MightySat II and Earth Observing-1 (EO-1) missions by the US Air Force Research Lab and NASA, the PROBA

mini-satellite mission, and ENVISAT by the European Space Agency (ESA). Other missions include PRISMA by the Italian Space Agency (ISA), HISUI by the Japan Aerospace Exploration Agency, and most recently, EnMAP by the German Aerospace Center [119], [120]. The rapid advancement in the design of hyperspectral imaging systems has been driven by the synergy between improvements in optical technology—leading to more reliable, faster, and compact devices—and the development of powerful models and algorithms for analyzing hyperspectral data. These advancements have significantly enhanced the performance and accessibility of hyperspectral imaging across various fields.

III. SPECTRAL IMAGING ACQUISITION TECHNIQUE

Spectral imaging encompasses a range of acquisition methodologies playing a pivotal role in the detailed characterization of the object within a scene. These approaches, ranging from instantaneous acquisition to various scanning techniques, facilitate the extraction of spectral signatures that are crucial for numerous applications. Spectral imaging acquisition simultaneously encodes spatial and spectral data, generating a per-pixel spectral signature that characterizes the material's interaction with incident light across multiple wavelengths. This spectral signature quantitatively represents the reflectance, transmittance, or absorbance properties of the material as a function of wavelength. The acquired spectral image can be represented as a function of spatial coordinates (x, y) and wavelength (λ) :

$$I(x, y, \lambda) = \Re(x, y, \lambda) \cdot \mathcal{S}(\lambda) \cdot \mathcal{L}(x, y, \lambda) + \mathcal{N}(x, y, \lambda) \quad (2)$$

where $I(x, y, \lambda)$ is the measured intensity at position (x, y) and λ , $\Re(x, y, \lambda)$ is the reflectance of the object at (x, y) for wavelength λ , $\mathcal{S}(\lambda)$ is the spectral sensitivity function of the camera sensor, $\mathcal{L}(x, y, \lambda)$ is the illumination intensity at each wavelength, and $\mathcal{N}(x, y, \lambda)$ represents the noise. Moreover, spectral imaging data can be mathematically represented and discretized as a third-order tensor:

$$\mathcal{K} = \{\mathbb{N}_{\mathcal{P}, \mathcal{Q}, \mathcal{R}}\}, \mathcal{P} = 1, \dots, \mathbb{P}, \mathcal{Q} = 1, \dots, \mathbb{Q}, \mathcal{R} = 1, \dots, \mathbb{R} \quad (3)$$

where $\mathcal{P}, \mathcal{Q}$ denote the spatial dimensions, and $\mathcal{R}$ represents the number of spectral bands, and $\mathbb{N}_{\mathcal{P}, \mathcal{Q}, \mathcal{R}}$ is the intensity recorded at pixel $(\mathcal{P}, \mathcal{Q})$ for spectral band $\mathcal{R}$. Moreover, the spectral signature of any given pixel $(\mathcal{P}, \mathcal{Q})$ can be obtained by normalizing the measured intensity with respect to illumination and sensor response:

$$\mathbb{S}_{\text{sig}}(x, y) = \frac{I(x, y, \lambda)}{\mathcal{L}(x, y, \lambda) \mathcal{S}(\lambda)} \quad (4)$$

This formulation in eq. (4) provides a way to normalize the spectral response, ensuring that the extracted spectral signature is independent of the illumination intensity and sensor characteristics. The normalization process is crucial

for accurate material identification and comparative spectral analysis. Without proper normalization, variations in lighting conditions and sensor properties could lead to inconsistencies in spectral measurements.

As summarized in Figure 3, spectral imaging data acquisition is primarily achieved through two principal methods:

- 1) **Snapshot technique:** acquires the entire spectral data instantaneously for all pixels.
- 2) **Scanning technique:** acquires the spectral data while spectrally or spatially moving along the scene continuously.

The scanning acquisition methodology is further subdivided into two categories:

- **Spectral scanning** (commonly referred to as staring), wherein a stack of images is acquired for each tunable scan of a wavelength at a time.
- **Spatial scanning**, where spectral information is gathered progressively across segments of the target.

The spatial scanning itself is further classified into:

- 1) **Point Scanning** (Whiskbroom) sequentially scans individual points within the spatial domain of the target, acquiring spectral information at each specific location.
- 2) **Line Scanning** (Pushbroom) scans the target line-by-line, enabling the simultaneous acquisition of both spatial and spectral data.

Additional scanning techniques include spatio-spectral scanning, which produces diagonal slices of the target, with each slice representing a two-dimensional spatial image encoded with spectral information [121]. Another important technique is time scanning, which involves capturing a series of images over a period, combining both spatial and spectral data. The spectral information is subsequently extracted through mathematical post-processing [122].

The following sections provide a detailed examination of these spectral imaging acquisition techniques. However, this review does not include discussions on spatiotemporal scanning and time scanning methodologies.

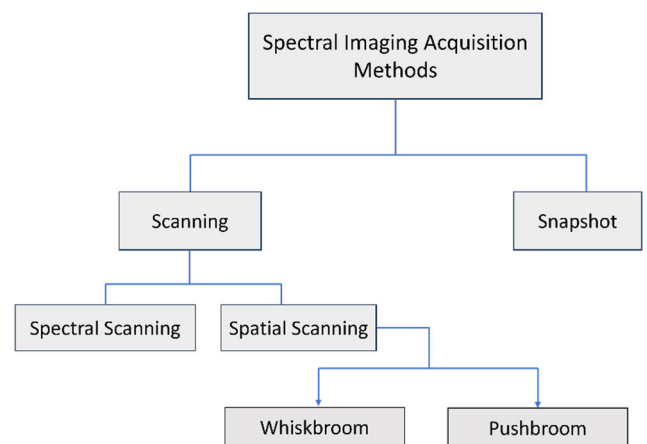


FIGURE 3. Classifications of spectral imaging acquisition techniques.

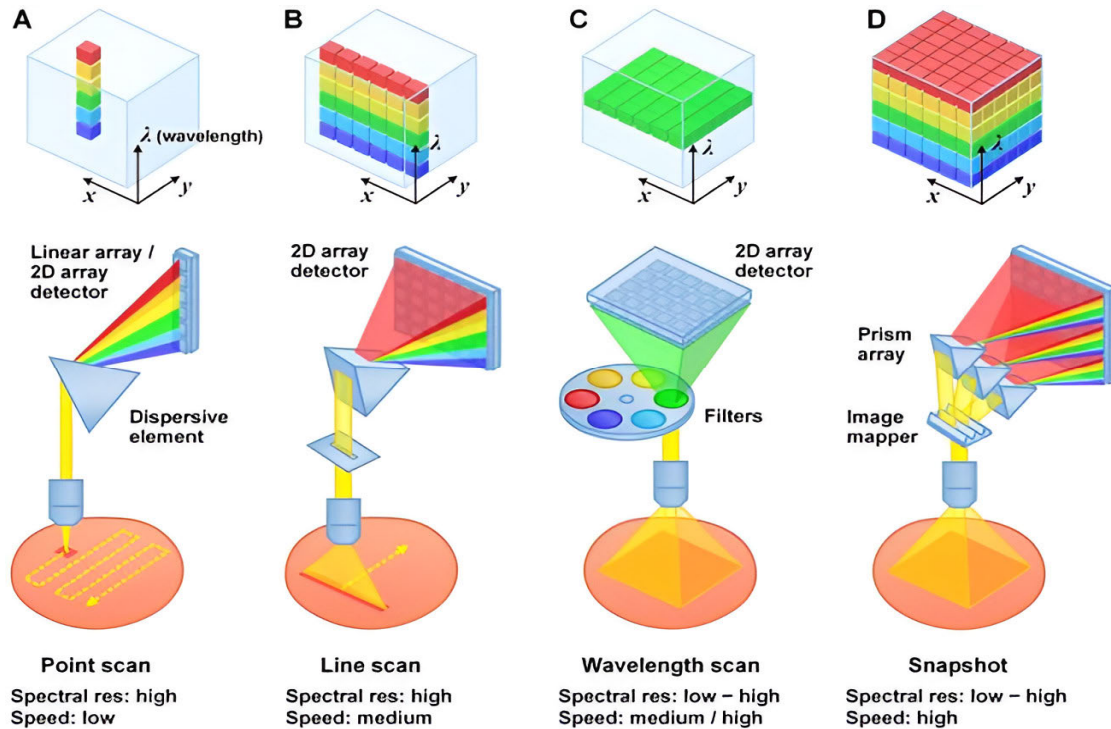


FIGURE 4. Typical spectral imaging acquisition techniques (a) Point scanning or whiskbroom, (b) Line scanning or pushbroom, (c) Spectral scanning (SC) or staring, and (d) Snapshot (SS) – which allows simultaneous acquisition of the entire 3-D data cube within a single integration period [123].

A. SNAPSHOT TECHNIQUE

This method involves capturing complete data of the target, encompassing both spatial and spectral information for all pixels instantaneously, as shown in Figure 4 (d). This approach, familiar in astronomy as integral field spectroscopy, spatially divides the scene onto a detector, allowing dispersion to gather spectral information. Technologies like integrated field spectroscopy with faceted mirrors (IFS-M), Coherent Fiber Bundles (IFS-F), and Lenslet Array (IFS-L) have emerged from this method, each offering nuanced spectral imaging capabilities. For detailed insights into these snapshot spectral imaging technologies and many others, refer to this work [27]. Another showcased approach integrates a detector with a tilted filter to achieve snapshot spectral imaging [124]. This design employs an optical duplicator and a lenslet array to replicate the scene onto the detector through specific filter tiles arranged in a two-dimensional matrix. This amalgamation creates a comprehensive data cube, facilitating simultaneous representation and analysis of multiple spectral bands, enabling a holistic comprehension of the scene's spectral attributes. Snapshot spectral imaging presents advantages such as rapid data acquisition in aerial imaging and real-time applications, efficient light collection as it involves no physical motion (unlike scanning techniques), and minimal motion artifacts. However, in dynamic scenes with spatial or spectral changes, this method may lead to blurring in the acquired image.

B. SPECTRAL SCANNING (STARING) TECHNIQUE

In the spectral scanning or staring approach, a stack of images is captured during each tunable scan of a wavelength by registering the optical power within the filter's bandwidth.

As seen from Figure 4 (c), unlike snapshot or conventional scanning methods, the staring technique captures images of the scene one wavelength at a time, which is subsequently combined to form spectral data cubes. One common implementation involves using a set of color filters with distinct wavelength transmission characteristics. Examples include liquid crystal tunable filters, circular variable filters, acousto-optic tunable filters, or filter wheels, which are typically placed either before the light source or in front of the detector. This configuration allows for the acquisition of a predefined set of two-dimensional images of the target at different wavelengths or spectral bands [125], enabling the construction of a spectral data cube by stacking the wavelength-specific images. The primary advantage of the spectral scanning technique is its compatibility with conventional imaging systems, such as microscopes, allowing for seamless integration with appropriate filters for spectral microscopy. The spectral scanning methodology facilitates flexible image acquisition, allowing for the capture of diverse spectral channels as needed in various scenarios. However, attaining optimal outcomes from spectral scanning imagers significantly depends on fine tuning the filter, which inherently tends to be slow, resulting in low throughput and

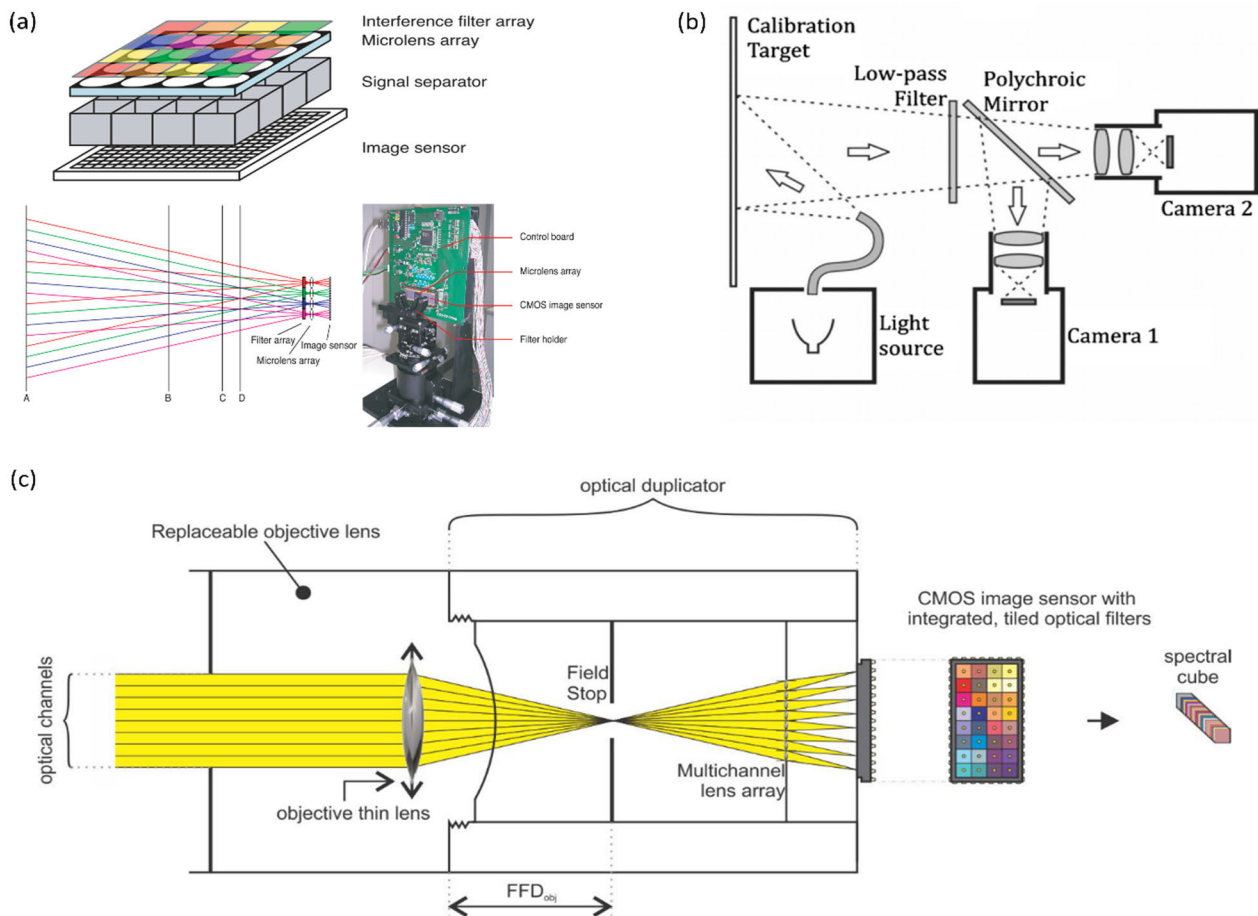


FIGURE 5. (a) The schematic representation outlining the multispectral imaging capturing using TOMBO system, depicting the filter architecture, the schematic diagram of the system, and the Photograph of the TAMBO system prototype [156]. (b) The schematic depiction of the multi-bandpass filter based MSI reported in [158]. (c) The schematic layout of the snapshot multispectral imager reported in [124].

limited compactness. Moreover, challenges such as polarization sensitivities, extended tunability time, and physical thickness contribute to its suboptimal performance. Many attempts have been made to implement faster staring spectral imagers based on different types of filters with compact and faster spectral scan [126], [127], [128]. However, despite the remarkable improvement in the spectral scanning speed they still suffer from lower throughput, thus, limiting their performance [125].

C. SPATIAL SCANNING TECHNIQUE

The spatial scanning technique operates by leveraging the relative motion between the imager and the object, with spatial movement originating from either the imager or the object itself. Unlike traditional filter-based methods, this approach utilizes diffractive elements such as prisms or gratings for wavelength separation. These elements disperse the light into its constituent wavelengths, and the detector captures the corresponding intensities. Spatial scanning techniques are generally categorized into two primary classifications:

1) POINT SCANNING (WHISKBROOM)

The point scanning or whiskbroom is done by moving across the target, capturing the spectral data at individual points. This is achieved by $x - y$ spatial scanning while simultaneously measuring the full spectral domain of the nearby regions of the target (see Figure 4 (a)). Whiskbroom scanning mechanisms were employed in a variety of airborne imaging spectrometers. For example, missions like AVIRIS and HyMAP utilize whiskbroom scanning technology, where a rotating mirror performs side-by-side scans perpendicular to the imager's direction of movement [129]. As the mirror sweeps from one edge of the field of view to the other, it directs light beams from different parts of the scene onto a dispersive element, such as a prism or grating. The dispersed light is then captured by a linear detector array, where each detector element corresponds to a specific wavelength range. Whiskbroom scanning requires fewer detectors, simplifying the calibration process and providing greater control over the depth of field, which helps reduce background noise around the target. Despite the advantage of recording both

spatial and spectral data at each pixel, whiskbroom scanning is constrained by longer acquisition times due to the mechanical movement required to scan both spatial dimensions. Consequently, its use is largely limited to specialized imaging techniques like confocal microscopy, where point-by-point measurements are taken while rejecting out-of-focus signals [130], [131].

2) LINE SCANNING (PUSHBROOM)

Line scanning, also known as pushbroom scanning, is one of the most widely used techniques in spectral imaging. It enables the acquisition of an entire row or line of spectral imagery simultaneously, as illustrated in Figure 4(b), with the aid of a detector array and a slit [132]. This method employs a narrow rectangular aperture or slit to capture the spatial information of the target along one direction, while a diffraction grating or prism disperses the incoming light into its constituent wavelengths in the orthogonal direction. A two-dimensional detector array is used, with one dimension capturing spectral information and the other capturing spatial data, generating three-dimensional data cubes from consecutive line acquisitions. In remote sensing applications, the movement of the carrier (aircraft or satellite) facilitates scanning along the track. The inverse of the line frequency corresponds to the pixel dwell time, allowing efficient data capture. Compared to whiskbroom scanning, pushbroom scanning is significantly faster [133], and more compact, due to the reduced number of moving parts. In fact, pushbroom systems are typically more than 100 times faster than whiskbroom systems, allowing for longer pixel dwell times and higher spatial and radiometric resolution. Additionally, pushbroom scanning offers superior spectral resolution, sensitivity, signal-to-noise ratio (SNR), and angular resolution, making it a preferred choice for bench-top systems and various industrial applications [111].

IV. ADVANCES IN MULTISPECTRAL IMAGING SYSTEM DESIGN

Multispectral imaging stands as a pivotal domain offering insights into material characterization, object identification, and beyond. By capturing and analyzing how objects reflect, absorb, or emit light across different spectral bands, MSI uncovers information about the composition, structure, and properties of a scene or object. This allows for an understanding of the scene or object that goes beyond what is possible with traditional imaging techniques. The continuous advancements in multispectral imaging systems have propelled innovative designs, pushing the boundaries of spectral sensitivity, spatial resolution, and overall performance. The use of multispectral imaging systems enables capturing and analyzing data across multiple discrete spectral bands that become indispensable tools across many disciplines, transcending traditional imaging capabilities by providing insights beyond the visible spectrum. From biological sampling [134], analysis and detection of some chemical elements in living plants [135], management of fertilizer in

some specific farming sites [136], to facial identification systems in an unfavorable scenario with for example poor illumination, co-operative objects, and short standoff distance generally experienced by military and other law enforcement agencies [137], meat Quality Assessment [138], [139], UAV based remote sensing for the detections of pathogens on leaf surface [140], and woodlands [141], and several other [142], [143], [144], [145], [146], [147], [148], [149], [150], [151], [152], [153], [154], [155] and more. Progress in multispectral imaging system design continues to push the boundaries of spectral imaging capabilities, making it an essential tool for scientific research, industry, and so on. As MSI systems continue to evolve, their impact will further expand, offering new possibilities for spectral analysis across a wide range of disciplines.

A. TWO DECADES OF PROGRESS: FROM EARLY CONCEPTS TO RECENT DEVELOPMENTS

The literature on multispectral imaging systems (MSI) encompasses numerous reports detailing various design concepts and technological advances. While some of these advancements are summarized in Table 1, the following section provides an overview of earlier design concepts, followed by a comprehensive review of more recent developments. Among the early contributions over the past two decades is the work by Shogenji et al. [156], who in 2004 proposed a thin observation module by bound optics (TOMBO) for MSI, inspired by multi-aperture optical imaging systems [157]. The design consisted of a microlens array, an interference filter array, and an image sensor. The TOMBO-based MS imager captures multiple sub-images through a series of micro-lenses arranged in a grid, which are then digitally processed to reconstruct the original image. Figure 5 (a) illustrates the filtering scheme, the proposed design, and the prototype, respectively. Four years later, Duempelmann et al. [166] expanded on this concept by introducing a diversity plate positioned in front of 18 separate lenses. The diversity plate, containing distinct optical elements, modified the optical information of each sub-image, effectively increasing the number of spectral channels to 18, spanning a spectral range of 460 to 889 nm.

The use of filters and camera technologies to capture MS data was reported by Themelis et al. [158] in 2008, where they introduced an MSI system that incorporated color CCD cameras coupled with a multiple-bandpass filter. Their setup, depicted in Figure 5 (b), enabled the simultaneous acquisition of images across several predefined spectral bands, allowing the concurrent capture of eight spectral bands within the visible range using two color CCD cameras and a polychroic mirror. In 2010, Gupta et al. [159] employed multilayer dielectric mirrors to create a Fabry-Perot filter integrated with an InGaAs camera, achieving 16 spectral channels within the short-wavelength infrared region. Similarly, in the same year, Hadar et al. [160] reported a real-time MSI system utilizing a freestanding metallic subwavelength grating-based bandpass

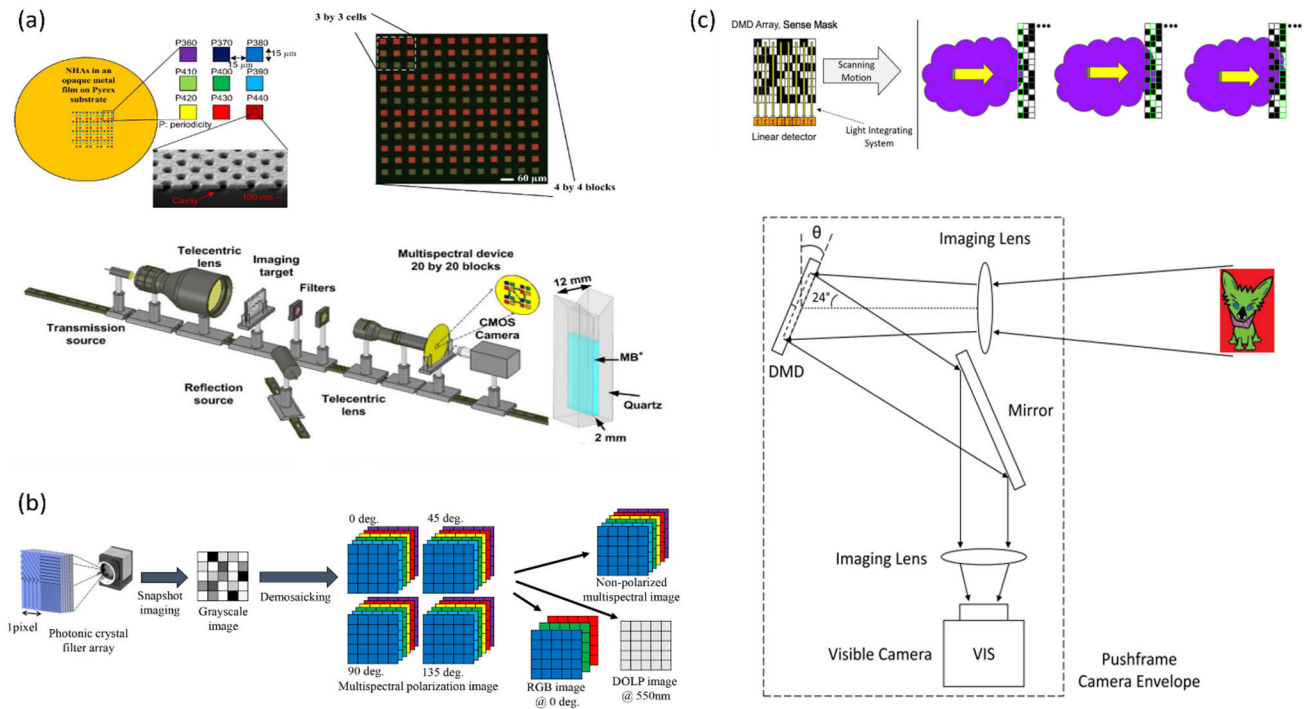


FIGURE 6. (a) Nanohole array based multispectral imager reported in [162] features a 4×4 block array, where each block contains a 3×3 grid cells with each of cell representing the nanohole (NHA). The imaging setup supports snapshot multispectral imaging in both reflection and transmission mode. (b) The outline of the multispectral polarization filter array imaging proposed in [163]. (c) Push-frame multispectral imaging system as presented in [167] featuring the schematic of the Push-frame MSI acquisition process, and the schematic of the experimental setup of the system.

filter, paired with a HgCdTe camera to cover 10 spectral channels across a wavelength range of 500 nm to 2500 nm. In 2013, Grahn and Geladi [131] introduced a novel approach for a compact snapshot MSI. Their design integrated a tilted filter onto a CMOS detector, enabling 32 spectral channels within the 600 nm to 1000 nm range. Illustrated in Figure 5 (c), the design included an optical duplicator and objective lens, which facilitated high spatial resolution by positioning a field stop at the focal plane. A multichannel lens array was employed to relay the field stop onto the center of each filter tile, aiming to create a more compact spectral imaging system, closer to the realm of hyperspectral imaging due to its extensive spectral coverage. Further innovations include a design integrating vertical silicon nanowires as filters with a monochrome camera, reported in [161], which enabled an 8-channel MSI system spanning 400 nm to 1000 nm. Moreover, multi-element mosaic filters, leveraging extraordinary optical transmission through closely spaced nanohole arrays, were proposed by Najiminaini et al. [162].

The nanohole arrays, fabricated within a 100-nm gold film, facilitated a two-dimensional MSI system functioning in both reflection and transmission modes (see Figure 6 (a)). A concept for a polarization filter-based MSI system was introduced by Shinoda et al. [163], as summarized in Figure 6 (b). The filter design utilizes a wavy, thin-film filter array fabricated via an auto-cloning process, resembling the structure of photonic crystals. The filter is integrated with a

monochrome camera and exploits its lattice pitch and wavy geometry to modulate polarization characteristics and spectral sensitivity. The system covers a spectral range of 420 to 720 nm, with image recovery and analysis achieved through a demosaicing technique. The evolution of these designs has been further exemplified by subsequent developments in plasmonic filter arrays [164], [165] and tunable plasmonic filters [166] demonstrating ongoing advancements in the field.

1) SINGLE PIXEL DETECTORS TECHNOLOGY

The single pixel detection (SPD) captures information from a scene using a single-point detector rather than a traditional array of pixels. This is particularly advantageous in spectral bands where high-efficiency, low-noise detector arrays are either unavailable or cost-prohibitive. Single-pixel detectors work by sequentially projecting structured patterns onto the target, with the detector measuring the intensity of light reflected or transmitted through each pattern. By using computational techniques, the collected signals can be processed to reconstruct a full image, even in low-light or specialized spectral conditions. Nonetheless, a primary constraint inherent to SPD is that it often requires longer acquisition times, making it best suited for static or slow-moving scenes. This engenders a trade-off between image resolution and acquisition speed, thus restricting the utility of these imaging systems to scenarios where either time or resolution

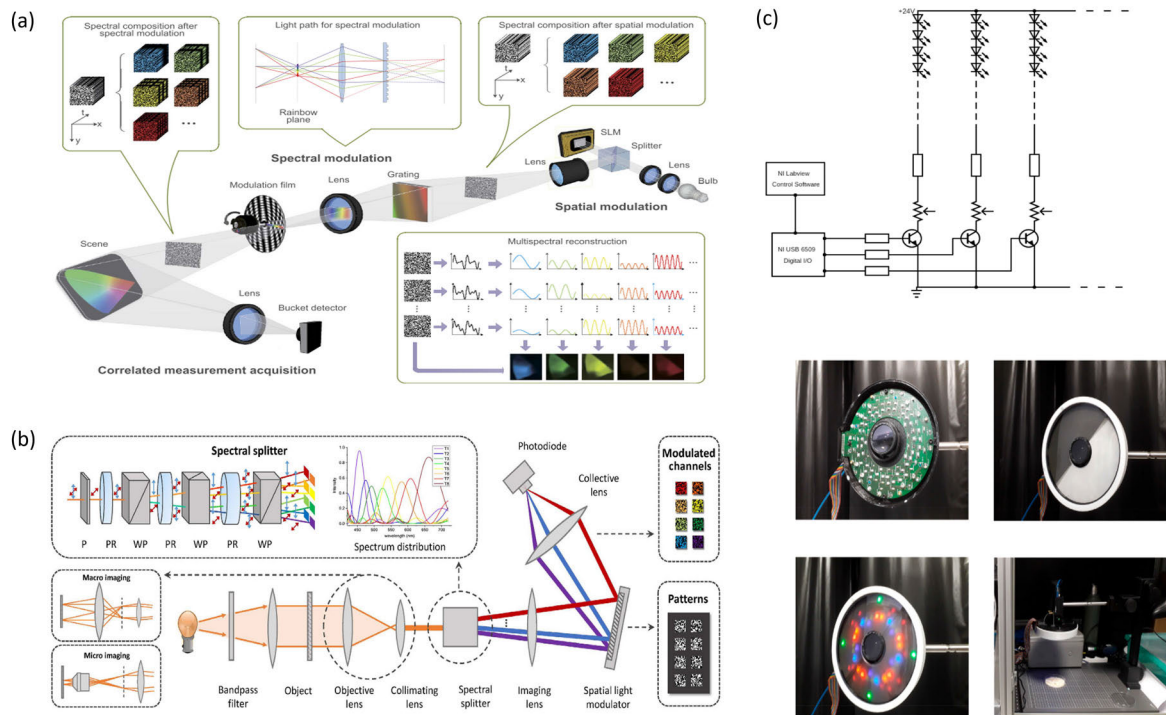


FIGURE 7. Illustrates three distinct multispectral imaging systems. (a) The schematic diagram of the proposed multispectral single-pixel imaging system reported in [168]. (b) The proposed MSI system based on spatio-spectral modulation reported in [169]. The system begins with light emitted from the source, which illuminates the target. This light then traverses through a spectral filter, dividing it into 8 distinct spectra. The spectra undergo encoding via a spatial light modulator present on each channel, and the encoded patterns are detected by a single pixel detector. The components of the spectral filter shown in the inset: P-polarizer, PR-Phase Retarder, and WP-Wollaston prism. (c) Schematic of the LED-based Multispectral imager reported in [171], including the illustrative diagram of the proposed system's control scheme. A photograph showing the LED PCB after the removal of the front cover. The system with an added polarizer and diffuser. The assembled system showing the illumination of the LED arrays, and the assembly of the multispectral imaging system is observed from an edge-on perspective. The camera is affixed to an optical post, while the light-head is positioned on the camera body using a 3D printed structure.

constraints are not critical. To address this, Noblet et al. [167] propose the design of a push-frame single pixel camera (SPC) MS imaging system incorporating a digital micromirror device (DMD), achromatic doublet lens, mirror, and a camera enveloped fitted with Bayer filter. Hadamard patterns are encoded onto the DMD, and as the scene traverses across these patterns, the single-pixel linear detector selectively captures a specific column of reflected light emanating from the scene, facilitated by the cylindrical lenses. The iterative acquisition of signals from each discrete linear detector element is synchronized with the progressive movement of the scene across each column of the pattern, thereby enabling a rigorous reconstruction of each scene column within the short-wave infrared (SWIR) region. The schematic of the experimental setup and the mechanism of their proposed push-frame image acquisition are illustrated in Figure 6 (c).

Additionally, a proof of concept for the design of multispectral single-pixel imaging has been proposed by Bian et al. [168]. Summarized in Figure 7 (a), the proposed system employs a spatial light modulator (SLM) to spatially modulate the broadband light emitted by a bulb, resulting in a sequence of 2D random illumination patterns. Subsequently, the 2D patterns undergo spectral stretching, accomplished

through a diffraction grating and a series of lenses, resulting in the formation of a rainbow stripe that is modulated before being spatially transformed into a two-dimensional pattern. The 2D-spatial pattern and the sinusoidal spectral pattern are recorded by a bucket detector. The fast response of the bucket detector allows multiplexing of the 3D data cubes of the scene to a dense 1D data collection series. The data series is then demultiplexed using computational techniques within the framework of single-pixel imaging. The subsequent image reconstruction involves decoding spectral response signals via Fourier decomposition and demodulating spatial information using a compressive sensing-based algorithm. Following that, is the work of Li et al. [169] where by harnessing the high spatial resolution of SLMs, they proposed and designed an SPD-based MSI system. The system is based on an all-optical spectral splitting device that can separate the incoming light from the target into distinct spectral channels.

Each channel is concurrently and uniquely encoded with an independent pattern using the SLM, and the SPD simultaneously records the encoded signals. This innovative methodology aims to enhance spectral imaging capabilities through optical modulation instead of mechanical one while leveraging the advantages offered by 3D compressive sensing

techniques. The schematic of the proposed design is shown in Figure 7 (b) depicting the operation of the system and the optical architecture of the spectral splitter shown in the inset.

2) LEDS ARRAYS TECHNOLOGY

Recent advancements have highlighted the use of LED arrays as light sources for multispectral imaging (MSI) systems, providing a cost-effective approach to spectral data acquisition. These systems emit light across specific wavelengths, either sequentially or parallel, to capture wavelength-specific information and cover multiple spectral channels. The technique involves illuminating the target with precise wavelengths and capturing images at each wavelength using a monochrome camera, which are then combined to form spectral data cubes. One of the early implementations was reported by Parmar et al. [170] in 2012, where an MSI system was developed to perform rapid sequential captures using a range of LEDs with distinct colors. The system featured a printed circuit board (PCB) with 216 LEDs, organized into groups of eight different types, along with a diffuser board. Their setup enabled high-quality multispectral data acquisition with improved temporal resolution. A more recent design featuring 18 different LEDs was introduced in [171], using PCB-based LEDs with a control scheme depicted in Figure 7 (c), a monochrome camera, and a 3D-printed housing. Their system covered a wider spectral range from 365 nm to 1065 nm across 18 spectral channels, offering a cost-efficient design. Similarly, Herrera-Ramírez et al. [172], reported a compact MSI system utilizing LED-based illumination across 23 bands within the UV (370 nm) to IR (1630 nm) range, synchronized with two imaging sensors. Recently, a similar design was reported by Taylor-Williams et al. [173] for MSI acquisition of nailfold capillaries to extract important features like the hemoglobin oxygenation of the blood. Thereby enhancing the potential to have an improved sensitivity in diagnosing and measuring the progression of microvascular diseases.

In 2018, Liu et al. [174], introduced a tunable LED-based underwater MSI system named TuLUMIS (Tunable LED-based Underwater Multispectral Imaging System). Their proposed TuLUMIS system represents a significant advancement in innovative underwater MSI solution that enhances spectral resolution using pressure-neutral LEDs instead of traditional optical filters. By synchronizing a monochrome camera with tunable LED illumination, it captures spectral information across eight wavelengths (405–660 nm), significantly improving color discrimination (76.66% over RGB cameras). Unlike bulky HSI systems, their TuLUMIS is compact and cost-effective, operating at depths of 6000 meters without requiring pressure housing for LEDs. Field tests on the RV Maria S. Merian (MSM61 cruise) (see Figure 8 (a) (ii)) confirmed its ability to capture multispectral images of pelagic fauna, though motion artifacts and illumination inconsistencies were noted. Future improvements, such as optimized LED spacing and real-time intensity tuning, could

further enhance its capabilities. TuLUMIS offers a scalable, efficient solution for applications like deep-sea exploration, coral reef monitoring, and marine habitat classification. Bolton et al. [175], proposed a compact and cost-effective MSI system using multi-spectral LED arrays for illumination, covering 13 spectral channels within the 403 nm to 948 nm range. The detection module comprised a low-cost monochrome camera paired mechanically with a vision lens and a linear polarizer, while in-house software managed the light control and image analysis. The block diagram of the system architecture, depicted in Figure 8 (b), features three main components: an illumination module, a detection module, and a tablet computer equipped with control software linked to the LED drive via an Arduino Nano board. Additionally, several studies have explored MSI systems based on LED arrays combined with dielectric multichroic filters (sinusoidally varying filters in terms of transmission spectra) [176], [177], [178], [179]. These systems, known as Fourier MSI systems, draw inspiration from Fourier transform spectroscopy, acquiring spectral samples through a series of interferometric measurements while manipulating the optical path difference (OPD).

3) SPECTRAL TUNING AND FILTER WHEELS TECHNOLOGY

Spectral filtering and tuning are critical components in the design of multispectral imaging (MSI) systems, enabling precise selection and control of specific wavelength bands for image acquisition. In MSI systems, spectral filters serve the purpose of isolating light at desired wavelengths, allowing the detector to capture detailed spectral information from each channel. Traditionally, this has been achieved using fixed band-pass filters arranged in arrays, tunable filters, or filter wheels. Recent advancements in MSI systems have introduced a variety of innovative approaches to enhance spectral tunability, spatial resolution, and system integration via spectral and filter wheels' approaches. One such development was reported by Ding et al. [180] in 2019, where a configurable coded aperture snapshot spectral imaging (CASSI) system was designed using a dual-prism assembly as the dispersive element. This system features precise spectral wavelength tuning achieved by controlling the air gap within the dual-prism configuration. It demonstrated a spectral tunability of 1 nm in the wavelength range of 400 nm to 550 nm, and 2–12 nm at longer wavelengths (550–700 nm). The proposed design offers simplicity and cost-effectiveness; however, practical implementation may require additional optical components to address chromatic aberration and ensure linear dispersion. Further advancement has been achieved through the deployment of polarization and filter-based designs. The work in [181], showcased a system that utilized a color polarization sensor, an imaging lens with a polarized-type divided aperture color-coding method (P-DACC), and an array of spectral filters. This compact MSI system could acquire nine distinct spectral bands across the visible to near-infrared (NIR) regions. The design also

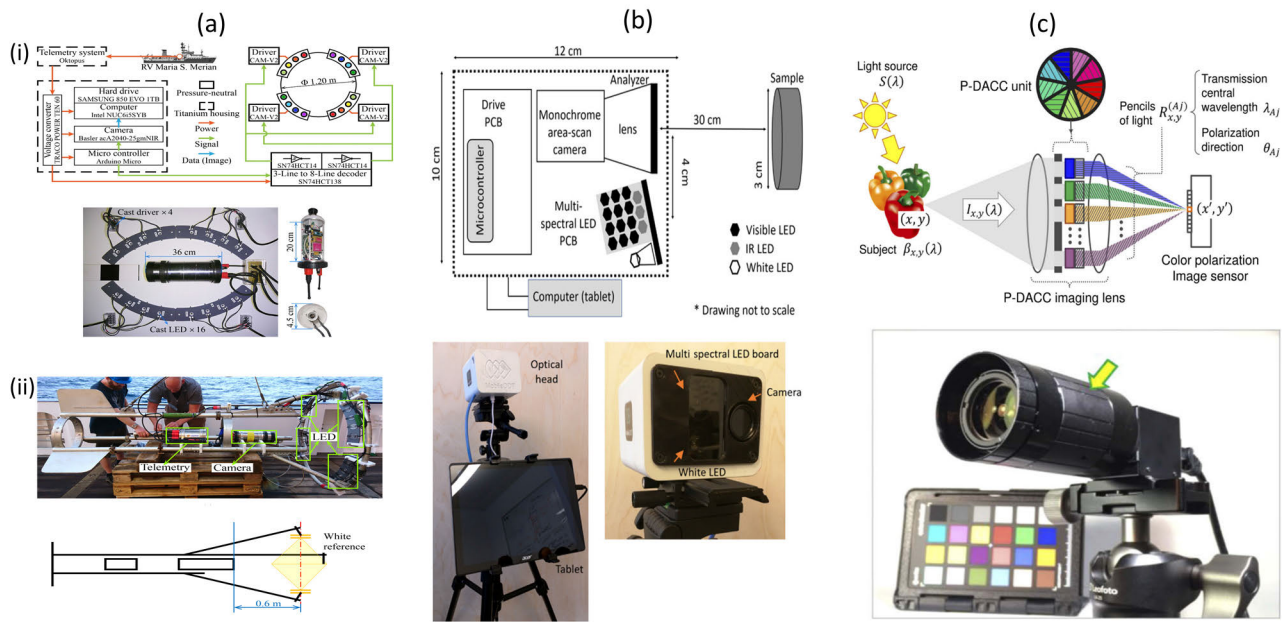


FIGURE 8. (a) The experimental setup of Tunable LED-based Underwater Multispectral Imaging System (TuLUMIS) proposed in [174], includes: (i) (Top) A schematic representation of the TuLUMIS system, detailing its key components, including LEDs, a camera, and essential modules for power management, synchronization, control, data storage, and signal transmission. (Below) is the close-up views providing deeper insights into its modular design and optical architecture. (ii) The photograph depicting the mounting of TuLUMIS on PELAGIOS frame and deployment during cruise MSM 61, along with the schematic representation of the LED array arrangement and the camera's focal length configuration. (b) The schematic depicting the block diagram of the portable and low-cost LED based multispectral imager proposed in [175], showing a detailed breakdown of the system's components and their interactions. Additionally, the figure includes the complete physical setup, showcasing the imager mounted with a tablet for real-time data visualization and an optical head designed for multispectral data acquisition. This setup highlights the system's compact design, emphasizing its portability and ease of deployment for field applications. (c) The schematic detailing the key optical components and their arrangement in the snapshot MSI system based on pixel-wise polarization sensor technique proposed in [181]. Additionally, the figure includes a photograph of the prototyped camera head, providing a visual representation of its physical design.

enables real-time spectral video capture while maintaining an operational noise level below 8%. The incorporation of a variable band-pass filter integrated into the imaging lens allowed for efficient spectral band selection and quantification. The schematic shown in Figure 8 (c), illustrates the integration of the filter and polarizer onto the detector, along with the process of the polarization and wavelength coding and decoding.

One notable approach involves the filter wheel mechanism, which was initially deployed in the Euclid mission in [182]. This mechanism was essential for imaging photometers, enabling precise measurements of light distribution to analyze distant galaxies. Adapted for MSI, similar systems have been developed, incorporating focusing optical components and motorized filter wheels to achieve spectral selectivity [183], [184], [185]. These innovations have expanded MSI applications, including document analysis [186], airborne and underwater MSI acquisition [187], [188], [189], environmental monitoring [190], as well as the observation of dynamic Earth processes [191]. Despite their adaptability, traditional filter wheel-based MSI systems are typically larger due to the need to accommodate multiple filters, which often results in a significant physical footprint. To address these size constraints, Yu et al. [192] proposed a customized filter array incorporating an additional panchromatic

channel to enhance spatial resolution. This design utilizes an optical interference filter mechanism that regulates specific wavelengths through the manipulation of constructive and destructive interference. A specialized filter holder was developed to accommodate the modular filter array onto the imaging sensor, allowing for independent replacement or reconfiguration of system components. The optical architecture of this system, shown in Figure 9 (a), includes a microlens array positioned in front of the imaging module, utilizing a two-axis positioner. The system spans a spectral range from 400 nm to 800 nm across five spectral channels, offering an adaptable and flexible configuration for efficient MSI. Further miniaturization was achieved by Wang et al. [193], who introduced a miniaturized MSI system employing printed circuit board (PCB) technology. This approach significantly reduced the overall size of the system by integrating filtering channels into an in-house developed ultrasonic motor. The motor minimized the spacing between channels, allowing for the inclusion of only the necessary filters specific to the imaging targets. The system, depicted in Figure 9 (b), incorporates a CCD camera with an objective lens, integrated multichannel filters, and a hollow ultrasonic motor for spectral band extraction, covering four channels from 400 nm to 1000 nm. In another innovative design, Hagen et al. [194], proposed integrating an optical

rotator-based filter with a conventional RGB polarization camera, achieving a 12-channel MSI system covering wavelengths from 400 to 700 nm. These systems illustrate the ongoing efforts to optimize multispectral imaging through advancements in compactness, spectral tunability, and improved system design.

Multispectral imaging systems and analytical methodologies have proven to be valuable non-invasive tools for a range of biomedical applications. This has spurred significant efforts towards the development of portable MSI devices, particularly in the near-infrared (NIR) region, which penetrates deeper into tissues, revealing diagnostic information not visible to the human eye or conventional RGB imaging systems. Kim et al. [195] reported one such development: an MSI device integrated with a smartphone, designed for diagnosing skin lesions. The system's design, summarized in Figure 9(c), uses the smartphone's flashlight as a light source, which is directed towards the skin through a plano-concave lens and mirror after passing through an optical bandpass filter and a vertical linear polarizer. The light is then captured by the smartphone's built-in CMOS camera for further processing. Another proposed design of a compact multispectral imaging system based on multiple apertures and a slanted linear variable filter has been recently reported in [196]. This system comprises a vertical stack of micro-structured optical layers, integrated near an image sensor. The LVF is positioned at the top, followed by a glass substrate with an aperture array on its upper surface and a micro-lens array (MLA) on the lower surface. The bottom of the stack consists of a baffle array and a cover glass. This configuration enables the system to capture images with narrow, distinct spectral content, facilitating fine-tuned spectral analysis. More details on the recently proposed designs of multispectral imaging systems using multi aperture imaging configuration can be found in [192] and [197]. Remarkably, there have been notable developments and emergence of multispectral imaging system designs tailored for specific applications. This specialized design ensures that each system, with its unique optical elements, is optimized for its intended use, thus, improving efficiency and data quality. For detailed information on these designs, the reader can refer to [190], [198], [199], [200], [201], [202], and [203].

Table 1 provides a comprehensive summary of the key advancements in MSI system design over the past two decades. The overall trends identified in this review highlight a growing emphasis on two main areas: portability and specialization for specific applications, spanning domains such as medicine, agriculture, space exploration, and more. This shift reflects the increasing demand for purpose-built MSI systems that cater to a wide range of applications. Additionally, a notable trend in Table 1 is the increase in the number of spectral bands captured by these systems. Expanding the spectral range allows for more comprehensive data acquisition, facilitating finer-grained analysis. This is particularly advantageous for applications like remote sensing and geological exploration, where richer datasets are essential

for detecting subtle variations and nuances in the environment. The continuous advancements in sensor technology and innovative filter designs have significantly improved MSI systems, leading to higher resolution and reduced noise levels. These improvements enhance the accuracy and reliability of acquired multispectral data, which is crucial for applications like medical imaging and art restoration. Moreover, the development of new algorithms for image reconstruction, as summarized in Table 1, has enabled the extraction of subtle details, further enhancing the capabilities of MSI systems.

Despite these remarkable advancements, it is important to acknowledge certain limitations. As indicated in Table 1, MSI systems are constrained by the relatively low number of spectral bands per pixel and the discretization of spectral bands during data acquisition. These limitations pose challenges in achieving high spectral resolution and reducing the sensitivity of MSI systems to subtle variations in emitted or reflected radiation. Consequently, this can hinder the ability to detect and identify differences in materials, especially in complex environments like water, land, or the atmosphere [12]. This review underscores the ongoing evolution of MSI systems, driven by diverse applications and the pursuit of more efficient and compact solutions. While significant progress has been made, particularly in enhancing portability and specialization, challenges related to spectral resolution persist. Continued innovation in sensor technology, optical components, and data processing algorithms holds the potential to expand the utility of MSI systems across various scientific, industrial, and research domains.

B. REMARKS AND FUTURE OUTLOOK FOR MSI SYSTEMS

Over the past two decades, significant advancements have transformed MSI systems, extending their application across scientific and industrial domains. Early MSI designs were largely limited by the number of spectral bands, slow acquisition speeds, and bulky setups. However, through continuous efforts in optical components, detectors, and computational techniques, MSI systems have evolved into more compact, and highly specialized devices. The development of technologies such as single-pixel detection, LED arrays, and advanced spectral tuning mechanisms, including filter wheels, has contributed to this transformation. Single-pixel detection techniques, though constrained by acquisition speed, have proven particularly effective in scenarios where high-resolution detector arrays are either unavailable or too expensive. By using spatial encoding patterns, single-pixel detectors can reconstruct high-resolution images while minimizing the hardware complexity, thus offering cost-efficient solutions for specialized spectral imaging tasks. However, the trade-off between acquisition time and spatial resolution remains a challenge, and future research must address this limitation to broaden the applicability of such techniques, especially in dynamic or real-time scenarios.

The use of LED arrays for MSI systems has also been a game changer, particularly due to their ability to provide

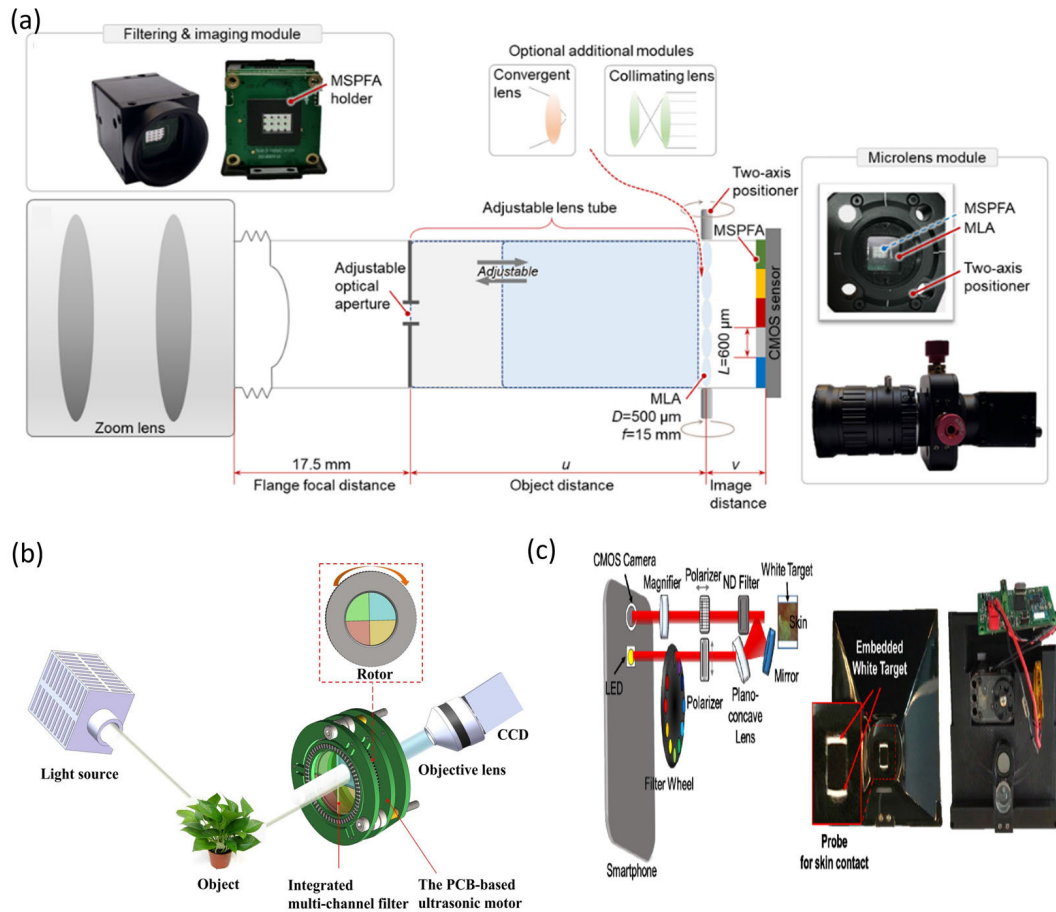


FIGURE 9. Illustrates three different multispectral imaging systems: (I) The schematic illustration of the assembly of the modular MSI system proposed in [192], including (a) The integration of the filter array onto the imaging sensor with the in-housed holder, (b) The positioning of the microlens array on a two-axis positioner, and (c) The overall optical framework of the proposed system. (II) Schematic overview of the proposed miniaturized customized filtering Wheel-based multispectral imaging (MCFW-MSI) reported in [193]. (III) (a) Schematic depicting the smart phone based multispectral imager proposed in [155], and (b) The photographic of the imager, and the magnified photo of the skin contact probe region indicated by a red rectangle and a white target that has been embedded [170].

rapid and wavelength-specific illumination in a compact and energy-efficient manner. The modular nature of these arrays, combined with the control over spectral tuning, has enabled MSI systems to achieve high temporal and spatial resolution at reduced costs. Recent designs incorporating a wide range of LED wavelengths have demonstrated improved spectral coverage, further expanding the realization of MSI systems at a reduced cost in fields like agriculture and environmental monitoring. On the other hand, spectral tuning through filter wheels and similar mechanisms has long been employed to provide high-resolution MSI systems. Although these systems often involve complex mechanical components, advancements in miniaturization and automation have enabled their integration into portable devices. However, filter-based approaches, while effective in certain applications, are typically limited by physical constraints, such as the number of filters or channels that can be incorporated. Addressing these limitations

through hybrid optical and computational techniques, such as compressed sensing, presents an exciting avenue for future research.

Looking ahead, the future of MSI systems lies in the convergence of innovative hardware, such as LED arrays and advanced filters, with sophisticated computational algorithms. This includes the application of artificial intelligence, machine learning, and compressed sensing techniques to enhance spectral resolution, reduce acquisition time, and improve data processing efficiency. Furthermore, the trend toward the miniaturization of MSI systems will continue, to develop portable, field-ready devices for applications such as remote sensing, medical diagnostics, and industrial inspection. In conclusion, the advancements in MSI over the past two decades have opened numerous possibilities for both research and practical applications. However, challenges remain, particularly in balancing the trade-offs between resolution,

TABLE 1. Advances in the field of multispectral imaging system design.

Year	Light Source	Filter Type	Detector Type	NOC	WR (μm)	AT	FOV	SP	AM	IRM	Ref
2004	-	Interference filter array	CMOS	7	0.4 – 0.7	-	-	1040 × 960	SS	M-PRM	[155]
2008	-	Narrow Bandpass filters	M-CCD	18	0.5 – 0.9	-	-	4008 x 2672	SS	-	[341]
2008	LED	Multiple Bandpass filter	C-CCD	8	0.4– 0.7	4 ms	-	-	SS	-	[157]
2010	-	FP filter	InGaAs	16	1.5 - 1.8	-	-	48×70	SS	-	[158]
2010	-	Mosaic BP filter	FPA	10	3 - 5	-	$\pm 3^\circ$	320x240	SS	-	[159]
2012	LED Array	-	CMOS	15	0.4 – 1.1	-	-	1024 x 1024	SS	-	[169]
2013	-	Filtering Wheel	CCD	8	0.35 – 1.1	-	-	-	-	-	[185]
2013	-	Tilted filter	CMOS	32	0.6 - 1	30 C/s	-	400x400	SS	-	[123]
2013	LED	Si Nanowires based filter	CCD	8	0.4 - 1	-	-	4x4	SS	-	[160]
2013	Halogen Lamp	NHAs based filter	CMOS	9	0.5 - 0.9	-	-	-	SS	-	[161]
2014	LED	-	M-CCD	23	0.37-1.63	94s	-	1392 × 1040	SS	PIM & SI	[171]
2015	LED Array	Dielectric Multichroic filter	CMOS	8	0.45-0.9	-	-	-	SS	FFT	[175]
2016	LED	Bandpass filter	CMOS	9	0.44-0.69	0.65s	-	-	SS	-	[194]
2016	Bulb	Diffraction Grating	BD	10	0.45 – 0.65	1 min	-	64x64	-	CS	[167]
2016	-	Plasmonic Filter Array	LWIR	6	0.75- 1.1	-	-	10 × 10	-	-	[164]
2017	-	Filter Array	SPD	8	0.42-0.72	-	-	100 × 100	-	-	[168]
2017	Laser Sources	Line Pattern Grating	-	5	0.405-0.55	-	5.9x 5cm	10	SS	FRAME	[342]
2017	LED	Tunable Plasmonic Filter	MC	-	0.4 -0.7	-	50 ⁰	100 × 100	SS	IH-Code	[165]
2018	LED	-	MC	8	0.4 – 0.7	35ms	-	-	SC	SAM & SID	[173]
2018	LED	Fourier filters	CCD	4	0.45-0. 9	-	-	36	SS	-	[178]
2018	LED	-	MC	13	Vis - NIR	300ms	-	-	SS	-	[174]
2018	-	Filter Array	MC	8	0.42 – 0.72	-	-	100 × 100	SS	DS	[162]
2019	LED	Filter Array	MC	9	0.5 -0. 8	-	-	16	SS	-	[163]
2019	-	Filtering Wheel	FPA	6	7.5 - 13	1s	45°	640x512	-	-	[190]
2020	Multi- λ LEDs	Filtering Wheel	MC	18	0.4 – 1.1	-	-	-	SS	-	[170]
2020	SC light Source	Multiple BPF and PLF arrays	PC	9	0.4 – 0.9	20 fps	-	1224x1024	-	DS	[180]

TABLE 1. (Continued.) Advances in the field of multispectral imaging system design.

2020	-	Bayer filter	SPD	-	SWIR	-	-	256 x 256	SC	HSP	[166]
2020	LED	Bandpass filter	MC	3	0.5 – 0.62	-	-	-	SS	-	[343]
2020	-	Multilayer filter	CMOS	3	0.4 – 0.9	-	-	160x120 per band	-	IH-Code	[200]
2020	-	Filter wheel	-	6	0.4 – 0.7	7.5s	-	-	-	-	[182]
2021	-	SLVF	CCD	66	0.45-0.85	-	68°	400x400	SS	-	[195]
2021	-	-	CMOS	11	0.47-0.828	-	-	50 x 50	SC	-	[344]
2022	LED	-	CMOS	3	0.5 – 0.6	5pfs	2.5x2. 5 mm	-	-	IH-Code	[172]
2022	LED	Filter wheel	CCD	7	0.4 – 0.1	-	-	-	-	IH-Code	[189]
2023	-	Bandpass filters	CMOS	9	0.4- 0.95	-	-	-	SS	SCSR	[196]
2023	-	Filter Wheel	CCD	4	0.4 - 1	6fps	47°	2592 ×1944	SS	-	[192]
2023	-	Filter Array	CMOS	5	0.4 – 0.8	-	-	330x320	SS	-	[191]
2023	-	-	LWIR	4	0.32 - 12	-	6°	3x3	-	-	[199]
2023	-	-	-	-	0.35 – 15.5	1s	-	-	-	-	[201]
2023	-	-	-	5	0.4 – 0.1	-	164°	2592 ×1944	-	-	[198]
2024	-	Filter Array	CMOS	8	0.38 – 0.95	-	-	4x4	-	-	[345]
2024	-	Filter Array	-	16	-	-	-	-	-	DSFNet	[346]
2024	-	Filter Array	CMOS	9	0.4 – 0.9	-	-	320 × 240	-	-	[347]
2024	-	Filter Array	-	8	0.45 – 0.7	-	-	-	-	DGRN /CACM	[348]
2024	-	Filter Array	CCD	8	0.4 – 0.8	-	-	16x14	-	-	[349]
2024	-	-	-	-	8 - 12	-	-	-	SS	MST - NN	[350]
2024	-	Filter Wheel	SWIR Camera	6	1.51 – 1.56	-	-	-	-	-	[351]

NOC: Number of Channels, **AM:** Acquisition Mode, **TA:** Acquisition Time, **FOV:** Field of View, **SP:** Spatial Points, **WR:** Wavelength Range, **IRM:** Image Reconstruction Methods, **SS:** SnapShot, **SC:** Scanning, **CCD:** Charged Couple Detector, **M-CCD:** Monochrome Charged Couple Detector, **C-CCD:** Color Charged Couple Detector, **FPA:** Focal Plane Array, **CMOS:** Complimentary Metal-oxide Semiconductor Detector, **BD:** Bucket Detector, **MC:** Monochrome Camera, **SPD:** Single Pixel Detector, **PC:** Polarization Camera, **CPF:** PushFrame, **M-PRM:** Modified Pixel Rearranging Method, **LP:** Line Pair, **PIM & SI:** Pseudo-inverse Method and Spline Interpolation, **IH-Code:** In house Code, **CS:** Computational simulation, **PD:** Photodetector, **FFT:** Forward Fourier Transform, **CS:** Compressive Sensing, **FRAME:** Frequency Recognition Algorithm for Multiple Exposures, **SAM:** Spectral Angle Mapper, **SID:** Spectral Information Divergence, **Vis-NIR:** Visible to Near infrared Spectrum, **DS:** Demosaicing, **BDF:** Bandpass filter, **PLF:** Polarization filter, **SWIR:** Short-Wave Infrared, **HSP:** Hadamard Sampling Pattern, **SLVF:** Slanted Linear Variable Filter, **SCSR:** Spectral Clustering Super-Resolution Reconstruction, **FPS:** Frame Per Second, **FPA:** Focal Plane Array, **DSFNet:** Dual Space Fusion Network for Occlusion-Robust Dense, **DGRN:** Depth-guided reconstruction network, **DGM:** Depth-guided model, **CACM:** Channel Adaptive Convolution Model.

Units:

ms: milli-seconds, s: seconds, C/s – Sube per second, lp/mm: line pairs per millimeter fps: Frame Per Second, p: Pixel,

acquisition speed, and system complexity. Future developments will likely focus on optimizing these parameters,

leveraging the synergy between hardware innovations and advanced computational methods to create more efficient,

accessible, and specialized MSI systems for a wide range of applications.

V. ADVANCES IN HYPERSPECTRAL IMAGING SYSTEM DESIGNS

HSI systems provide detailed spectral profiles for each pixel in an image, allowing for precise material or object identification and characterization in more complex scenes. While MSI systems have long provided a balance between spectral information and ease of implementation, the HSI systems have pushed the boundaries in terms of providing deep spectral insight. HSI systems generate massive datasets due to their large number of spectral bands, requiring more advanced data analysis techniques and algorithms to extract meaningful information. These systems often employ machine learning or compressive sensing algorithms to handle the high-dimensional data, making them computationally intensive but capable of providing superior analytical precision. MSI systems, by comparison, produce smaller datasets that are easier to handle and require less processing power, but they trade off some spectral resolution and accuracy in exchange for greater efficiency and speed. The journey of HSI systems began in the late 20th century, initially spurred by the need for more detailed remote sensing data for environmental monitoring and resource management.

Early HSI systems were bulky, expensive, and primarily deployed on large platforms such as satellites and aircraft. Over the decades, significant strides have been made in the design and miniaturization of these systems, making HSI more accessible. The evolution of HSI can be traced through several key milestones, including the development of the first imaging spectrometers, the integration of advanced sensor technologies, and the introduction of sophisticated data processing algorithms. These advancements have not only improved the performance and capabilities of HSI systems but have also expanded their applicability across diverse fields such as agriculture, environmental monitoring, defense, medical diagnostics, and industrial inspection. This review section aims to explore the key advancements in HSI systems design, focusing on the innovative technologies and methodologies that have driven the evolution of HSI systems. Specifically, this review section will focus on the four primary types of HSI systems: staring, snapshot, whiskbroom, and pushbroom.

A. SPECTRAL SCANNING (STARING) TECHNIQUE

The staring technique is distinguished by its capacity to capture high-resolution spectral data over a fixed field of view, making it a valuable approach in HSI. This method operates by employing a tunable filter or a series of discrete filters to sequentially capture images at different wavelengths. Each filter allows a narrow band of wavelengths to pass through to the detector array, which then records the intensity of light at that specific wavelength across the entire scene. By repeating this process across multiple wavelengths, a complete

hyperspectral data cube is generated, with each pixel containing a comprehensive spectral profile. Unlike pushbroom and whiskbroom techniques, which sequentially scan the field of view either line by line or point by point, the staring technique captures the entire scene at once for each wavelength. This snapshot-like approach eliminates the need for mechanical scanning components, thereby reducing mechanical complexity and enhancing temporal resolution. However, due to the inherent limitations in filter bandwidth, the spectral data captured at each wavelength must be interpreted considering the filter's spectral response. This relationship can be mathematically expressed as follows [56]:

$$I_i = \int_0^{\infty} m_i(\lambda) r(\lambda) d\lambda \quad (5)$$

where I_i signifies the intensity of the spectral data-cubes captured by the i -th filter, with $m_i(\lambda)$ representing the spectral response of the filter and $r(\lambda)$ denoting the aggregated spectrum reaching the imager.

This section of the review provides a detailed exploration of the designs of staring HSI systems, with a focus on the optical components and filters that have been instrumental in their advancement. Key optical components driving the development of this technique include diffractive optical elements, which enable precise wavelength dispersion, acousto-optic tunable filters (AOTFs) and liquid crystal tunable filters (LCTFs), which allow rapid wavelength tuning across a broad spectral range, as well as Fabry-Perot interferometers and linear variable filters (LVFs), which offer narrowband spectral filtering.

1) DESIGNS BASED ON DISPERSIVE AND DIFFRACTIVE OPTICAL COMPONENTS

Early designs of spectral scanning-based HSI systems predominantly relied on diffractive and dispersive optical components, such as prisms and gratings. One of the pioneering works in this area was reported by Okamoto and Yamaguchi [204], who developed a staring spectral imaging system utilizing a transmission grating to diffract collimated light from the target. This method allowed for the simultaneous capture of spectral and spatial data using a monochromatic camera, and subsequent reconstruction of both spectral and spatial distributions was achieved using multiplicative algebraic reconstruction methods. Building upon this, Mooney et al. [205] introduced a variable-pitch grating design aimed at capturing spatial and spectral projections, with three-dimensional images reconstructed using computed tomography (CT) techniques. Similarly, Bernhardt [206] proposed a hybrid staring HSI system, which employed a combination of rotating and variable dispersive elements to capture data, reconstructing three-dimensional data cubes using the same CT-based techniques. Hinnrichs and Massie [207] provided a detailed review of the operational principles and imaging capabilities of an advanced HSI system developed by Pacific Advanced Technology for the

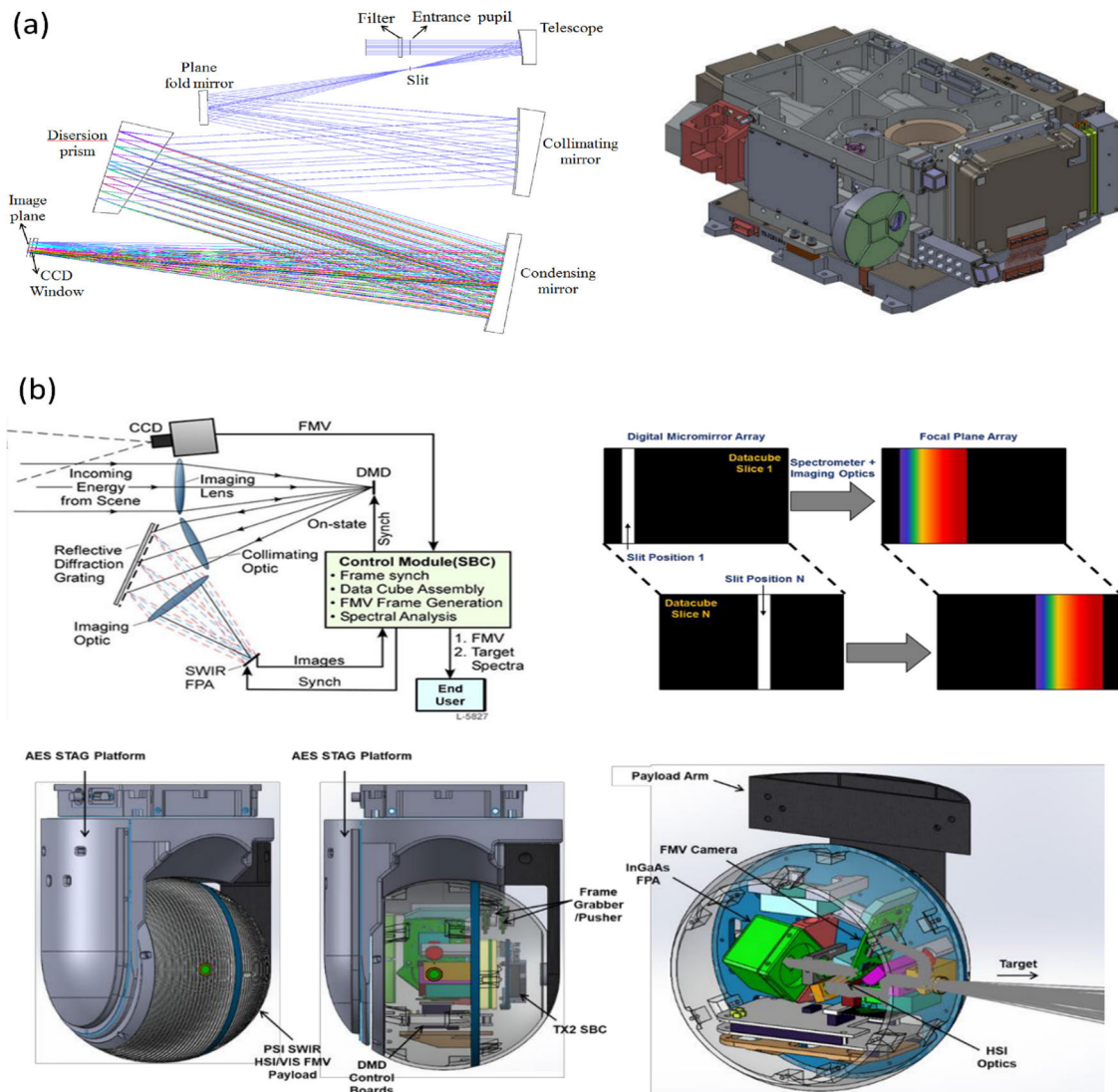


FIGURE 10. Illustration of two advanced spaceborne hyperspectral imaging system designs, each employing distinct optical configurations to optimize spectral and spatial imaging performance (a) The first design proposed in [211], integrates an off-axis parabolic telescope with a prism-dispersive off-axis aspheric spectrometer. The ray tracing schematic (left) demonstrates how light is collected by the telescope, then dispersed through a prism and focused by an aspheric spectrometer to achieve high-resolution spectral imaging. This system's mechanical structure is also depicted in the right section of the figure, showcasing its compact yet efficient layout optimized for spaceborne applications. (b) The second design proposed in [212] presents an ultra-compact HSI system tailored for integration within a gimbal platform. The schematic (top left) diagram illustrates the imager's optical concept, emphasizing how incident light from the scene is directed on to a digital micromirror device (DMD) slit. The system captures incoming light and projects it onto a digital micromirror device (DMD), where a programmable slit selectively scans the scene. Reflected light from the slit is collimated and passed through a diffraction grating, which disperses spectral components along the vertical axis. The combined spatial and spectral data is then focused onto an InGaAs focal plane array (FPA) for hyperspectral imaging. A conceptual schematic (top right) illustrates how the DMD entrance slits are mapped onto the FPA, ensuring precise spectral and spatial data acquisition. A three-dimensional model (bottom left) of the system within the gimbal platform highlights its compact structure, demonstrating its feasibility for deployment in space-constrained environments. Additionally, the middle and right sections of the figure reveal the internal components of the system, detailing the arrangement of optical and electronic elements necessary for its operation.

US Office of Naval Research. This system, built around novel dispersive components, was engineered specifically to capture both temporal and infrared spectral signatures. It utilized an InSb focal plane array detector, covering a spectral range of 3000 to 5000 nm with a spectral resolution of 6 cm^{-1} , a 2.5° field of view, and an acquisition time of 2 seconds.

In another notable work, Mooney et al. [208] proposed an alternative design for a staring HSI system using a direct-vision prism, which allowed for the selective passage of one spectral band while other bands deviated along different paths. Spectral scanning was achieved by rotating the prism within a bearing, capturing a sequence of camera

frames at different angular rotations. A few years later, diffractive lenses were also demonstrated for use in HSI systems, capable of covering a wide spectral range [209], [210]. Xue et al. [211] presented the design of a spaceborne HSI system intended for observing ozone distribution in the Earth's upper atmosphere. The system, depicted in Figure 10 (a), combines an off-axis parabolic telescope with a prism-dispersive off-axis aspheric spectrometer. It offers a wide spectral range (280–1000 nm) and operates with a large dynamic range to simultaneously capture ultraviolet (UV), visible, and near-infrared (NIR) radiation.

In 2022, Giblin et al. [212] introduced a compact HSI system design based on a Czerny-Turner spectrometer integrated with a digital micromirror device (DMD), which functions as a programmable entrance slit. This proposed design allows for staring mode operation, flexible revisit times, and reduces reliance on the speed of the host system's movement. The DMD consists of an array of micro-mirrors that can be individually controlled via an industrial single board computer by selectively directing incoming light through individual micro-mirrors, the light is passed through the spectrometer and then onto a detector. As the mirrors are electronically manipulated, HS data cubes can be acquired using an InGaAs focal plane array detector. The design, with a total volume of approximately 1000 cm³, can capture full-motion video over a spectral range of 900–1605 nm. The schematic of the proposed configuration and the system's gimbal-based functionality, which enables integration into airborne or ground-based platforms, is shown in Figure 10 (b). In the same year, a hyperspectral-depth imager with dual-mode four-dimensional (4D) functionality was demonstrated in [213], capable of simultaneously acquiring three-dimensional spatial information, fluorescence, and reflectance spectra. The proposed system is a fusion of a Prism-Grating-Prism (PGP)-based Staring HSI system and a binocular line laser stereo vision system. The use of diffractive optical components in HSI systems offers several advantages, including lower costs and ease of use. Additionally, they do not require extra electronics, unlike tunable filters such as acousto-optic and liquid crystal filters.

The development of staring HSI systems based on dispersive and diffractive optical components has marked a significant advancement in the field of spectral imaging. These systems offer a practical and cost-effective solution for acquiring spectral data without the need for complex mechanical scanning mechanisms, making them particularly advantageous for real-time and compact spectral imaging applications. Dispersive elements, such as prisms and gratings, have proven effective for spectral separation, while diffractive components provide a simpler and more economical alternative to tunable filters, eliminating the need for additional electronics. Despite these benefits, challenges persist, particularly regarding spectral resolution and alignment sensitivity. Dispersive and diffractive systems often face limitations in their ability to cover a broad spectral

range with high precision, and they can be prone to image mis-registration and low throughput, especially in environments requiring rapid spectral tuning. These shortcomings highlight the need for continued innovation in the design of these optical components, aiming to improve spectral tunability, resolution, and robustness.

2) DESIGNS BASED ON ACOUSTO – OPTIC TUNABLE FILTERS

Acousto-optic tunable filters (AOTFs) operate based on the acousto-optic effect, which induces changes in the refractive index of certain crystalline or amorphous materials, typically crystals such as tellurium dioxide (TeO₂) or quartz, when subjected to an acoustic wave. As the acoustic wave propagates through these materials, the interaction between the light beam and the density compression waves creates a periodic modulation of the refractive index, effectively producing a diffraction grating effect. AOTFs are electronically controlled spectral filters, consisting of piezoelectric transducers driven by a radio frequency (RF) signal, which generates the density compression waves that modulate the refractive index. Depending on the nature of the interaction between the incident light and the acoustic wave within the crystal, AOTFs can be classified into collinear and non-collinear types [214]. In collinear AOTFs, the incident light and acoustic waves propagate along the same axis within the crystal, whereas in non-collinear AOTFs, the interaction occurs at non-collinear angles, which affects the diffraction properties and overall filter performance. The theoretical framework governing AOTF operation is based on anisotropic Bragg diffraction theory, as detailed in [215]. The primary advantages of AOTF based designs include their high transmission efficiency, rapid scan capabilities, broad spectral range coverage, and controllability, making them highly suitable for applications requiring fast and flexible spectral tuning.

The design and development of acousto-optic tunable filter (AOTF)-based staring hyperspectral imaging (HSI) systems have garnered significant attention over the past two decades, with various configurations and improvements reported in the literature. These advancements have primarily focused on enhancing spectral resolution, improving light throughput, and expanding spectral coverage. A more detailed discussion of these advancements can be found in [128], [216], [217], [218], and [219]. In 2018, Wang et al. introduced the utilization of all-reflective fore-optics employing a novel afocal three-mirror optics to expand the spectral coverage of AOTF-based spectral imagers [220]. This innovative design is engineered to utilize a shared objective for two AOTF channels, ensuring uniform scene imaging. In the same year, a double filtering design using a single AOTF for HSI was proposed [221], showcasing a marked improvement in spectral resolution. This configuration allowed for better separation of spectral bands, enhancing the system's ability to discern subtle spectral variations. By employing

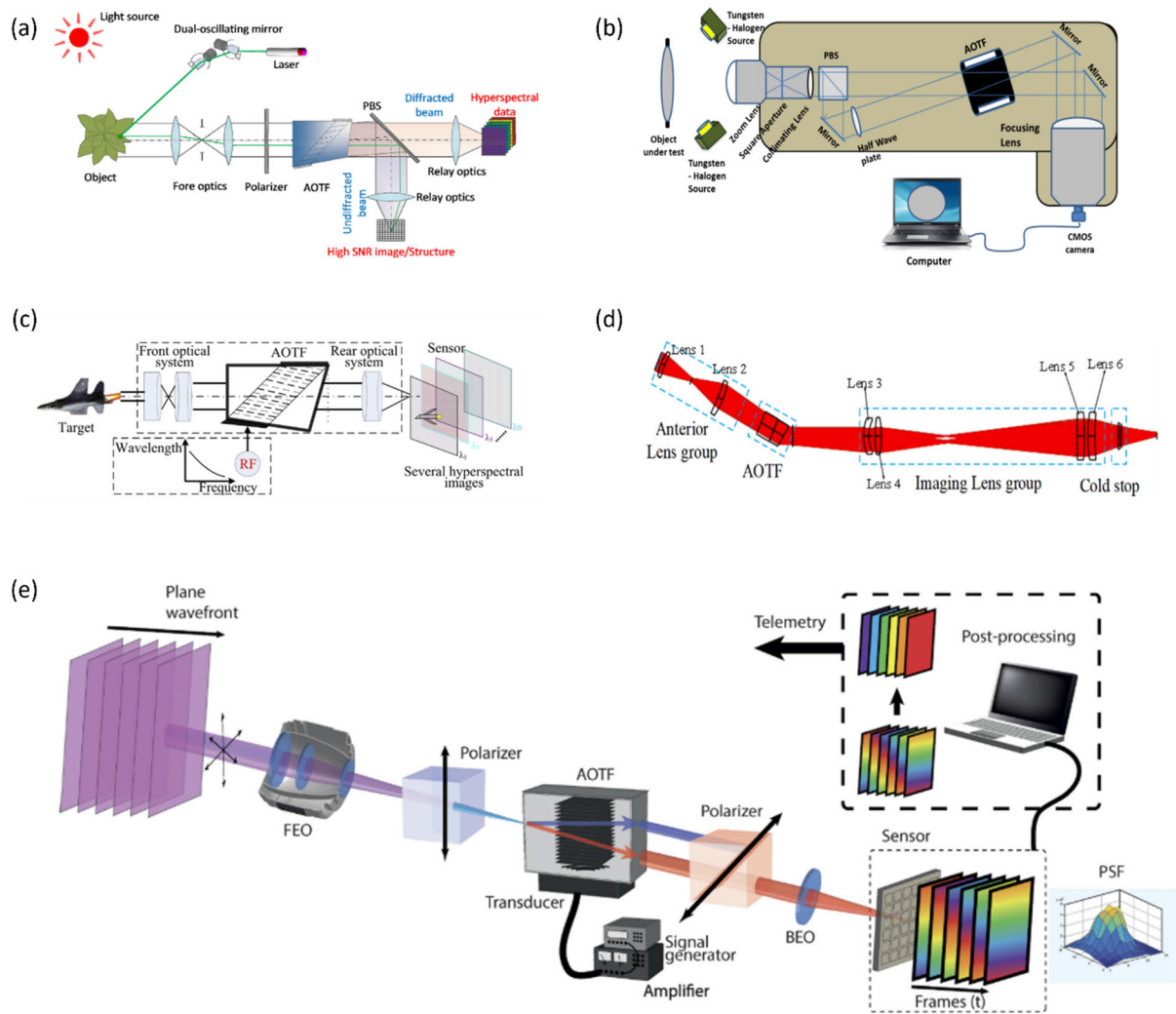


FIGURE 11. (a) Schematic layout of the proposed AOTF-based three-dimensional hyperspectral imaging system proposed in [222] (b) Schematic layout of an AOTF-based hyperspectral imaging system presented in [224] (c) The setup used in [227] for the proposed spatial-spectral discrimination scheme for spectral imaging based on AOTF. (d) The ray tracing layout of the proposed AOTF-based MWIR HSI presented in [228]. (e). The schematic of the proposed idea of spatio-spectral HSI with an AOTF in [229].

double-pass filtering, this approach also reduced crosstalk between adjacent spectral channels, addressing one of the common limitations in AOTF-based systems.

Furthermore, Zhao et al. [222] proposed an AOTF-based dual-path configuration for an HSI system, which integrated laser triangulation for sub-pixel resolution. As depicted in Figure 11 (a), the proposed system can acquire panchromatic, hyperspectral, and three-dimensional structural data simultaneously. The dual-path setup enables the system to separate spectral and spatial data acquisition, allowing for more precise measurements of structural details at the sub-pixel level. This multi-modal approach demonstrates a significant leap forward in AOTF-based spectral imaging, making the system highly adaptable for various applications, including remote sensing, biomedical imaging, and industrial inspection. In early 2019, Yushkov et al. [223]

validated a novel protocol enabling HSI acquisition using an acousto-optic tunable filter (AOTF). This approach involved adaptively configuring the non-collinear AOTF transmission passband, adjusting for both the illumination source spectrum and equalizing the imager's spectral sensitivity. This adaptive tuning allowed for more precise spectral data acquisition, optimizing the filter's performance across various illumination conditions. Following this, Abdlaty and Fang [224] proposed an alternative design for a staring HSI system operating in the visible to near-infrared (NIR) spectrum, utilizing off-the-shelf optical components. The system employed an AOTF, a CMOS detector, and an in-house image analysis program. As illustrated in Figure 11 (b), the design incorporates a polarizing beam splitter that divides the collimated beam. A half-wave plate is positioned in the path of the reflected beam before it enters the AOTF aperture, ensuring

that the polarization of the beams matches the AOTF's polarization requirements, thus enhancing diffraction efficiency. To address the path length difference between the transmitted and reflected beams, they introduced two mirrors, ensuring that both beams reach the detector simultaneously. Their setup is controlled by a computer, which also manages data acquisition and storage, ensuring seamless operation.

In 2021, a study presented the use of a new high-performance AOTF specifically designed for airborne HSI systems [225], marking a significant improvement in performance for remote sensing applications. Furthermore, a comprehensive study on various optical schemes for implementing spectral imaging through AOTF was presented in [226], offering insights into the practical configurations of AOTF-based HSI systems and their performance under different operating conditions. The study offers valuable insights into practical implementations, comparing the strengths and limitations of each optical arrangement, and serves as a useful guide for researchers seeking to further advance the field. By exploring the interplay between optical components, filter configurations, and system architecture, the study lays the groundwork for works that could enhance the performance, resolution, and application range of AOTF-SI systems.

Zhao et al. [227] introduced an innovative approach utilizing an acousto-optic tunable filter (AOTF)-based spectral imager to mitigate the effects of jamming signals in infrared imaging systems (see Figure 11 (c)). Operating in spectral scanning acquisition mode, this method leverages the AOTF's inherent ability to perform spatial-spectral discrimination, effectively reducing interference and ensuring clearer imaging in the presence of jamming signals. The technique demonstrates the versatility of AOTF technology in enhancing the resilience of infrared imaging systems, particularly in complex environments. Li et al. [228], proposed a design that exploits the non-collinear AOTF to acquire HSI in the thermal infrared range, specifically targeting swiftly moving objects. As illustrated in (Figure 11 (d)), their design consists of an anterior lens group, an AOTF, an imaging lens group, and a mid-wavelength infrared (MWIR) cooled detector. The front lens group captures the target image onto a field stop, creating a primary image plane. The beam then passes through a collimating lens group to form a parallel beam, where the AOTF selects the desired spectral band by diffracting the light based on the chosen imaging center wavelength. Finally, the imaging lens group focuses the diffracted beam onto the infrared detector, enabling fast spectral acquisition. This system showcases the rapid tuning speed and lightweight construction that make AOTFs particularly advantageous for real-time hyperspectral imaging of dynamic scenes, all while benefiting from the absence of moving parts.

The use of quasi-collinear AOTF designs as primary diffractive components in achieving spatio-spectral HSI systems has been explored in [229]. As depicted in Figure 11 (e), the configuration diffracts the incoming light wavefront into multiple spectral lines, each satisfying the Bragg condition

at a defined wavelength and tuning frequency of the AOTF. While these quasi-collinear AOTF designs offer significant advantages in spatial-spectral resolution, they face limitations in terms of throughput. The intricate nature of AOTF structures often leads to low optical throughput, particularly when operating across broad spectral ranges or under low-light conditions, which can hinder overall system efficiency. Royer et al. [230] presented an overview of a pioneering HSI system designed for space exploration. Developed by a team at the Institut d'Astrophysique Spatiale, this system is based on a wide-aperture AOTF covering a spectral range from 950 nm to 3600 nm. The system offers significant potential for planetary observation, particularly in analyzing surface compositions of celestial bodies. However, obtaining accurate temperature data and HS data cubes from fast-moving targets presents a substantial challenge, particularly due to the required short integration times.

The integration of acousto-optic tunable filters (AOTFs) into HS) systems has proven to be a significant technological advancement, providing benefits such as rapid spectral tuning, compact design, and the elimination of mechanical components. Over the past two decades, AOTF-based HSI systems have evolved considerably, with innovations in optical design, spectral resolution, and system adaptability. The capability of AOTFs in spatial-spectral discrimination, particularly in complex imaging scenarios like infrared jamming mitigation and airborne remote sensing, has demonstrated their growing importance in modern imaging applications. However, despite the numerous advantages, AOTF-based HSI systems face challenges such as low optical throughput, particularly when dealing with broad spectral ranges or complex environmental conditions. Additionally, ensuring polarization control and managing optical alignment are critical factors that continue to affect system performance. Addressing these challenges, while continuing to improve spectral resolution and imaging speed, remains a priority for future research.

3) DESIGNS BASED ON LIQUID CRYSTAL TUNABLE FILTERS

A Liquid Crystal Tunable Filter (LCTF) uses liquid crystal molecules sandwiched between transparent conductive plates to adjust its spectral characteristics. The orientation of the liquid crystal molecules can be manipulated by applying an external voltage which changes the alignment of the molecules, thereby, altering the wavelengths of the transmitted light through the filter [231]. The operational principles of the LCTF are derived from the multistage Lyot-Ohman filter. The key innovation in LCTF technology lies in replacing the stationary wave plates of the original Lyot-Ohman filter with tunable liquid crystal wave plates, providing enhanced flexibility in wavelength selection. The filtering mechanism of the LCTF relies on a series of critical components: polarizers, a retarder, and the liquid crystal waveplate, which together defines the spectral filtering characteristics. The adjustable

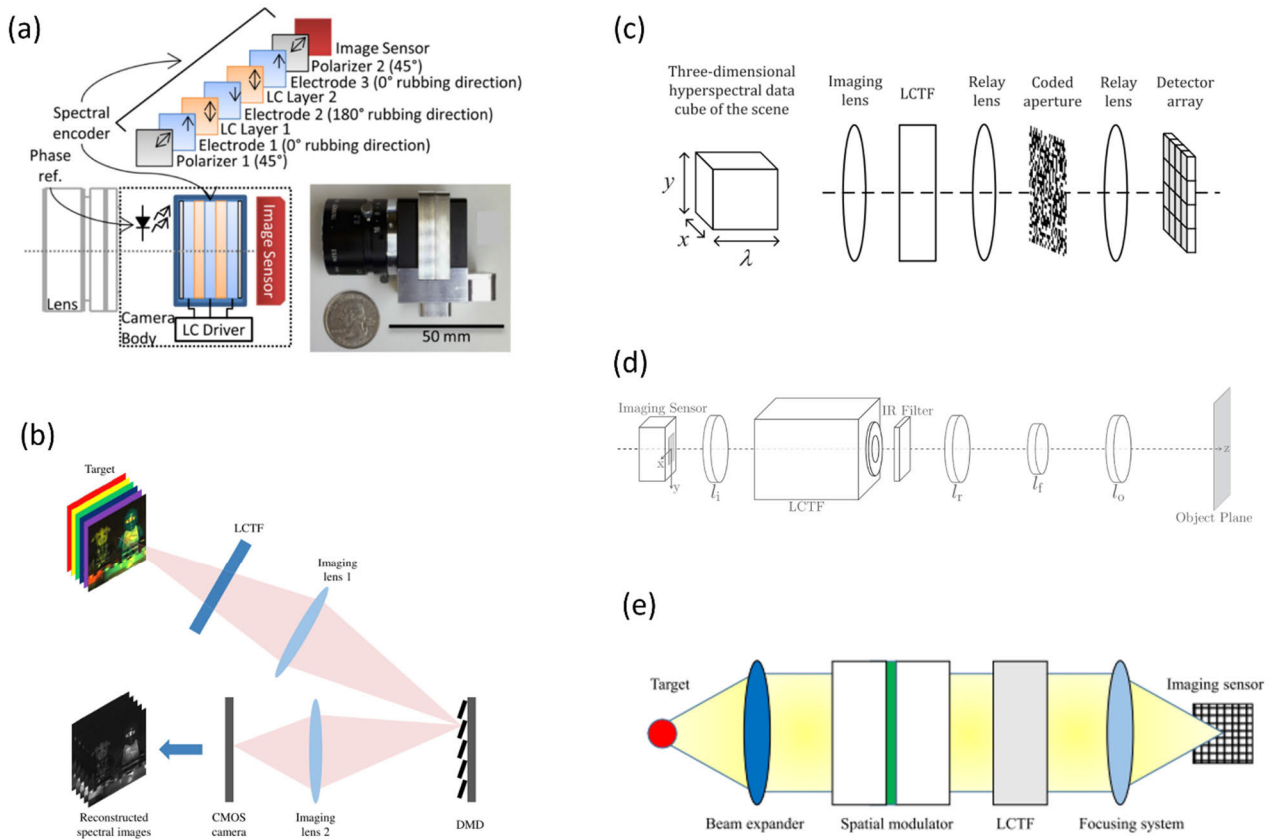


FIGURE 12. (a) LC polarization interferometer-based HSI system proposed in [238] depicting the placement of the liquid crystal spectral encoder inside the camera body, along with the corresponding photo of system (b) A schematic illustration of the proposed LCTF based HSI using two-tone adoptive compressive sensing scheme presented in [241], demonstrating its operation framework and configuration. (c) The illustrative sketch of the coded aperture LCTF based spectral imager proposed in [242], highlighting its structural configuration. (d) LCTF based HSI system presented in [247] with detailed optical layout. The system consists of objective lens (l_o), l_f is an added lens for enhanced functionality, l_r is a relay lens for infinite focusing of the real image of the target on to the LCTF, l_i is an imaging lens for focusing the beam on to the imaging sensor, and IR-filter is a NIR filter to ensure only the desired spectrum within the spectral range reaches the LCTF. (e) The schematic illustration of polarization-based HSI system proposed in [248], detailing its key components and their integration within the imaging framework.

retardance of each liquid crystal element allows for the precise selection of narrow spectral bands. The liquid crystal waveplates are aligned with their fast axes positioned at 45° relative to the axis of the first polarizer, enabling polarization decomposition of the incoming light. This decomposition causes the light components aligned with the principal axes of the liquid crystal waveplates to experience distinct phase shifts, thus influencing the transmitted wavelength. Under normal input incidence, the phase delay introduced by the liquid crystal waveplates in the Lyot-Öhman configuration determines the polarization modulation, while the transmission characteristics of the filter, consisting of M stages, are governed as described in [232]. This multistage design enables finely tuned spectral filtering, allowing selective transmission of specific wavelengths. Such precise control makes LCTF highly effective for applications requiring precise wavelength selection. In addition, LCTs combine high transmission efficiency, broad spectral coverage, and precise wavelength controllability without mechanical components. Their compact, robust design ensures durability

and integration flexibility, while their temperature sensitivity allows additional control in regulated environments. These features make LCTFs ideal for applications such as hyperspectral imaging, spectroscopy, optical communications, and so on.

Over the past decade, the application of LCTF in HSI systems has experienced significant advancements [233], [234], [235]. LCTFs have gained widespread adoption due to their advantageous characteristics, including simple design, minimal wavefront distortion, throughput control, and the ability to accommodate large apertures with wide fields of view. These qualities have established LCTFs as essential components in staring HSI systems, enhancing spectral selectivity and facilitating the efficient acquisition of detailed spectral data [236]. One of the primary challenges in LCTF technology is achieving rapid wavelength tunability without sacrificing performance. To address this, Schundler [237] introduced an approach utilizing holographic polymer-dispersed liquid crystal (H-PDLC) filters for HSI system design. The H-PDLC filters, which combine polymer-dispersed liquid

crystal with holography, offer a hybrid structure that significantly enhances the tunability and spectral control of LCTFs. The proposed design improves not only the spectral resolution but also the switching speed, making these filters suitable for fast and adaptable HSI applications. In 2015, Hegyi and Martini [238] introduced a polarization interferometric HSI system based on liquid crystal technology. This design utilizes the orthogonal polarization states of liquid crystal variable retarders to modify the optical path delay between these states. As shown in Figure 12 (a), the system's optical path includes a liquid crystal component that adjusts the path delay, generating an interferogram at each pixel. By capturing a series of images while dynamically modifying the path delay, the system can construct a detailed spectral dataset. This approach not only enhances the spectral fidelity of the system but also allows for simultaneous spectral and spatial acquisition. In 2017, Jullien et al. [239] proposed the design of a polarized HSI system, employing two successive liquid crystal cells. These cells, when dynamically controlled via voltage steps, function as a Fourier spectrometer, enabling precise spectral filtering. The introduction of liquid crystal cells as dynamic Fourier spectrometers represents a significant advancement in polarized HSI. This design opens new possibilities for multi-dimensional spectral imaging, where polarization data can be captured alongside spectral information, providing deeper insights into material properties and surface characteristics. In 2018, a comparative study was conducted to analyze the performance of AOTFs and LCTFs in HSI systems [240]. This comparison brings attention to the trade-offs inherent in tunable filter technology. The study examined several critical metrics, including image quality, spectral resolution, and target-to-background discrimination. The finding highlighted that while both filter technologies offer distinct advantages, AOTFs tend to outperform LCTFs in terms of spectral resolution due to their rapid tunability and broader wavelength coverage. However, LCTFs excel in simplicity and ease of integration, especially in systems requiring fewer moving parts and more robust designs.

In 2020, Xu et al. [241] introduced a two-tone adaptive compressive sensing (TACS) coding technique for an LCTF-based HSI system. Their proposed design, illustrated in Figure 12 (b), demonstrates how the incoming light from the target interacts with the LCTF for spectral selectivity. The resulting monochromatic image, defined by the LCTF's central wavelength, is then subjected to spatial modulation through a coded aperture. The TACS methodology leverages a high-resolution coded aperture implemented using a digital micromirror device (DMD) array. This array, composed of individually adjustable micromirrors, generates various binary-coded patterns to refine the spatial modulation of the monochromatic image, and a CMOS camera forms the basis for reconstructing the high-resolution spectral images at each tunable wavelength of the LCTF. Similarly, Xi et al. [242] applied compressive sensing (CS) principles to enhance both the spatial and spectral resolutions of the LCTF-based

HSI system. By formulating and solving a convex optimization problem, they successfully reconstructed high-resolution images from fewer measurements, thus addressing traditional constraints related to spatial and spectral resolutions of LCTF-based HSI systems. The schematic representation of their design, depicted in Figure 12 (c), shows the optical architecture employed in their proposed system. Additionally, several other studies in the literature have employed CS techniques to improve the spatial and spectral resolution of LCTF-based HSI systems [243], [244]. One notable study addressed the challenges of underwater environments by proposing a specialized underwater hyperspectral imaging system [245]. Their design incorporates modifications to accommodate the optical properties and light attenuation characteristics of water, ensuring high-quality spectral data collection in submerged settings. Furthermore, the use of three-dimensional convolutional neural networks (3D-CNNs) combined with compressive sensing for HS image classification has also been reported [246]. Blaksley et al. [247], developed a prototype LCTF-based HSI system designed for real-time capture of HSI, focusing on dynamic facial observations in living subjects (see Figure 12 (d)). Most recently, Zhang et al. [248] proposed a novel design combining LCTF and spatial modulators, including a half-waveplate and Savart polarizer, to create an HSI system with polarization control capabilities. As depicted in Figure 12 (e), this system integrates five key components—LCTF, spatial modulator, beam expander, focusing optics, and imaging sensor—allowing for rapid HSI data acquisition alongside comprehensive polarization information of the target.

The evolution of LCTF-based HSI systems over the past decade has demonstrated significant advancements, particularly in terms of tunability, compactness, and spectral selectivity. LCTFs offer distinct advantages such as rapid wavelength tuning, broad spectral range coverage, large aperture capabilities, and minimal wavefront distortion, making them highly capable for HS imaging. However, despite these strengths, LCTF-based HSI systems face inherent challenges that can lead to reduced performance. While compressive sensing and three-dimensional convolutional neural networks (3D-CNNs) have helped to address spatial and spectral resolution limitations, further improvements are needed to optimize the balance between image quality and system complexity. The recent efforts to integrate polarization control, adaptive compressive sensing, and hybrid designs using DMDs highlight the potential for future LCTF-based HSI systems to overcome current limitations. These advancements suggest a continued trend toward multifunctional systems, capable of capturing richer, more detailed spectral information in challenging environments. Moreover, as LCTF technology continues to grow, its combination with machine learning algorithms, enhanced optical components, and smarter modulation techniques will likely define the next generation of LCTF-based HSI systems. These innovations will enable LCTF-based HSI to expand further into fields

like remote sensing, industrial inspection, and precision agriculture, offering more efficient and adaptable solutions to complex imaging challenges.

4) DESIGNS BASED ON FABRY PEROT INTERFEROMETERS

A Fabry-Perot interferometer (FPI) filter consists of two parallel reflective surfaces separated by a defined gap, through which light passes and undergoes multiple reflections. As light reflects between these surfaces, it experiences constructive and destructive interference, which is dependent on both the wavelength of the light and the length of the cavity. This interference pattern enables the selective transmission of specific wavelengths while blocking others, resulting in a narrowband spectral filter. The transmitted irradiance in an FBI is derived from the collective contributions of multiple light beams that reflect and transmit within the cavity. This summation process is mathematically described by the Airy function, leading to equation (7) as shown in [249] and [250].

$$I_{transmitted} \propto \frac{1}{1 + \frac{4R}{(1-R)^2} \sin^2 \left(\frac{2\pi d}{\lambda} \right)} \quad (6)$$

where R represents the ideal reflectivity of the mirrors, while d refers to the separation between these mirrors. When the distance between the two reflective surfaces is adjusted mechanically, the system is referred to as an interferometer. However, if the surfaces are fixed and aligned for parallelism by fastening a spacer—typically made from materials such as invar or quartz—it is classified as an etalon [249]. The key advantages of the FPI include its high optical throughput and its ability to achieve high and adjustable spectral resolution, making it a versatile tool in spectral imaging and analysis.

Fabry-Perot (FP) interferometric techniques for HSI emerged as early as the 1980s and 1990s, primarily due to their lower angle sensitivity and temperature insensitivity compared to other tunable filters. For instance, the Digital Array Scanned Interferometer (DASI) HSI system combines the advantages of a grating spectrometer with a two-beam interferometer to achieve enhanced spectral resolution [251], [252]. The development of two-dimensional focal plane arrays with improved quantum efficiency and extensive linear dynamic ranges has significantly contributed to the reported success of DASI. In 2001, NASA developed and tested surface reflectance mapping using interferometric spectral imagery to simultaneously measure the spectral properties of a single target using both ground-based and aircraft-mounted imagers [252]. About a decade later, thermal HSI systems for space application employing Fourier Transform (FT) algorithms coupled with scanning low-finesse FP interferometers were explored in [253]. For further insights on related works, please refer to the detailed discussions and analyses provided in [254] and [255], as they offer additional context and expand on the topics. The work of Pisani and Zucco [256] represents a significant advancement over traditional HSI by enhancing both luminosity and

compactness (see Figure 13 (a)). Their design incorporates an FPI, which enables high optical throughput, thereby reducing the required exposure time for image acquisition. The FPI's ability to efficiently transmit light through its interferometric structure enhances overall system performance, particularly in low-light environments. Additionally, Fourier Transform (FT) algorithms are employed to extract spectral information from the interferograms generated by scanning the distance between the interferometer mirrors.

Moreover, variants of FPI-based HSI systems have been reported in the literature, including the Sagnac or triangular path interferometer, as demonstrated by Yoshihara and Kitada [257]. This configuration generates equal inclination fringes without the need for collimating optics, offering highly reliable light flux. Barnes [258] presented a photodiode array Fourier transform spectrometer that employs a self-scanned photodiode array, which mitigates systematic background noise by first measuring it in the absence of fringes and subsequently using this data to adjust the interferogram. Additionally, birefringent interferometers have been incorporated into the design of photodiode array-based Fourier transform spectrometers in [259], where a Wollaston prism is used to create spatial interferograms detected by a linear photodiode array. Moreover, Dupuis et al. [260], demonstrated a Fourier transform spectrometer wherein the target (biological sample on a surface) is illuminated using an expanded laser light. A dichroic beam splitter then separates the reflected light, passing the fluorescence spectrum while reflecting the fundamental wavelength. The fluorescence enters the interferometer, where a wavelength-dependent amplitude modulation is imposed via interference, encoding the spectrum at electrical frequencies. They employed a CCD camera and imaging lens to spatially resolve the fluorescence source.

P. Gatteringer et al.'s integration of FP tunable filters with a digital micrometer in 2019 highlights the importance of precise spatial encoding to enhance single-pixel HSI systems [261]. Their approach (see Figure 13 (b).), which spans the near-infrared spectrum (1500 to 2200 nm), stands out for its effective light concentration and high spatial resolution, enabling applications in fields requiring detailed spectral analysis, such as biomedical imaging or remote sensing. Their configuration spans a spectral range of 1500 to 2200 nm, offering a 25mm field of view with a spatial resolution of 0.4mm. Similarly, Perri et al. [262] presented a variation of the Translating-Wedge-based Identical Pulses Encoding Systems (TWINS) [263] birefringent interferometer to achieve HSI through the generation of delayed images. Their system combined Fourier Transform HSI with the TWINS time-scanning interferometer, as depicted in Figure 13 (c). Their approach enabled the production of high-contrast, high-coherence interferograms at the single-pixel level, resulting in improved target discrimination and reduced acquisition time. Their system operates within a wavelength range of 400nm to 1000nm. Chandra et al. [264] proposed an adaptive HSI system incorporating a reflective Spatial Light

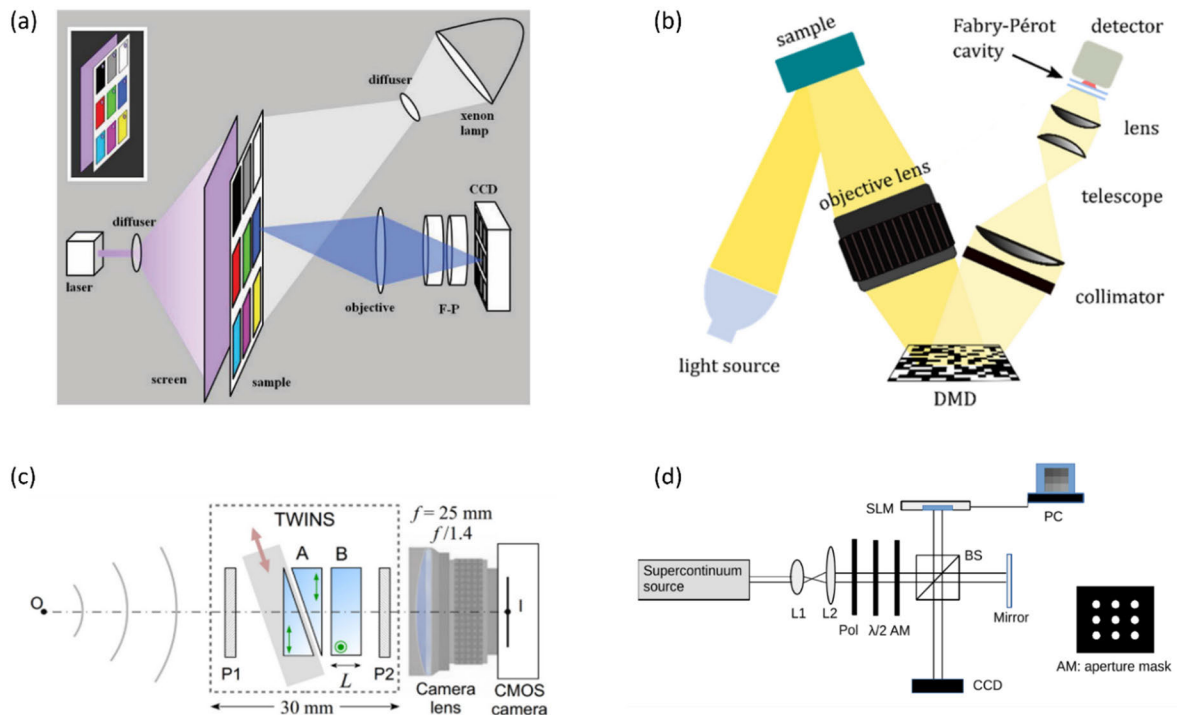


FIGURE 13. (a) A schematic of the FPI-based spectral imager presented in [256] (b) Broadband NIR HSI system based on FP tunable filter reported in [261] (c) A Schematic setup of the hyperspectral camera with TWINS birefringent interferometer. P1 and P2: wire-grid polarizers; A block: α -BBO wedges, with the optical axis parallel to the plane (green double arrow); B block: α -BBO plate, with the optical axis normal to the plane (green circle) [262]. (d) Schematic diagram showing incident structured illumination in an SLM-based Michelson interferometer. Key: L1, L2: Lenses; Pol: Polarizer; $\lambda/2$: Half wave-plate; AM: Aperture mask having 3×3 holes; BS: Beam splitter. The plane mirror is mounted on a translation stage. For measuring the spectra from the Michelson interferometer, the CCD is replaced by a fiber spectrometer whose tip size is small enough to sample a single fringe pattern at a time [264].

Modulator (SLM) into one arm of a Michelson interferometer. By altering the pattern displayed on the SLM using a checkerboard phase mask, tunable scanning was achieved. The schematic of their method ideal for laboratory-based spectral imaging or industrial quality control, is illustrated in Figure 13 (d). The design leverages the electrically tunable refractive index of the liquid crystal structure within the SLM to induce varying phase delays in individual beamlets, allowing for the capture of nearly contiguous spectral information over a wavelength range of 450nm to 900nm in a single acquisition. Most recently, in 2023, a computational HSI system utilizing a superposition FP resonance cavity filter with a transfer matrix method encoding layer was proposed [265]. Additionally, Zhou et al. [266] demonstrated the integration of micro-electromechanical systems (MEMS)-based FP filter onto CMOS/CCD detectors, enabling the development of a compact spectral imager. Their work represents a breakthrough in miniaturization, enabling the development of compact, high-performance spectral imagers suitable for a wide range of applications, including portable devices.

Looking at the trajectory of FPI-based staring HSI systems, there is a clear movement towards increasing the compactness and portability of these systems. As seen with the MEMS-based FP filters, FPI systems are no longer confined to large, stationary platforms but are becoming

increasingly accessible for on-the-go and field-based applications. The integration of computational methods such as Fourier Transform algorithms and compressive sensing has revolutionized how spectral data is processed and analyzed. These techniques allow for the rapid reconstruction of HS data cubes from fewer measurements, significantly reducing data acquisition times while maintaining high spectral resolution. Despite the advancements in FPI-based HSI systems, challenges remain. One of the primary limitations of current designs is the delicate nature of the Fabry-Perot interferometers, which can result in reduced throughput under certain operational conditions. Additionally, misalignment in the optical setup or mechanical instabilities in the moving parts of the interferometer can compromise the accuracy and reliability of spectral data acquisition. Overcoming these challenges will be crucial for the further advancement of FPI-based systems, particularly in applications requiring high durability and robustness, such as space exploration and harsh environmental monitoring.

5) DESIGNS BASED ON LINEAR VARIABLE FILTERS

Linear Variable Filters (LVFs), also known as wedge filters, operate as band-pass filters where the thickness of their filter coating changes linearly across their length. This linear variation in thickness results in a directional variation in

the transmitted spectral content. This variability enables the LVF to essentially function as an infinite series of distinct filters, offering a continuum of selectable wavelengths along its surface [119]. LVFs were originally explored in [267], [268], and [269] and applied in spectral imaging applications [126], [126], [270], [271], and current research focuses on miniaturizing them to match the size of spectral imager detectors. This adaptation allows for direct placement of the LVF directly in front of or near the imager's focal plane. Consequently, this design approach enables the creation of HSI system characterized by moderate spectral resolution alongside exceptionally high spatial resolution.

To this end, extensive literature reports have explored the integration of LVFs in the design of HSI systems. LVFs are favored for their mass-producibility, achieved through dynamic or static masking during the deposition process. For example, in 2018, Gustafsson et al. [272] reported the integration of an LVF-based HSI system by mounting the LVF with almost full octave transmission in front of a large FPA sensor. The schematic illustration of fitting the filter onto the FPA is depicted in Figure 14 (a), with images acquired over the wavelength range of 450 nm to 900 nm through a step-and-stare technique. The technique involved rotating the camera or translating it relative to the target, aligning minute shifts in the scene's position with the linearly evolving properties of the LVF along its spectral axis. In 2020, Rahmloew et al. [273] demonstrated an HSI system by integrating an LVF in the casing of an off-the-shelf monochrome camera. This integration facilitated the smooth capture of temporal, spatial, and spectral data in a staring acquisition mode. In the same year, Son et al. [274] introduced a novel design for a portable and cost-effective HSI system prototype using LVFs created through static masking. Tailored specifically for agricultural technology purposes, their design relied on a stepper motor paired with a custom rack and pinion system, enabling movement of the LVF across the monochromatic imaging camera. This setup resulted in the creation of distinct vertical strips within each image, corresponding to linearly spaced intervals across various wavelengths. The schematic of their proposed design is depicted in Figure 14 (b), illustrating the fundamental components of the setup. Additionally, Figure 14 (b) shows a picture of their prototyped design (bottom right), portraying the internals on the left and the compact system on the right, with a weight of approximately 650 grams. In 2021, Cheng et al. [275] demonstrated a reduced-size version of the prototype originally proposed by Son et al. [275], specifically designed for the monitoring of large-area power system equipment. In [276], Fan et al. demonstrated an LVF-based HSI camera using low-cost, high-performance CMOS sensors, each connected to an imaging module circuit board. Their calibration device uses an integrating sphere to provide uniform illumination for the camera's field of view, and digital domain time delay integration (TDI) is employed to process the images, enhancing the signal-to-noise ratio (SNR) significantly.

A comparative analysis conducted by Mahendra and Chandra [277] on the design of all-dielectric, metal-dielectric, and metal-dielectric-metal (MDM) based LVFs, underscores the need to carefully weigh fabrication complexity, cost, and spectral performance when choosing LVOF designs for specific HSI applications, particularly targeting the VNIR (450–900 nm). Their analysis revealed that all-dielectric LVOFs offer superior spectral resolution but come at the cost of significant fabrication complexity. The MDM-based LVOFs, while less complex to fabricate, present lower transmission rates and a broader full width at half maximum). However, the metal-dielectric-based induced transmission filter offers a balanced and more cost-effective solution. There are also recent advancements in the use of LVFs for the design of miniaturized spectral sensing systems with future applications particularly in distributed sensor systems, Internet of Things devices, mobile autonomous platforms, and various other applications. For instance, in 2022, Kobylinskiy et al. [278] demonstrated the use of a standard LVF in a spectral sensor, showcasing its capability to capture a continuous spectrum. Their work highlighted a notable efficiency improvement in the use of LVF for spectral sensing, compared to conventional systems employing an upright illumination principle. Moreover, another design on the deployment of LVF for spectral sensing was demonstrated in 2023 by Labaj et al. [279]. Their design is based on the fusion of LVF and an uncooled HgCdTe detector array, effectively covering the spectral range of 3000 to 4000nm. Their setup enables the analysis of spectral absorption of the target, particularly for environmental monitoring.

Table 2 offers an in-depth review of the recent advancements in spectral scanning acquisition techniques employed over the five years. As seen from the table, the choice of light sources has diversified significantly, moving from standard halogen lamps to more controlled illumination like Xenon lamps, which are advantageous for their broad emission spectrum and high intensity. The filter types employed alongside these light sources have also advanced, with a shift towards more sophisticated filters enabling dynamic wavelength selections and rapid tuning over wide wavelength ranges. In addition, the detectors have also seen significant evolution, with Charged Couple Device (CCD) detectors being supplemented by Complementary Metal-oxide Semiconductor (CMOS) and Indium Gallium Arsenide (InGaAs) detectors. Acquisition times (AT) have seen considerable reductions, enabling faster imaging capabilities. Some systems, like the one in [230], boast acquisition times as low as $0.65\mu s$, facilitating near-instantaneous data collection. Field of view (FOV) is also an important consideration, with systems like that in [247] providing a 32 cm^2 field of view, suitable for applications requiring wide-area coverage. It is worth noting that compressive Sensing (CS) has emerged as a popular image reconstruction method due to its ability to reconstruct images from under-sampled data, making it ideal for real-time applications. Feature Extraction and

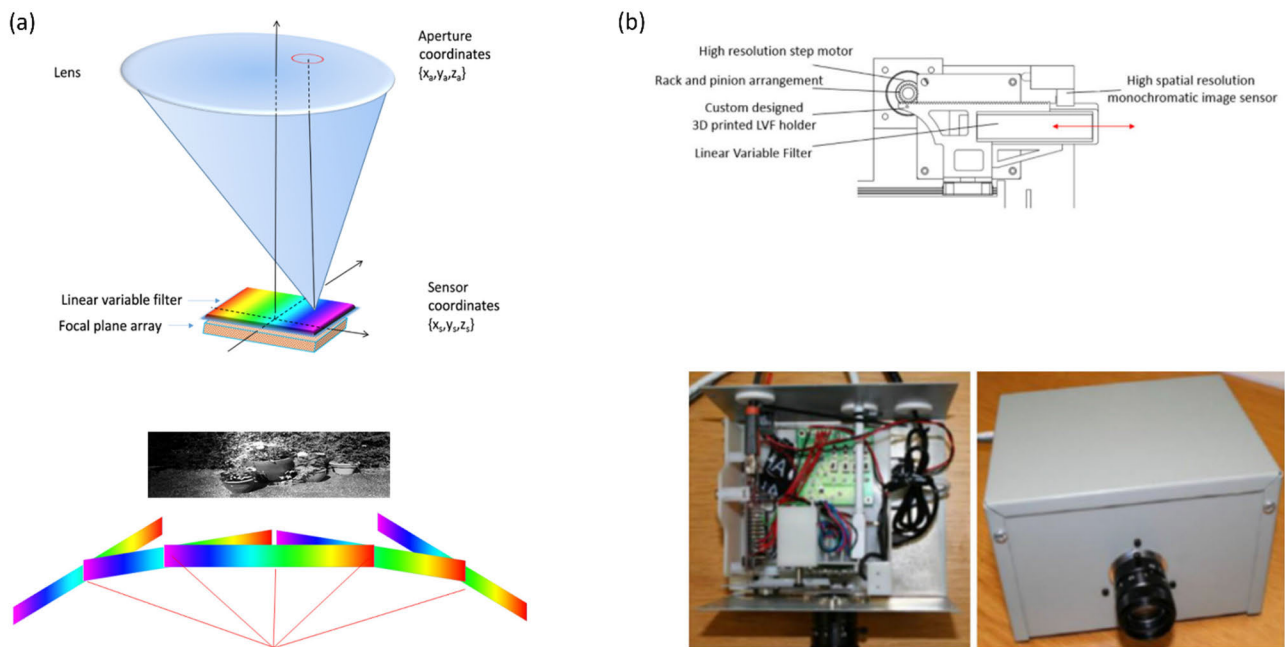


FIGURE 14. (I) Schematic of the LVF-based HSI system proposed in [272] featuring the integration of the LVF with a lens and the focal plane array detector, and the target images acquisition at different angular rotation which are combined to form the data cubes at different wavelength (bottom left). (b) The LVF-based HSI system proposed in [274] including the schematic of the basic components (top right), and the prototype assembly with the internals on the left and the compact system on the right (bottom right).

Classification Algorithms (FE and CA) are also noted, highlighting a trend towards incorporating machine learning and pattern recognition into hyperspectral imaging for applications such as agriculture technology (AgriTech) as mentioned in [274].

6) REMARKS AND FUTURE OUTLOOKS FOR STARING BASED HSI

The evolution of Staring HSI systems has been marked by significant technological advancements, each driven by the development and integration of various types of filtering schemes. Each of these schemes offered distinct benefits, yet also presented unique challenges in terms of performance, resolution, and practicality. Dispersive and diffractive optics laid the foundation for early spectral scanning HSI systems by leveraging prisms and gratings to achieve wavelength separation, allowing for the acquisition of high-resolution spectral data. Despite their early success, the mechanical complexity and slower spectral tunability of these systems limited their adoption in more modern applications requiring speed and compactness. The introduction of AOTF marked a significant leap forward, offering electronically tunable filters with rapid wavelength switching, wide spectral coverage, and high spectral resolution. AOTF systems, particularly in non-collinear configurations, have proven invaluable for airborne and spaceborne applications, where speed and minimal moving parts are critical. However, their sensitivity to mechanical and thermal stability, as well as alignment challenges, have led researchers to explore alternative tunable filter designs.

LCTF-based HSI systems, on the other hand, have gained prominence due to their simplicity and low power consumption. By manipulating the alignment of liquid crystal molecules to control the transmitted wavelength, LCTFs offer flexible, electronically tunable filtering with minimal wavefront distortion. While these systems tend to lag behind AOTFs in terms of spectral resolution, their ease of integration, particularly in compact, portable systems, has made them a popular choice for biomedical and real-time imaging applications. FPI technology stands out for its ability to deliver exceptionally high optical throughput and tunable spectral resolution. By adjusting the gap between two parallel reflective surfaces, FPI filters can achieve precise wavelength selectivity. Recent advancements have focused on miniaturizing FPIs for integration into compact HSI systems, enabling applications in space exploration, remote sensing, and high-resolution industrial inspections. However, FPI systems are often constrained by their mechanical tuning requirements, which can affect their operational speed and long-term stability. LVF has emerged as a versatile filter option, capable of offering continuous spectral selectivity along its length. By integrating LVF directly onto detector arrays, modern systems have been able to achieve hyperspectral imaging with moderate spectral resolution but high spatial resolution. This technology is especially advantageous in applications requiring lightweight, mass-producible, and cost-effective solutions, though the spectral resolution is often limited compared to more sophisticated tunable filter designs.

TABLE 2. Advances in the field of staring HSI system design.

Year	Light Source	Filter Type	Detector Type	NOC	WR (μm)	AT	FOV	SP	IRM	App. Target	Ref
2019	-	Prism	CCD	-	0.28 - 1		2.4°x 0.02°	-	-	Space Application	[210]
2019	Halogen Lamp	-	InGaAs	-	1.5 – 2.2	4 mins	25x25	64x64	CS	Chemical	[260]
2019	Halogen Lamp	AOTF	CMOS		0.45 – 0.85	-	-	-	-	-	[223]
2019	Xenon Lamp	Birefringent Interferometer	CMOS	-	0.4 - 1	10ms	15.4°	-	-	-	[261]
2020	-	LCTF	CMOS	24	0.45 – 0.65	-	-	400× 400	CS	-	[240]
2020	-	LVF	CMOS	-	0.45 – 0.9	-	-	-	FE & CA	Agritech	[273]
2021	Xenon Lamp	LCTF	CCD		0.42 – 0.73	10s	32cm ²	100 × 100	-	-	[246]
2022	-	PGP	CMOS	-	0.4 – 0.8	180s	20°	-	-	-	[212]
2022	-	AOTF	HgCdTe	200	0.95 – 3.6	0.65 μs	2.7°	-	-	Space Application	[229]
2022	-	Grating	CMOS		0.9 – 1.605			44 × 40			[211]
2022	SC	Interferometer	CCD	30	0.45 – 0.9	-	-	400	-	-	[263]
2023	-	AOTF	-	-	0.37 – 4.8	22ms	8°	30 μm	-	-	[227]
2023	-	LCTF	CCD	-	0.42 – 0.95 0.9 – 1.7	-	-	11 μm	-	-	[247]
2024	-	Grating	DA	7	0.36 – 1	-	-	-	-	-	[352]

NOC: Number of Channels, **AM:** Acquisition Mode, **AT:** Acquisition Time, **FOV:** Field of View, **SP:** Spatial Points, **WR:** Spectral Range, **IRM:** Image Reconstruction methods, **CCD:** Charged Couple Detector, **FPA:** Focal Plane Array, **DA:** Detector Array, **CMOS:** Complimentary Metal-oxide Semiconductor Detector, **MC:** Monochrome Camera, **FE & CA:** Feature Extraction and Classification Algorithm, **CS:** Compressive Sensing, **BDF:** Bandpass filter, **PLF:** Polarization filter, **SWIR:** Short-Wave Infrared, **HSP:** Hadamard Sampling Pattern **FPA:** Focal Plane Array, **PGP:** Prism Grating Prism, **LCTF:** Liquid Crystal Tunable Filter, **AOTF:** Acousto-Optic Tunable Filter, **LVF:** Linear Variable Filter

Each filter technology presents a trade-off between spectral resolution, tunability, mechanical complexity, and system integration. The ongoing advancements in HSI filter design reflect the increasing demand for systems that balance high performance with compactness, and speed. Future developments are likely to focus on hybrid approaches that combine the strengths of multiple filter technologies, leveraging computational techniques and advanced signal processing algorithms to further enhance the efficiency and capability of Staring HSI systems. The exploration of new materials, miniaturization techniques, and integration strategies will undoubtedly continue to push the boundaries of staring HSI, expanding its applicability across scientific, industrial, and commercial domains.

B. SNAPSHOT TECHNIQUE

The snapshot (SSI) technique is designed for the simultaneous acquisition of the entire 3-D data cube within a single integration period, circumventing the constraints associated with the use of tunable filters in conventional staring methods [280]. It revolutionized the way spectral data is captured and analyzed by allowing the acquisition of an entire spectral data cube in a single shot. This advancement is critical for applications where the subject cannot be exposed to prolonged imaging times, such as in dynamic environments or when observing transient phenomena. Among the various SSI techniques, each has unique characteristics that cater to specific requirements regarding resolution, spectral range, and complexity. An ideal snapshot Spectral

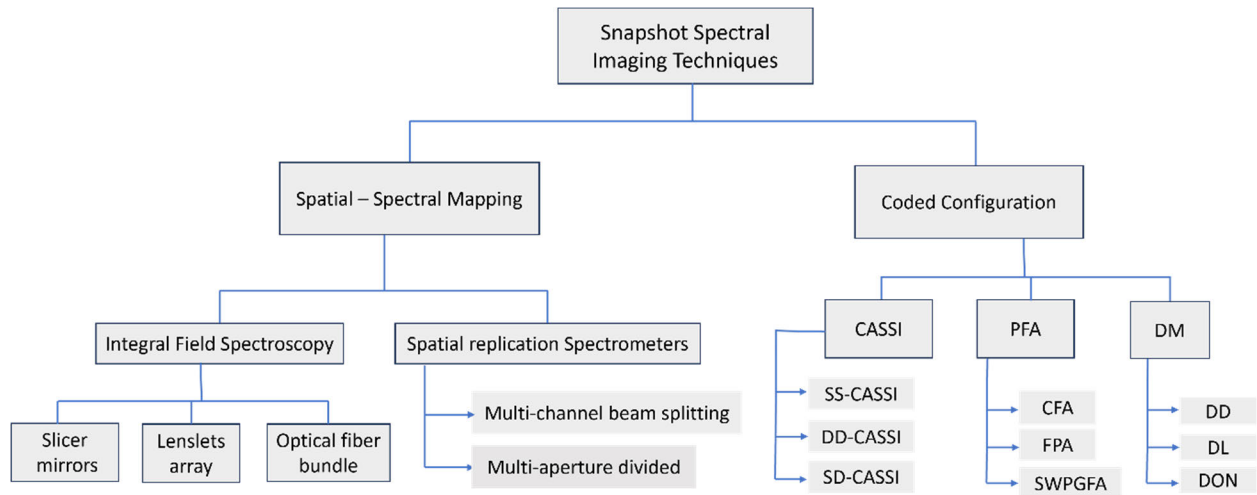


FIGURE 15. An overview on the various classifications of Snapshot Spectral Imaging Acquisition techniques reported in the literature. CASSI: Coded Aperture Snapshot Spectral Imaging, SS-CASSI: Spatial-Spectral CASSI, DD-CASSI: Double Dispersion CASSI, SD: Single Dispersion CASSI, PFA: Pixelated Filter Array, CFA: Color Filter Array, FPA: Fabry-Perrot Filter Array, SWPGFA: Sub-wavelengths Patterned Grating Filter Array, DM: diffraction Modulation, DD: Diffractive Diffuser, DL: Diffractive Lens, DON: Diffractive Neural Network.

Imaging system would seamlessly integrate high spatial and spectral resolution, compactness, and efficient computational processing. However, achieving such an optimal system involves navigating a series of trade-offs concerning image acquisition speed, resolution, and computational requirements, among other factors. A comprehensive summary of various classifications of snapshot spectral imaging acquisition methods explored in the literature are depicted in Figure 15.

Typically, snapshot spectral imaging commonly acquires spectral data by spatially dispersing the 3D spectral cube of a scene and aligning it onto a 2D detector array via the spectral dimension. This approach, termed spatial-spectral mapping, primarily offers the advantage of creating a direct one-to-one mapping between the detector's pixels and the spectral cube's voxels. Moreover, this simplifies the reconstruction process significantly, as there's no need for intricate post-processing algorithms to interpret the data. To this end, spatial-spectral mapping can manifest as either integral field spectroscopy (IFS) or spatial replication spectrometers (SRS). As seen in Figure 15, the IFS employs an Integral Field Unit (IFU) array, realized through various optical elements, including arrays of mirrors, Lenslets, or fiber optics. Each of these configurations captures light from a two-dimensional field of view and then spatially divides it into discrete elements, each representing a tiny segment of the target image. This segmentation allows for the simultaneous acquisition of spectra from multiple points across the field of view, thereby creating a 3D data cube: two spatial dimensions and one spectral dimension. On the other hand, the SRS spatially operates on a target scene using either a lenslet array or beam splitters, and it can be categorized into two primary types: multi-channel beam-splitting spectral imaging and multi-aperture divided spectral imaging (see Figure 15).

In the coded configuration approach, custom-made optics are utilized to encode spectral characteristics such as amplitude, phase, or wavelength. This method of encoding effectively acts as a sensing matrix, which is crucial for developing a robust mathematical model. This model accurately depicts the way spectral information is encoded before its detection. Utilizing this mathematical framework, in conjunction with compressed sensing principles, allows for the reconstruction of a spectral data cube from the encoded inputs. The coded configuration can be implemented through various methods including coded apertures, pixelated filter arrays, or direct modulation techniques [62]. A concise explanation of the techniques is summarized in Figure 15.

1) SPATIAL – SPECTRAL MAPPING

• Integral Field Spectroscopy (IFS)

IFS techniques capture a spectrum at every spatial location of an image simultaneously, often used in astronomy.

- 1) **Slicer mirrors:** These are a series of reflective surfaces or 'slicers' that take an input image and slice it into strips, which are then rearranged side by side to form a long, narrow, two-dimensional image. This image is then passed through a spectrograph to produce a spectral data cube.
- 2) **Lenslets array:** An array of micro-lenses, each focusing light from a different part of the scene onto a detector array. The light from each lenslet is spread into a spectrum, and these mini spectra are tiled across the detector. Each point of the image has associated spectral data.
- 3) **Optical fiber bundle:** A bundle of optical fibers is used to collect light from each spatial element of the scene and route it to a spectrometer. The fibers' ends are arranged in a two-dimensional format at the input

(scene-facing side) and a one-dimensional format at the output (spectrometer-facing side), effectively mapping a 2D scene into a 1D spectral array.

- **Spatial Replication Spectrometers**

These methods utilize multiple paths to simultaneously capture different parts of the spectrum.

- 1) **Multi-channel beam splitting:** Uses beam splitters to divide the incoming light into multiple beams, each analyzed by its own spectrometer, effectively multiplying the system's throughput.
- 2) **Multi - aperture divided:** Similar to multi-channel beam splitting but uses physical apertures to split the scene into different sections. Each section is then analyzed separately, enabling the simultaneous capture of various spectral regions.

2) CODED CONFIGURATION

- **Coded Aperture Snapshot Spectral Imaging (CASSI)**

Utilizes a coded aperture to spatially modulate the light before it goes through a dispersive element. The resulting two-dimensional image contains both spatial and spectral information and needs computational reconstruction to separate the spectral data cube. The key advantage of CASSI is its ability to capture high-dimensional spectral information with a single shot. Spatial and spectral data are multiplexed through coding and dispersion, allowing for compact system design. However, the reconstruction of spectral data from compressed measurements requires complex computational methods.

- 1) **Spatial-Spectral (SS) - CASSI:** Enhances CASSI by modulating both spatial and spectral aspects of the scene, improving the richness of the captured data and potentially allowing for higher-resolution reconstructions.
- 2) **Double-Dispersion (DD) - CASSI:** Introduces a second dispersion to the CASSI setup, leading to a more complex system that can offer improved spectral resolution by better separating overlapping spectral features.
- 3) **Single-Dispersion (SD) - CASSI:** Utilizes a single dispersive element to encode spectral information onto a detector, simplifying the CASSI system and reducing complexity.

- **Pixelated Filter Array (PFA)**

Comprises an array of filters, each corresponding to a different spectral band, placed directly in front of the sensor pixels. The captured image provides spectral data at every pixel.

- 1) **Color Filter Array (CFA):** Enhances CASSI by modulating both spatial and spectral aspects of the scene, improving the richness of the captured data and potentially allowing for higher-resolution reconstructions.
- 2) **Fabry-Perrot Filter Array (FPA):** Uses a series of tunable Fabry-Perrot interferometers, which can be finely controlled to transmit specific wavelengths. FPAs can offer high spectral resolution across a broad range.

- 3) **Sub-wavelength Patterned Grating Filter Array (SWPGFA):** Employs subwavelength grating structures that act as spectral filters, diffracting different wavelengths in specific directions, thus spatially separating the spectral components.

- **Diffraction Modulation (DM)**

Incorporates diffractive optical elements to modulate an incoming light field in such a way that its spectral components are spread out over the detector in a controlled manner.

- 1) **Diffraction Diffuser (DD):** Utilizes a diffuser with diffractive properties that scatters the incoming light while imprinting spectral information onto the spatial profile of the light.
- 2) **Diffractive Lens (DL):** Involves lenses with diffractive features that focus light based on wavelength, enabling the capture of different spectral components at different positions on the detector.
- 3) **Diffractive Neural Network (DON):** A cutting-edge concept that combines diffractive optics with the computational power of neural networks to process light through learned diffractive layers, providing a unique approach to imaging and analysis. This technique enables ultra-fast spectral imaging and analysis but is still in the early stages.

3) REMARKS AND FUTURE OUTLOOKS FOR SNAPSHOT HSI

The evolution of snapshot imaging technology over the years showcases a remarkable variation in the approaches to capturing spectral images. This evolution is characterized by innovations in computational algorithms, aimed at improving spatial and spectral resolution, reducing acquisition times, and expanding the applicable spectral range. A summary of the recent advancements in snapshot spectral imaging system design is shown in Table 2. The table presents various aspects of these systems, including light sources, device mechanisms, detector types, spectral ranges, acquisition times, fields of view, spatial resolutions, image reconstruction methods, and application targets. The choice of light source has transitioned from traditional halogen lamps to more specialized sources like plasma illumination and fiber-coupled sources, reflecting a push towards optimizing the illumination for specific imaging tasks. The device mechanisms have also seen a shift from simple lenslet arrays to more complex arrangements involving diffractive modulation and coded apertures, indicating a trend towards leveraging optical and computational techniques to extract more information from each captured frame. The table further highlights a shift from basic RGB and CCD cameras towards more sophisticated CMOS detectors, suggesting a move towards higher resolution, faster acquisition speeds, and better noise performance. This is crucial for applications requiring fine spatial detail and rapid imaging, such as UAV surveillance and medical diagnostics. Moreover, the trends for the improved spectral resolution, as shown in the table, underscore the continuous effort to enhance image quality. Notable improvements include the reduction of acquisition times and the expansion

TABLE 3. Advances in the field of snapshot HSI system design.

Year	Light Source	Device	Detector Type	NOC	WR (μm)	AT	FOV	SP	IRM	Ref
2019	Plasma illumination	Diffraction Modulation	RGB Camera	25	0.4 – 0.7	-	-	-	DNN	[353]
2019	-	Spatial-Spectral Mapping	CMOS	-	0.515 – 0.842	-	-	-	-	[354]
2019	Halogen Lamp	Lenslet array	CCD	61	0.45 – 0.75	100-400 ms	-	2x28	Six-step Algorithm	[355]
2020	FC- Light Source	Lenslet array	-	-	0.4 – 0.7	-	-	40		[356]
2020	-	Filter Array	CMOS	64	0.35 – 0.9	-	-	448 × 320	CS	[357]
2020	Halogen Lamp	Lens Array	CCD	-	0.5 – 0.6	-	400x1 40 μm		-	[358]
2021	-	Coded Aperture	-	-	-	-	-	256 × 210	CS	[359]
2021	Halogen Lamp	Color Coded Apertures	CMOS	49	0.42 – 0.66	-	-	482 × 512	SVM	[360]
2021	-	Diffraction Modulation	-	-	0.4 – 0.64	-	-	224x224	S & GIF	[361]
2021	-	Diffraction Modulation	CCD	-	-	-	3mm	11x12	-	[362]
2021	-	Lenslet array	-	24	0.45 – 0.65	-	14x3 mm	112×24	-	[363]
2022	LED	Coded Aperture	-	-	0.4 – 0.7	-	-	128x128	CS	[364]
2023	-	Diffraction Modulation	RGB Camera	-	-	-	-	174x174	MLEM	[365]
2023	-	Diffraction Modulation	-	31	-	16 ms	-	64x64	CTIS-GAN	[366]
2023	-	Pixelated Filter Array	CCD	28	0.41 – 0.8	-	60°x5 0°	-	ADMM	[367]
2023	Broadband LS	Diffraction Modulation	RGB Camera	-	0.46 – 0.9	-	-	256×256	CAFormer	[368]
2023	Halogen Lamp	Coded Aperture	Fundus Camera	35	0.445 – 0.602	-	-	1180 × 1100	DNN	[369]
2023	Photography Lamp	Coded Aperture	CCD	-	0.45 – 0.65	-	-	256× 256	PINN	[370]
2023	Halogen Lamp	Diffraction Modulation	CMOS	64	0.4 – 0.9	-	-	600 × 400	-	[371]
2023	-	Lenslet Array	-	25	0.4 – 0.7	-	9x9 mm	622×432	CNN	[372]
2023	Xenon Source	FP Filter	-	-	-	-	-	512 × 640	ML Algorithm	[373]
2023	Halogen Lamp	Coded Aperture	CCD	25	0.4 – 0.65	-	8x8 mm	–	CS	[374]
2023	Halogen Lamp	Fiber bundle	MVCam	23	0.4 – 1.4	30 ms	20mm	–	-	[375]

TABLE 3. (Continued.) Advances in the field of snapshot HSI system design.

2023	Pump Laser	Lenslet array	CCD	-	0.65 – 0.7	-	3x3mm	100× 100	-	[376]
2023	LED	Coded Aperture	FPA	25	0.4-0.7	-	-	256x25 6	-	[377]
2024	Halogen Lamp	Diffraction Modulation	CCD	12	0.43 – 0.65	-	-	950 × 976	CS	[378]
2024	-	Lenslet	FPA	-	0.45 – 0.65	-	-	162x9	-	[379]
2024	-	Prism Grating Prism	-	22	0.4 - 1	-	-	409x40 9	-	[380]
2024	-	-	-	-	0.465 – 0.7	-	36°	600x60 0	-	[381]
2024	LED	Prism	-	27	0.45 – 0.7	1.83 s	-	-	-	[382]
2024	-	Filter Array	-	18	7 - 14	-	-	-	-	[383]

NOC: Number of Channels, **AM:** Acquisition Mode, **AT:** Acquisition Time, **FOV:** Field of View, **SP:** Spatial Points, **WR:** Spectral Range, **IRM:** Image Reconstruction methods, **SS:** SnapShot, **SC:** Scanning, **CCD:** Charged Couple Detector, **M-CCD:** Monochrome Charged Couple Detector, **C-CCD:** Color Charged Couple Detector, **FPA:** Focal Plane Array, **DA:** Detector Array, **CMOS:** Complimentary Metal-oxide Semiconductor Detector, **BD:** Bucket Detector, **MC:** Monochrome Camera, **SPD:** Single Pixel Detector, **FC-Light Source:** Fiber Coupled Light Source, **PC:** Polarization Camera, **CPF:** PushFrame, **M-PRM:** Modified Pixel Re-Arranging Method, **LP:** Line Pair, **PIM & SI:** Pseudo-inverse Method and Spline Interpolation, **CS:** Computational simulation, **FE & CA:** Feature Extraction and Classification Algorithm, **PD:** Photodetector, **FFT:** Forward Fourier Transform, **CS:** Compressive Sensing, **FRAME:** Frequency Recognition Algorithm for Multiple Exposures, **SAM:** Spectral Angle Mapper, **SID:** Spectral Information Divergence, **Vis-NIR:** Visible to Near infrared Spectrum, **DS:** Demosaicing, **BDF:** Bandpass filter, **PLF:** Polarization filter, **SWIR:** Short-Wave Infrared, **HSP:** Hadamard Sampling Pattern, **SLVF:** Slanted Linear Variable Filter, **SCSR:** Spectral Clustering Super-Resolution Reconstruction, **FPS:** Frame Per Second, **FPA:** Focal Plane Array, **PGP:** Prism Grating Prism, **SVM:** Shift Variant Model.

of the spectral range, catering to a broader array of applications from medical imaging to food quality assessment. There is a clear trend towards more advanced image reconstruction methods, including machine learning algorithms like DNNs and physics-informed neural networks (PINNs) for their ability to handle complex and nonlinear data, offering superior image quality and enabling new applications in fields like medical imaging and remote sensing. Future trends seem to point towards further integration of computational imaging techniques, and a continual push towards better resolution and faster imaging speeds. For a more detailed exploration of snapshot spectral imaging techniques, the reader can consult some of the earlier reviews on snapshot spectral imaging [27], [281], [282] and a more recent work by Kaiyang et al. [62].

C. SPATIAL SCANNING TECHNIQUES

1) POINT SCANNING (WHISKBROOM) TECHNIQUE

The point scanning or whiskbroom approach involves capturing spectral data by traversing the target so that a single pixel (point) is scanned at a time. This hyperspectral imaging (HSI) scheme typically employs a moving mirror or rotating scanner to direct light from different points in the scene into the spectrometer. The light collected from each point is

dispersed by the spectrometer, typically using a diffraction grating or prism, which separates the light into its constituent wavelengths. The dispersed light is then captured by a 1D linear detector array, often a charge-coupled device (CCD) or complementary metal-oxide-semiconductor (CMOS) sensor, where each pixel of the detector records the intensity of a specific wavelength band. The evolution of whiskbroom spectral imaging has seen significant advancements since its inception. Initially, whiskbroom systems were primarily used in satellite-based remote sensing applications, such as the Earth resources technology satellites and the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). These early systems demonstrated the potential of whiskbroom scanning for detailed spectral and spatial data acquisition, but they were limited by their size, weight, and the complexity of their mechanical scanning mechanisms. With the advent of more compact and lightweight components, such as miniaturized spectrometers and more efficient optical systems, the application of whiskbroom imaging expanded to airborne platforms, including UAVs. These advancements made it possible to conduct low-altitude, high-resolution observations, which are particularly beneficial for environmental monitoring, precision agriculture, and other applications requiring detailed ground-level data. The main strength of whiskbroom lies in

its ability to capture data point-by-point, allowing for very high spatial resolution since each point is measured individually. This ensures detailed spatial information, which is crucial for applications requiring precise mapping. Moreover, the use of a spectrometer allows for a flexible selection of spectral bands, accommodating a wide range of wavelengths, making it suitable for various applications from agriculture to environmental monitoring. This section will explore point scanning in detail, highlighting both the strengths and drawbacks of the approach through a thorough examination of recent literature.

To this end, some of the early reports are the works by Uto et al. [283], [284] aimed to provide an affordable and practical solution for whiskbroom HSI systems, enabling frequent and accurate low-altitude Earth observations. In [283], they proposed an innovative point scanning based HSI system designed for use with unmanned aerial vehicles (UAVs). The proposed design features a lightweight whiskbroom scanner that uses a polygon mirror to direct light into a spectrometer. The system is built using off-the-shelf electronic components, enabling it to capture a broad spectral range from 340 to 750 nm with a 14-nm spectral resolution. Particularly suitable for low-altitude observations, their system offers enhanced accuracy, resolution, and data frequency compared to satellite-based counterparts. The system was tested on a UAV operating at low altitudes and speeds, where it scanned a checkered pattern to evaluate spatial consistency and spectral data accuracy. The flight speed and altitude were optimized for low-speed, low-altitude operations, with an exposure time of 3.2 ms per area. Figure 16(a) - (d) illustrates the photograph of the imager, the schematic of the system's block diagram, the arrangements of the devices, and the summarized interactive relationships between the linear actuator, the polygon mirror, the target, and sunlight, respectively. The key advantages of the proposed system lie in its light weight of 1200 g, making it easily integrated with lightweight UAVs, facilitating low-altitude and close-range observation with high spectral and spatial resolution. Moreover, it is also suitable for a variety of applications, including precision farming, environmental monitoring, and coastal ecosystem studies. Besides, the proposed design is affordable, making it accessible to a broader range of users, including smaller research institutes and private individuals. However, the whiskbroom scanning mechanism involves moving parts, which can introduce mechanical complexity and the motion between scans can lead to potential spatial misalignment in dynamic scenes. Moreover, while suitable for low-speed, low-altitude operations, the system may not be ideal for high-speed or high-altitude applications due to the slower acquisition speed inherent in whiskbroom scanning.

The design in [283] focuses on the use of off-the-shelf electronic components and compact spectrometers to create a system that is both cost-effective and lightweight. However, the design in [284] (see Figure 17 (a) - (e)), introduces a custom-manufactured optical fiber bundle and swing mirror

mechanism to enhance the spatial resolution and efficiency of the hyperspectral imaging process. This variant design transforms the target image projected by an objective lens into different sets of pixels using the optical fiber bundle, while the cross-track scanning is accomplished using the swing mirror. Spectral measurements at each pixel are conducted using a high-sensitivity spectrometer developed in collaboration with Hamamatsu Photonics. The system demonstrated high spatial and spectral accuracy in close-range measurements, and the aerial measurements conducted from a 20-meter altitude achieved a spatial resolution of 0.5 meters and an 8-meter swath. The system successfully classified three classes of marine macrophyte beds (agar, coralline, and sand) with overall accuracies of 75.5% and 71.9% under different illumination conditions. This design advances the previous design by integrating an optical fiber bundle and a swing mirror, achieving a spatial resolution of 0.5 meters and resolving the structural limitations of the polygon mirror-based system. Together, these studies represent a coherent effort to advance low-cost hyperspectral imaging technology for various remote sensing applications.

In 2017 K. Shibuya et al. [285] presented a novel whiskbroom HSI technique that combines dual-comb spectroscopy (DCS) with Hadamard-transform-based single-pixel imaging (SPI). This method enables the acquisition of 12,000 hyperspectral images of both amplitude and phase with a spatial resolution of 46 μm , without the need for mechanical scanning. The proposed design utilizes two mode-locked erbium-doped fiber lasers to generate optical frequency combs with slightly different repetition frequencies. A spatial light modulator (SLM) is employed to encode the spatial information of the sample using Hadamard patterns, which is then detected by a single-pixel detector, and the hyperspectral images are reconstructed using Fourier and inverse Hadamard transforms. The proposed design achieves a spectral resolution of 20 MHz and a comb mode interval of 100 MHz over a spectral range of 1.2 THz centered at 191.5 THz. It further demonstrated the capability of obtaining spectroscopic images of a test chart and determining the thickness of a chromium-coated layer using phase information. The key advantages of the proposed scheme include high spectral and spatial resolution without mechanical scanning, the capability of providing both amplitude and phase information, which can be used to determine optical constants and material thicknesses, and potentially applicable to various spectral regions including ultraviolet, visible, mid-infrared, and terahertz. However, the measurement time is relatively long due to the response time of the SLM, but this can be improved with faster modulators.

In 2023 Liu et al. [286], introduced a novel hyperspectral imaging (HSI) approach that processes spectral and spatial information independently using a unique coding scheme applied to a dispersed light source and a point scanning-based system. This method leverages the separation of spectral and spatial data to enhance compatibility with scanning systems

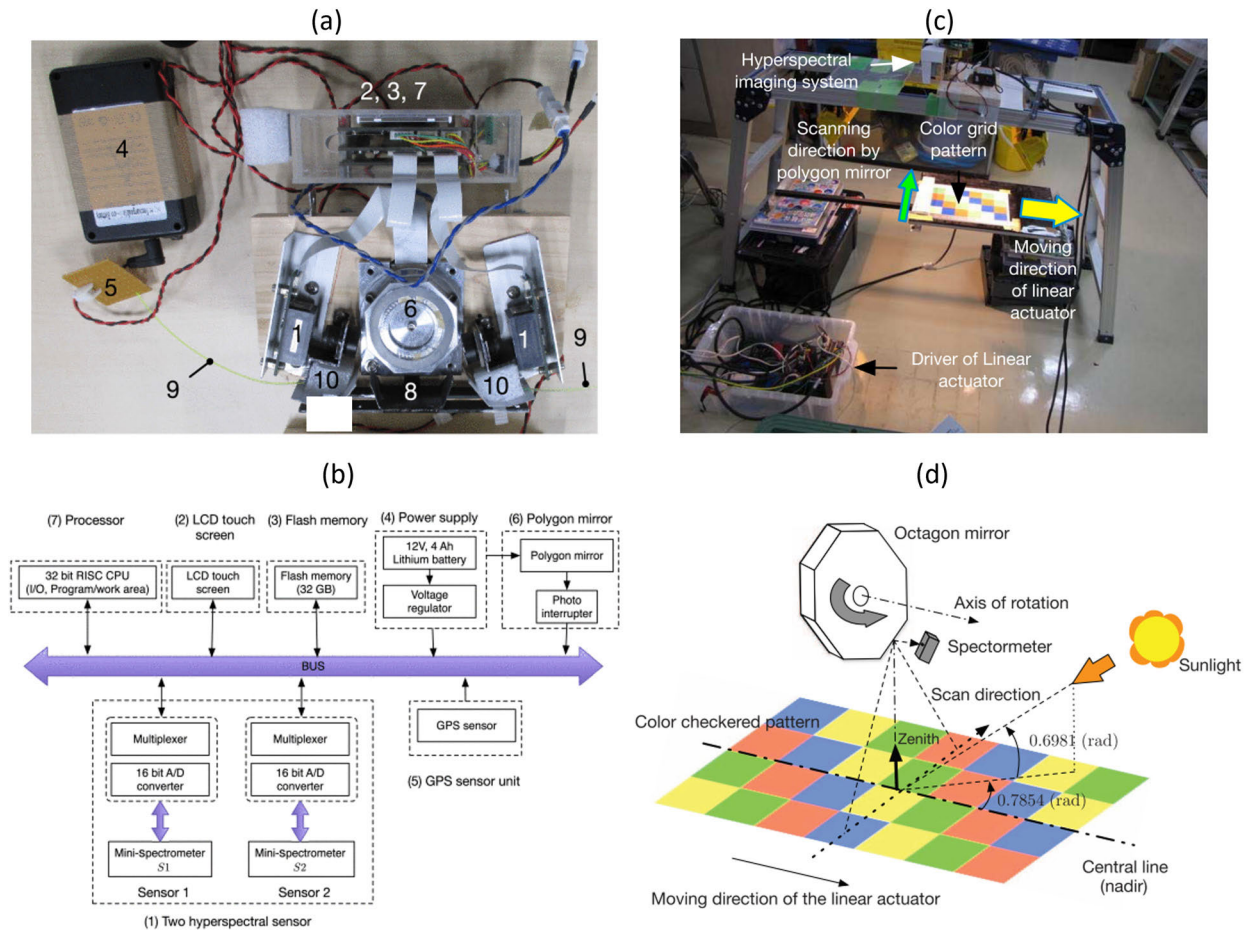


FIGURE 16. The proposed Uto's design of whiskbroom hyperspectral imagers based on a polygon mirror and a compact spectrometer, as reported in [283], includes (a) a photograph of the systems; (b) illustrates the schematic of the system's block diagram; (c) a photograph of the devices arrangements; and (d) a summary of the interaction between the polygon mirror, the target, the linear actuator, and sunlight.

like lidar, facilitating high-quality spectral data acquisition with improved efficiency. As illustrated in Figure 18 (a), the prototype employs common optical components, such as a prism and a spatial light modulator for spectral coding, along with a galvanometer for scanning. To reconstruct the full hyperspectral image from compressed data, the team developed an algorithm that capitalizes on the intrinsic properties of hyperspectral images—namely, the correlations between different wavelengths and the spatial similarities. While this scanning technique inherently slows image acquisition, the resultant image quality and compatibility with existing scanning systems present a compelling trade-off for various applications. This is followed by the work of Arad et al. [287], where they presented a novel, low-cost HS sampling device designed for agricultural and environmental monitoring. This device utilizes a dual-wedge prism scanner to capture a wide spectral range, significantly reducing the cost by simplifying the hardware design. To maintain high image quality, the system employs post-processing corrections to address chromatic distortions, such as spectral bias and spatial misregistration. The proposed design achieves a mismatch of 4%

when compared to ground-truth hyperspectral data, and the validation experiments demonstrated an RMSE (Root Mean Square Error) of 0.015 mm for spot position and 0.201 mm for spot diameter. However, the need for post-processing and potential environmental limitations must be considered when assessing its suitability for specific applications. The geometry of the proposed dual prism system and the conceptual schematic of the scanning system is shown in figure 18 (b).

Recently, Li et al. [288] proposed a method to achieve high spatial resolution images and spectral information using a single multi-aperture imaging system. By compensating for optical path differences, the sub-apertures are well-phased, enabling the system to produce high spatial resolution images comparable to those from a large single aperture. The system introduces controlled optical path delays between sub-apertures and scans them to obtain spectral information through Fourier transform imaging spectroscopy (FTIS) techniques. The team demonstrated the concept with simulations of point sources and extended objects and through experiments using a spatial light modulator to introduce precise phase delays. Properly setting the delay rates for

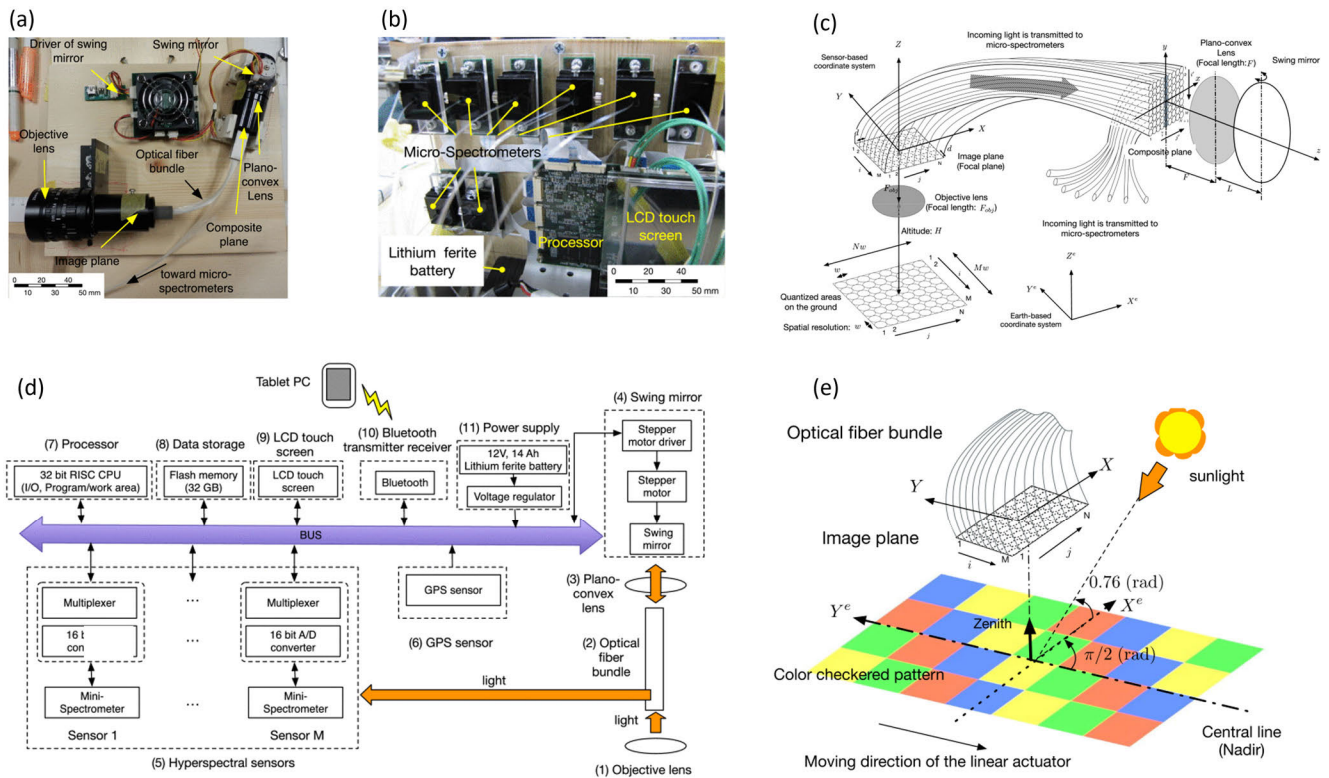


FIGURE 17. The proposed variant of Uto's designs, as reported in [284], features an alternative configuration based on an optical fiber bundle, a swing mirror, and compact spectrometer. Where (a) and (b) present photographs depicting the system's components; (c) Shows the schematic of the target observation configuration, including the objective lens, image plane, optical fiber bundle, composite plane, Plano-convex lens, and swing mirror; (d) Outlines the schematic of the system's block diagram, and (e) Provides illustrative summary of the interaction between the sensor, the target, the linear actuator, and sunlight.

each sub-aperture allows for the acquisition of non-overlapping spectral information from different frequency bands. Despite its promise, the practical implementation of this technique in real-world multi-aperture systems faces significant engineering challenges. Precision and stability for both high-resolution imaging and spectral information extraction is complex due to the need for precise control of optical path delays between multiple sub-apertures. Additionally, there is a trade-off between spectral resolution and acquisition time when scanning delays, and potential image degradation can occur when introducing path delays compared to the well-phased imaging condition. In [289], Son et al. presents a compact staring-type underwater spectral imaging system designed to overcome challenges in underwater imaging, such as system bulkiness and spectral resolution limitations. The system integrates a six-channel rotary filter camera, liquid lenses for autofocus, and a sealed chamber for operation at depths up to 300 meters. A k-Nearest Neighbor (KNN)-based interpolation method enhances spectral resolution from 6 to 31 channels within the 400–700 nm range. Features like auto-focusing using the Tenengrad function and image registration through the Scale-Invariant Feature Transform (SIFT) ensure accurate spectral reflectance reconstruction. Experimental validation using Munsell color cards

demonstrated high accuracy, with a Root Mean Square Error (RMSE) of 0.0651 and a Goodness of Fit Coefficient (GFC) of 0.9871. The system's compact design, cost-effectiveness, and advanced spectral capabilities make it suitable for various underwater applications, including marine monitoring and resource exploration. Despite its computational demands and reliance on artificial lighting, the system marks a significant step forward in underwater spectral imaging technology.

The primary strength of the whiskbroom scanning method, beyond its capability to capture data on a point-by-point basis for achieving very high spatial resolution, lies in its ability to optimize the signal-to-noise ratio for each individual measurement. This optimization results in higher-quality spectral data, which is crucial for accurate analysis and classification. Additionally, the single-point measurement approach minimizes optical aberrations typically present in wide-field imaging systems, thereby ensuring more precise spectral data and enhancing the overall reliability of the measurements. Renowned for its efficacy in circumventing non-uniform effects [283], this scanning method simplifies calibration procedures by reducing reliance on multiple spectrometers. There have also been further developments focused on reducing the size and cost of whiskbroom systems while maintaining or improving their performance. Innovations

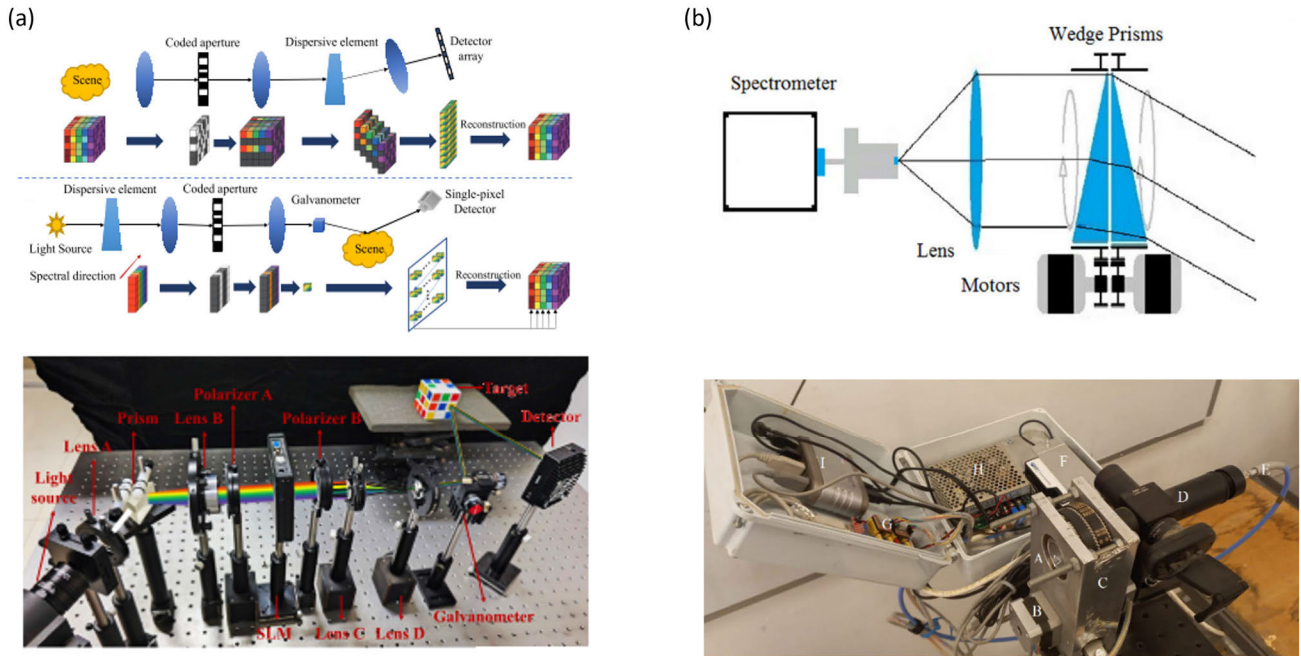


FIGURE 18. The two proposed designs for whiskbroom hyperspectral imagers; (a) The compressive HSI reported in [286], featuring a comparative between the conventional CASSI (up the dotted line) and the their proposed method, and the photograph of the hardware implementation of their proposed method (left bottom). (b) The schematic illustration of the dual-prism scheme reported in [287] featuring the conceptual illustration of the dual prism geometry, and the photograph image of the implemented design (bottom right).

such as the use of optical fiber bundles and advanced mirrors have enhanced the spatial resolution and efficiency of these systems. The integration of high-sensitivity compact spectrometers, as seen in the work by Uto et al. [284], has further improved the quality of spectral data obtained from whiskbroom imaging. Despite these advancements, whiskbroom imaging still faces challenges. Capturing data in both spatial and spectral dimensions at each pixel location necessitates mechanical scanning across two axes, resulting in longer data acquisition times. The field of view is limited by the scanning mechanism, which can be narrower compared to other hyperspectral imaging (HSI) systems, often requiring multiple passes to cover larger areas and further slowing down data acquisition. Additionally, the calibration of whiskbroom systems can be complex due to the moving parts and the need to ensure consistent alignment. This complexity can make the system more difficult to use and maintain, especially in field conditions, as any misalignment due to vibrations, mechanical wear, or scene dynamics can degrade the quality of the spectral data. However, in microscopic imaging, such as confocal microscopy, the whiskbroom scanning method is particularly advantageous. Its ability to measure a single point at a time while rejecting signals from other points improves image quality and spatial resolution, making it ideal for detailed imaging applications that require precise and accurate data [130].

2) LINE SCANNING (PUSHBROOM) TECHNIQUE

Line Scanning, also known as Pushbroom HSI, is a technique that involves the use of a linear detector array to

capture spectral data along one spatial dimension while the platform (such as a satellite, aircraft, or ground vehicle) moves across the scene to build the second spatial dimension. This method contrasts with point scanning, where a single detector element scans across both spatial dimensions and snapshot imaging, where the entire hyperspectral cube is captured simultaneously. Pushbroom HSI stands out for its efficiency and high-resolution spectral data acquisition, making it a preferred choice for numerous applications. The pushbroom HSI system's operational efficiency is largely attributed to its unique design, where a two-dimensional detector array is used to simultaneously capture spectral and spatial information along a single line of pixels. A pushbroom HSI system typically consists of an entrance slit, a dispersive element (such as a diffraction grating or prism), and a two-dimensional detector array. As the sensor platform moves, the line of pixels is swept across the target scene, and the dispersive element separates the incoming light into its constituent wavelengths. The detector array captures this dispersed light, producing a two-dimensional image where one dimension corresponds to spatial information along the scan line and the other dimension corresponds to spectral information. This design allows for rapid data acquisition without the need for mechanical scanning, minimizing motion artifacts and enabling high-speed imaging.

Moreover, the integration of cutting-edge sensor technologies, such as charge-coupled devices (CCDs) and complementary metal-oxide-semiconductor (CMOS) detectors, has significantly enhanced the performance of pushbroom HSI systems. These sensors offer high quantum efficiency, low

noise levels, and fast readout speeds, ensuring the accurate and efficient capture of hyperspectral data. Additionally, advancements in calibration techniques and data processing algorithms have further improved the accuracy and reliability of pushbroom HSI measurements. Despite its numerous advantages, pushbroom HSI also presents several challenges that need to be addressed. The large volume of data generated requires robust storage and processing capabilities, and the development of efficient algorithms for data compression, transmission, and analysis is crucial. Furthermore, the calibration and validation of pushbroom HSI systems demand meticulous attention to ensure the accuracy and consistency of spectral measurements. Environmental factors, such as atmospheric conditions and lighting variations, can also impact data quality, necessitating sophisticated correction techniques to mitigate their effects. In recent years, significant progress has been made in overcoming these challenges, driven by advancements in sensor technology, computational methods, and machine learning algorithms. The integration of artificial intelligence and machine learning with pushbroom HSI has opened new avenues for automated data analysis, feature extraction, and anomaly detection, enhancing the capability of HSI systems to deliver actionable insights in real-time.

There are numerous reports in the literature on the design of pushbroom HSI systems tailored for various applications. These designs are critical for enabling the high-resolution spectral and spatial data acquisition necessary for advanced analysis in fields ranging from environmental monitoring to industrial inspection. Among the early examples in the 70s and the 90s in this domain are the efforts to substantiate the design of pushbroom hyperspectral imagers using concentric spectrometer forms for earth observation and remote sensing [290], [291]. In the late 90s and 2000s, Mouroulis and colleagues made significant contributions in this area by developing a pushbroom HSI system that employs concentric surface forms [292], [293], [294]. In their design, gratings are placed on a spherical surface within a small catadioptric system. This configuration offers several advantages over traditional designs. By placing the gratings on a spherical surface, the system achieves more precise spectral dispersion, which enhances the spectral resolution and accuracy of the captured data. This placement reduces optical aberrations that can degrade image quality, thus ensuring that the system can produce high-fidelity spectral images. The use of a small catadioptric system further improves the design by combining reflective and refractive optical elements, which enhances the system's ability to focus and direct light. This combination allows for a compact yet powerful imaging system that can be deployed on various platforms, including satellites, aircraft, and unmanned aerial vehicles (UAVs) [295]. Another notable contribution in this area has been made by Davis et al., who designed a sophisticated portable pushbroom HSI system tailored for coastal environments. This system integrates a highly sensitive thinned CCD detector that is backside illuminated, enhancing its

ability to capture faint signals in complex and often turbid coastal waters. The backside illumination significantly improves the quantum efficiency of the CCD, allowing for more accurate and detailed spectral data collection even under challenging lighting conditions typically encountered in coastal regions [296]. In the same vein, Yang et al. showcased a CCD based HSI system incorporating an imaging spectrograph, focal plane scanner, image processing unit equipped with frame utility software, and a CCD camera [8]. This system was designed for both airborne remote sensing and stationary applications. The focal plane scanner facilitated pushbroom scanning by scanning the input image around the focal plane of the frontal lens, demonstrating the versatility and adaptability of pushbroom HSI systems in various operational settings. Moreover, Egloff et al. proposed an all-reflective hyperspectral imager utilizing a scanning micro-opto-electromechanical (MOEMS) mirror etched as a diffraction grating [297]. This compact system exhibited hybrid functionality, serving both as a directional scanning element and a wavelength dispersion component.

In the current decade, significant advancements have been made in pushbroom HSI systems, driven by innovations in sensor design and data processing algorithms. A notable example is the work of Arablouei et al. [298] in 2016, who proposed a pushbroom HSI system leveraging a digital micro-mirror device (DMD) to eliminate the need for macro moving parts. The key components of the proposed system include a DMD, a CMOS image sensor without a filter for hyperspectral data capture, a CMOS RGB image sensor for auxiliary imaging, and a transmissive diffraction grating. This prototype achieves a spatial resolution of 400×400 pixels and a spectral resolution of 500 bands in under thirty seconds, covering the visible near-infrared (VNIR) range (400-900 nm). The integration of dual image sensors and the innovative use of a DMD for scanning significantly enhance the system's performance and applicability across various fields, including agriculture, environmental monitoring, and biomedical imaging. Further advancements in pushbroom HSI systems have been achieved through spectral encoding using a single liquid crystal cell and compressive sensing, which provide more efficient and versatile hyperspectral data acquisition [243]. Additionally, the use of metal-polymer nanocomposites to produce a stretchable diffraction grating for compact systems was reported by Potenza et al. [299], where the design utilized a concave cylinder to curve the gratings, resulting in a catoptric device with minimal aberration. Between 2017 and 2018, several other innovative designs have been reported [300], [301], [302], [303], [304], [305], [306], contributing to the rapid evolution of pushbroom HSI technology and expanding its application potential across multiple disciplines. These advancements underscore the dynamic nature of HSI research and the continuous efforts to enhance imaging capabilities through technological innovation.

In 2019, Nevala and Baden [307], present a low-cost HSI system that achieves a balance between cost and

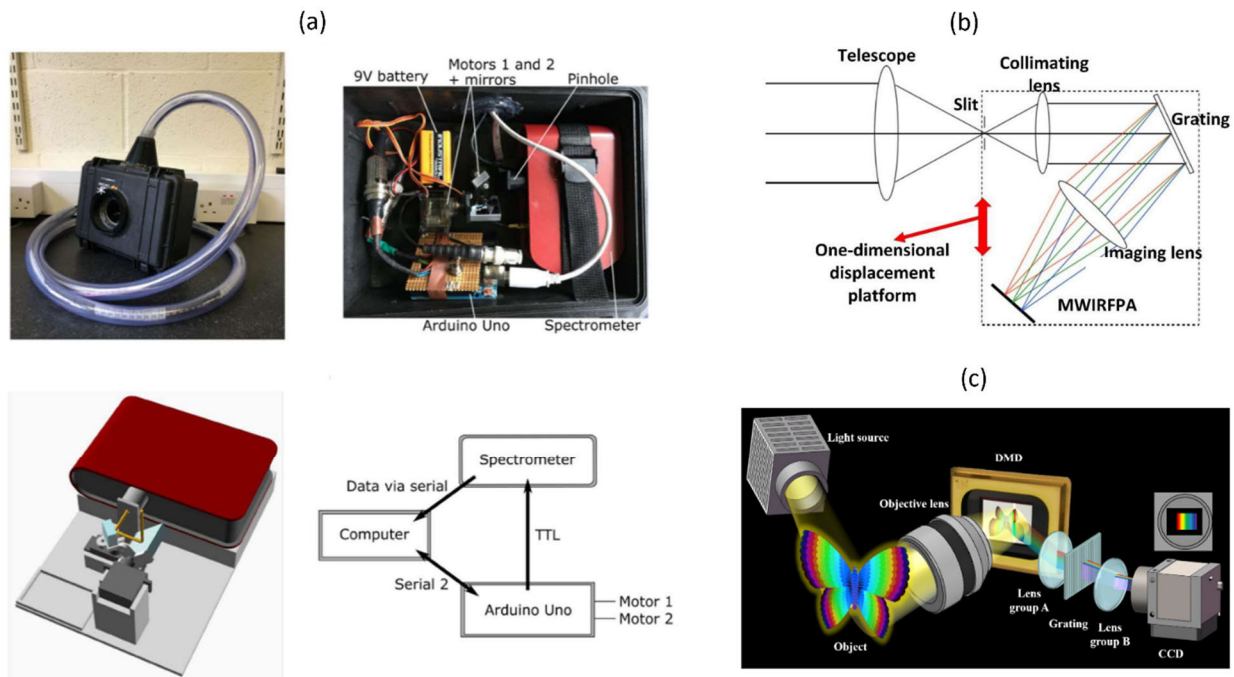


FIGURE 19. (a) The proposed designs for the pushbroom HSI system reported in [307] featuring the water proof packaging of the system, the internal packaging of the components, a simplified diagram of the optical pathway shows light (yellow lines) entering from above, striking a larger mirror inside the casing. This mirror reflects the light to a smaller mirror, which then directs it through a pinhole to the spectrometer's slit. Some light is partially blocked by the casing's edges due to deflection by the first mirror, causing dark bands at the horizontal edges of scanned images when the enclosure is closed. The Schematic illustration of the logical operation of the system. (b) The schematic diagram of the optical components of mid-IR pushbroom HSI system design proposed in [308]; (c) The conceptual schematic of the DMD-based pushbroom HSI system proposed in [309].

performance, providing sufficient spectral and spatial resolution for detailed analysis of natural scenes. The system is particularly useful for studying the color vision of animals, as it allows researchers to capture the spectral properties of natural habitats and underwater environments. By leveraging affordable materials and a straightforward optical setup as shown figure 19 (a) the proposed design demonstrates that high-quality spectral imaging does not necessarily require expensive and complex systems. This approach aligns with the ongoing trend of developing portable and cost-effective HSI solutions that can be deployed in a variety of field conditions, enhancing the versatility and applicability of hyperspectral imaging technology in contemporary research. Yu et al. [308], present a sophisticated design for a push-type mid-IR ($3.7 - 4.8 \mu\text{m}$), HSI system that employs a reflection planar grating. The system consists of a front optical system, a slit, a collimation system, a grating spectroscopic system, an imaging optical system, imaging electrical components, and a control module. The approach outlined by Yu et al. reflects a broader trend in the field toward developing more efficient, compact, and high-resolution hyperspectral imaging systems for applications in both civilian and military domains, such as geographical remote sensing, target detection, and thermal analysis. The schematic diagram of the optical components of the proposed design is shown in figure 19 (b), illustrating the arrangement of the reflection

planar grating, image sensor, and other optical components. In [309], Dong et al. presented a HSI system design leveraging a DMD for enhanced performance and flexibility (See Figure 19 (c)). This system bypasses the traditional limitations of pushbroom HSI system, such as the bulky slit mechanism and mechanical parts, by employing a DMD for on-chip scanning, and integrates a CMOS image sensor, a CMOS RGB sensor for auxiliary imaging, and a transmission diffraction grating. Although the proposed system achieves high tenability in spatial and spectral resolution while eliminating mechanical parts, the potential complexity in ensuring synchronization between DMD scanning and CCD recording may pose performance limitations.

The work by Jiang et al. [310] contributes to the field of HSI systems by adapting remote sensing hyperspectral spatial resolution (HSSR) techniques to HS microscopic applications, particularly in the field of biomedical research [311]. The proposed system (see Figure 20 (a)) employs a homemade pushbroom setup comprising a microscope, a spectrometer, and a scientific CMOS (sCMOS) detector. The authors used several HSSR algorithms to enhance spatial resolution, particularly focusing on the integration of high-resolution auxiliary images to improve the resolution of the hyperspectral data. The enhanced spatial resolution achieved through these methods and the improved imaging speed facilitate a more detailed analysis of biological tissues, which is

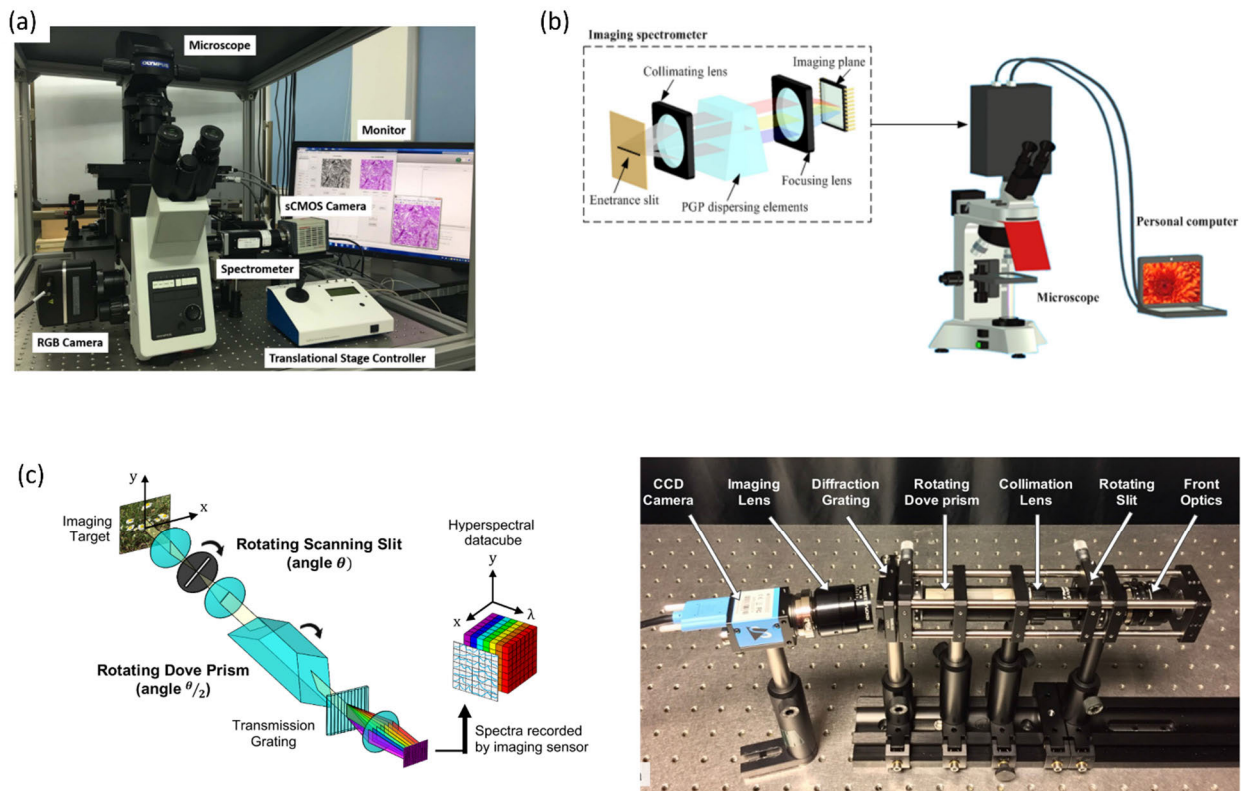


FIGURE 20. (a) The photograph of the proposed microscope based pushbroom HSI system reported in [310]. (b) The schematic diagram of the proposed Pushbroom HS microscopy system reported in [312]. (c) The proposed spatial scanning based HSI based on a rotating slit and a Dove prism reported in [313], including the conceptual schematic of the system, and the experimental setup depicting the system components.

essential for applications such as pathological diagnosis and biomedical research. This was followed by the work of Zhang et al. [312] where they proposed the integration of a prism-grating-prism spectrometer and an internal pushbroom mechanism for a compact, high-resolution imaging system capable of detailed tissue analysis (see Figure 20 (b)). The system captures data across 256 spectral channels within the 400 nm to 800 nm range. The use of a PGP imaging spectrometer ensures minimal image distortion (smile $< 6.4\%$ and keystone $< 1.6\%$ of a pixel), enhancing the accuracy of the spectral data.

A new concept for spatial scanning-based HSI system was proposed in [313] featuring a unique approach to hyperspectral imaging by utilizing a rotating slit and a Dove prism to achieve high spectral resolution through internal spatial scanning without relative motion between the system and target. The rotating slit selects image lines sequentially, while the Dove prism, rotating at half the slit angle, maintains vertical alignment with the diffraction grating. This approach facilitates spectral information extraction and maintains calibration across different slit positions. The system demonstrates potential for applications where conventional scanning methods are impractical, such as endoscopy. The schematic illustration of the system and the experimental setup are depicted in figure 20 (c), respectively. Hau and

Hou [314] demonstrated a MEMS-based self-referencing cascaded line-scan camera utilizing single-pixel detectors. The proposed system aims to provide a cost-effective alternative to traditional line-scan cameras, offering high spatial resolution and broad wavelength operability. The ability to achieve high spatial resolution and broad wavelength operability at a reduced cost opens new possibilities for hyperspectral imaging. This innovative approach aligns with current trends in making advanced HSI technologies cost-effective, more accessible, and adaptable, contributing to the ongoing evolution of hyperspectral imaging solutions. However, there is a potential complexity in aligning and integrating multiple units, and also the need for further optimization to maximize frame rates and efficiency.

In 2020, Yi et al. [315], present a novel hyperspectral imaging system that combines Hadamard transform-based multiplexing with single-pixel detection. This approach aims to enhance signal-to-noise ratio (SNR) and reduce system complexity and cost, particularly for visible near-infrared (VNIR) and infrared (IR) spectral bands. As shown in Figure 21(a), the hyperspectral imaging system employs a digital micro-mirror device (DMD) for spatial encoding and a resonant scanning mirror for spectral encoding. The setup includes a Czerny-Turner imaging spectrometer, which ensures minimal aberrations and high spectral resolution.

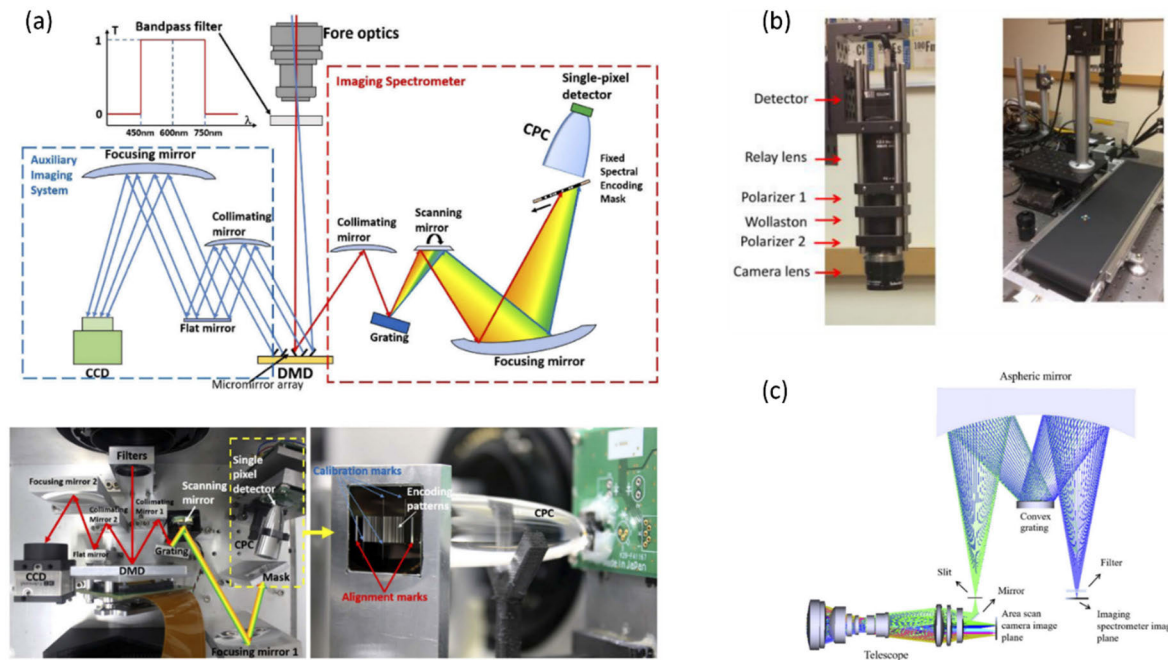


FIGURE 21. (a) The photograph of the pushbroom HSI system reported in [315], including the schematic of the hyperspectral imager showing the spatial and spectral encoding mechanisms and the optical pathway. The prototype photograph of their fabricated device, and the photo of the spectral encoding system, showcasing the encoding mask, a CPC (compound parabolic concentrator), and a single-pixel detector. The encoding mask features alignment marks, calibration marks, and encoding patterns. (b) The photograph of the proposed Wollaston prism-based pushbroom HSI system reported in [317]. (c) The optical pathway of the UAV-based pushbroom HSI presented in [318].

Similarly, Xu et al. [316] used a single DMD for projecting hyperspectral patterns to provide a low-cost, high-frame-rate, and compact solution for combining 3D imaging with hyperspectral data acquisition. The optical design includes a halogen light source, a cylindrical lens, a transmission diffraction grating, and achromatic lenses for focusing and collimating light. The DMD performs spatial and spectral modulation, projecting patterns that encode both 3D spatial and spectral information. Chamoun and Farhadi [317] employed a Wollaston prism placed between two polarizers to function as a spectral encoder. As light traverses the prism, it is divided into orthogonal polarizations, resulting in the formation of interference fringes overlaid on the image. This modified image, which includes both the scene and the fringes, is then projected onto a CMOS sensor. Spectral data is extracted by moving the sample across the scene (see figure 21 (b)), capturing a series of images that are subsequently used to create an interferogram for each pixel.

In 2021 Xue et al. [318] proposed a UAV-mounted hyperspectral imaging system that integrates a high-frame-rate area array camera and a hyperspectral camera sharing the same optical path, significantly reducing system weight and volume to 1.9 kg. The system uses a knife-edge mirror to divide the field of view between the area array camera and the hyperspectral camera, facilitating simultaneous high-resolution imaging and geometric rectification (see Figure 21 (c)). This system addresses challenges associated with UAV-based hyperspectral imaging, such as geometric distortions caused by UAV motion and limited payload capacities. Similarly,

Yang et al. [319], proposed a distortion correction method that utilizes linear features and the invariance of the cross ratio. Specifically, they use a polynomial model to correct distortions along the pushbroom direction and a radial distortion model for vertical distortions. They demonstrated distortion corrections in images captured by an off-axis Schwarzschild two-mirror hyperspectral imager. The off-axis Schwarzschild system is particularly valued in aerospace applications for its advantages such as no chromatic aberration, no blocking, a large field of view, and high numerical aperture. However, it has fewer free parameters, making it challenging to correct distortions directly through optical design. After applying the proposed distortion correction method, the maximum distortion along the pushbroom direction was reduced from 65 pixels to 5 pixels. In the vertical pushbroom direction, the distortion was reduced from 7 pixels to within 1 pixel.

In addition, Khezrabad et al. [320] later proposed another geometric correction method that offers a two-phase process to correct the distortions without relying on ground control data. The first phase focuses on removing compensable geometric distortions from the raw pushbroom image without the need for ground control data, producing a rectified version of the image. The second phase establishes mathematical models relating the rectified image to the 3D ground coordinate system, enhancing the ease of this relationship due to the improved geometric quality from the first phase. The proposed method showed superior performance, particularly in scenarios with high perturbations and low overlap between frames. In [321], Prentice et al. describe

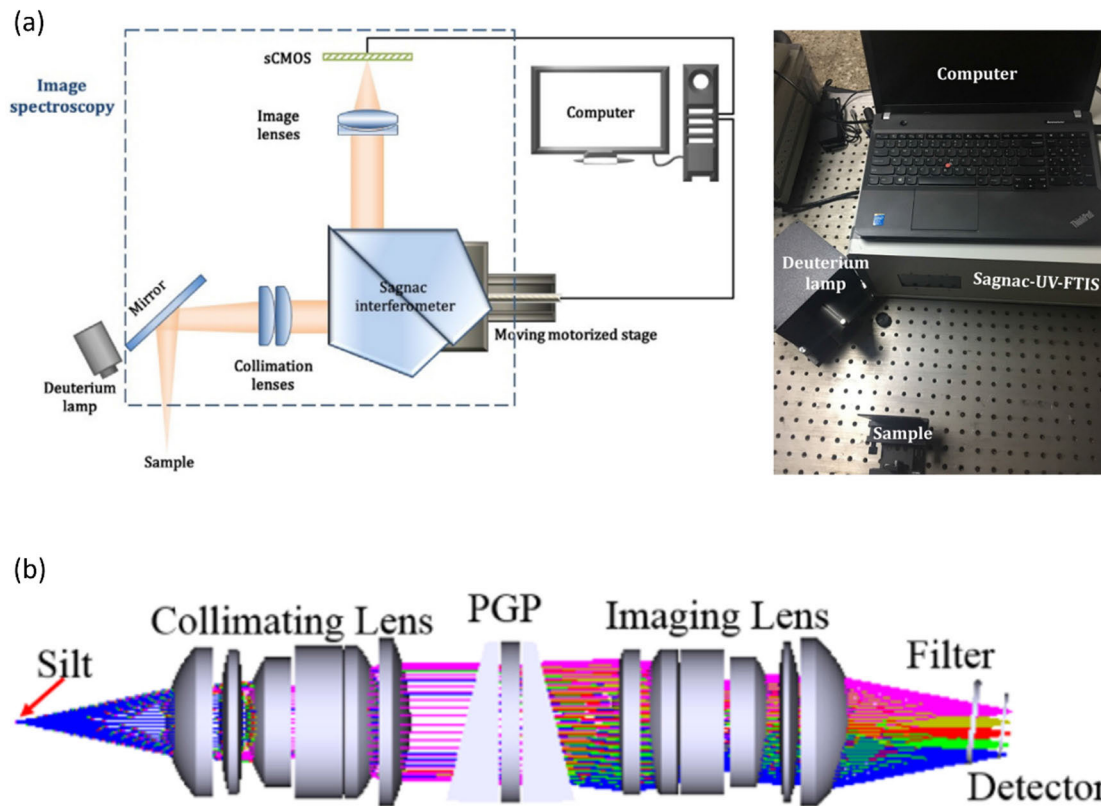


FIGURE 22. (a) The latent fingerprints detection system based on pushbroom HSI system reported in [322], includes a schematic of the system setup, and the prototype photograph of the system (b) The schematic of the optical structure of the proposed PGP based pushbroom underwater HSI system proposed in [323].

the development of a pushbroom hyperspectral imager for small satellite applications, utilizing commercial off-the-shelf (COTS) components to minimize cost and development time. The hyperspectral imager uses a series of three 50 mm lens objectives, a precision-cut $50\ \mu\text{m}$ slit, a blazed transmission grating, and a CMOS sensor. Although, the design is practical and an innovative solution for small satellite HSI using COTS components, the large hyperspectral data cubes require efficient onboard processing to reduce data size for downlinking.

Bai et al. [322] presented a technique for the detection of latent fingerprints and their residues utilizing a Sagnac ultraviolet Fourier transform imaging spectrometer (Sagnac-UV-FTIS). This system combines high-resolution pushbroom HS imaging with rapid data acquisition to produce detailed three-dimensional data cubes that capture both spatial and spectral information of the sample. The system offers a field of view of $59.98\ \text{mrad}$ and achieves wavelength resolutions of $1.59\ \text{nm}$ at $250\ \text{nm}$ and $4.09\ \text{nm}$ at $400\ \text{nm}$, providing approximately 59 spectral channels within the 250 to $400\ \text{nm}$ range. This proposed method presents significant advancements in forensic science, offering a non-destructive and rapid approach to latent fingerprint analysis. Figure 22 (a) illustrates the working platform setup of this advanced detection method. In [323], Xue et al. presented the development and implementation of an underwater hyperspectral imaging

(UHI) system that utilizes a prism–grating–prism (PGP). The system is designed to improve underwater detection capabilities by addressing the limitations of current technologies, such as low detection resolution and susceptibility to interference. Operating within a spectral range of $400\ \text{nm}$ to $800\ \text{nm}$, the proposed UHI system's optical design includes a large aperture front lens to enhance luminous flux and a PGP structure to achieve high spectral resolution and reduce spectral line bending. The PGP structure combines the dispersion characteristics of prisms and gratings, providing straight-in and straight-out light paths at the central wavelength. The schematic illustration of the optical structure of the system is depicted in Figure 22 (b) depicting the optical pathways of the configuration. Kubo et al. [324] proposed an HS two-photon excitation microscopy leveraging visible-wavelength two-photon excitation microscopy with line illumination and slit-confocal detection. This system employs a femtosecond pulsed laser at $530\ \text{nm}$ to enable the concurrent excitation of multiple fluorescent proteins. By integrating advanced linear spectral unmixing techniques, the proposed system localizes fluorescent probes within the cells, providing detailed imaging.

In 2022, Morales et al. [325], reported the development and validation of a custom-made scanning system for hyperspectral image acquisition in laboratory settings. The system integrates commercially available components into a flexible

TABLE 4. Advances in the field of pushbroom HSI system design.

Year	Light Source	Device	Detector Type	NOC	WR (μm)	AT	FOV	SP	IRM	Ref.
2019	-	Diffraction grating	-	-	0.35 – 0.95	4-6 mins	15°	-	PCA	[306]
2019	-	Blazed Grating	FPA	-	3.7 – 4.8	-	3.7°	30x25	ML Model	[307]
2019	Halogen Lamp	Transmission grating	CCD	500	0.45 – 0.65	30s	28°	400X400	Pixel Fusion	[308]
2019	Halogen Lamp	Prism –grating - Prism	CCD	256	0.4 – 0.8	-	-	-	ML Model	[311]
2019	White LED	Grating	CMOS	256	0.38 – 0.8	45 Hz	-	415x415	-	[309]
2019	Halogen Lamp	Dove prism	CCD	-	0.4 – 0.7	4-5 mins	-	-	-	[312]
2020	LED	Diffraction Grating	CCD	-	0.45 – 0.75	20s	-	359 × 45	-	[314]
2020	LED	Wollaston Prism	CMOS	-	0.4 – 0.7	-	-	-	-	[316]
2020	Halogen Lamp	Transmission Grating	Grayscale Camera	32	0.425 – 0.735	-	-	128x128	-	[315]
2021	-	Convex Grating	CMOS	-	0.4 - 1	-	40°x26°	-	-	[317]
2021	-	Blazed Transmission Grating	CMOS	124	0.3 - 1	-	70km	-	-	[320]
2021	-	Curved prism	LAD	90	0.4 – 0.9	-	37.2°	65x7	-	[318]
2021	Deuterium lamp	-	CMOS	59	0.25 – 0.4	-	59.98 mrad	-	-	[321]
2021	LED Array	PGP	CMOS	-	0.4 – 0.8	-	11x11°	640x640	-	[322]
2023	-	Kalman Filter	CCD	-	0.45– 0.65	-	-	-	-	[326]
2024	-	-	CMOS	12	0.4 - 1	-	36°	-	-	[327]

NOC: Number of Channels, **AM:** Acquisition Mode, **AT:** Acquisition Time, **FOV:** Field of View, **SP:** Spatial Points, **WR:** Spectral Range, **IRM:** Image Reconstruction methods, **LED:** Light Emitting Diode, **CCD:** Charged Couple Detector, **FPA:** Focal Plane Array, **CMOS:** Complimentary Metal-oxide Semiconductor Detector, **FPA:** Focal Plane Array, **PCA:** Principal Component Analysis, **ML:** Machine Learning.

infrastructure capable of obtaining data from any Generic Interface for Cameras (GenICam) compliant devices using the gigabyte Ethernet interface. In the same vein, Graham et al. [326], presented a novel method for measuring downwelling irradiance using a fiber optic sensor integrated into a pushbroom hyperspectral imager. The optical path starts with a cosine corrector, coupling light into an ultra-low-OH fiber with a core diameter of 100 μm and a numerical aperture of 0.22. The fiber is positioned at the upper edge of the spectrometer entrance slit, replacing the outermost spatial channels of the camera sensor with light transmitted by the fiber optic. The system is secured using a hypodermic needle and 3D printed guide, ensuring robustness for field use. A quick disconnect fitting allows for easy changes of

the external fiber patch cable. The integrated fiber optic spectrally resolved downwelling irradiance sensor provides an efficient method for measuring incident light levels using the same camera sensor as the main hyperspectral imager. In 2023, Hu et al. [327], presented an approach to pushbroom HSI that enhances signal-to-noise ratio (SNR) and throughput. Their method draws inspiration from time-delay integration (TDI) techniques commonly used in applications requiring high SNR, such as satellite imaging and industrial inspection. The system includes fore-optics, a digital micro-mirror device (DMD) for slit encoding, and a spectrometer. The DMD dynamically encodes the multi-slit entrance to the spectrometer, allowing flexible control of the slit patterns. The target is pushbroom scanned, with each slit sequentially

capturing the image, resulting in multiple exposures. The overlapped spectral images are captured on a CCD, and the Kalman filter reconstructs these images into a high-quality hyperspectral data cube. Their study demonstrates that fixed coding masks can replace DMDs in practical implementations, reducing system complexity and cost while maintaining high performance.

Recently, Liu et al. [328], introduces an internal scanning HSI system designed for deep-sea exploration, addressing the limitations of traditional push-broom systems reliant on platform movement. The system employs a rotating mirror mechanism within a compact, modular design, capable of operating at depths up to 6000 meters with a spectral range of 400–1000 nm and a resolution below 3 nm. A key innovation is the geometric distortion correction method, which ensures high-quality, undistorted imaging despite distortions caused by internal scanning. Experimental validation demonstrated accurate spectral calibration, effective geometric correction, and reliable imaging in both air and underwater conditions, with the watertight cabin withstanding the simulated deep-sea pressures. Compared to existing systems, the internal scanning mechanism offers enhanced flexibility and compactness, making it particularly suited for complex underwater environments. The system holds significant potential for applications in marine resource exploration, habitat mapping, and ecological monitoring, with future work focusing on increasing robustness and integrating with autonomous underwater vehicles. This work represents a major advancement in hyperspectral imaging for modern marine exploration.

A summary of recent advancements in pushbroom HSI system design from 2019 to 2024 is provided in Table 4. The table presents various aspects of these systems, including light sources, device mechanisms, detector types, spectral ranges, acquisition times, fields of view, spatial resolutions, image reconstruction methods, and application targets. The table highlights a significant improvement in system performance, image reconstruction methods, and application diversity. The integration of advanced optical components, and sophisticated reconstruction algorithms has expanded the capabilities of pushbroom HSI systems, making them suitable for a wide range of applications, including natural scene imaging, military, medical, UAV-based remote sensing, space, and forensics. The continuous evolution in the field is driven by the need for higher resolution, better SNR, and more efficient data processing methods. Thus, different methods were reported [329], [330], [331], [332], [333], [334], [335], [336], [337] for corrections such as geometric corrections, image registration errors, micro-vibrations in drones and small satellites, and so on.

3) REMARKS AND FUTURE OUTLOOK FOR SPATIAL SCANNING BASED HSI

Spatial scanning-based HSI systems have evolved from bulky satellite-based systems to lightweight UAV-compatible systems, enabling their use for high-resolution, low-altitude studies. While whiskbroom systems are advantageous for

capturing detailed spatial information, their reliance on mechanical scanning components introduces challenges related to calibration and stability, especially under dynamic conditions. Nevertheless, recent innovations, including the integration of fiber bundles and Hadamard-based single-pixel imaging, have mitigated some of these limitations, thereby improving system flexibility and spectral accuracy. Pushbroom techniques, known for their operational efficiency in capturing spectral and spatial data simultaneously, have also undergone notable development. The integration of advanced sensor technologies, such as CMOS detectors, along with machine learning for real-time data processing, has significantly enhanced the capabilities of these systems. Pushbroom HSI systems have further benefitted from innovative spectral encoding techniques that use components like DMDs, thereby reducing mechanical complexity and expanding their suitability across a range of platforms, from UAVs to satellites. This adaptability has enabled their effective deployment in both high-speed imaging scenarios and applications requiring detailed spectral analysis.

Looking to the future, ongoing research should prioritize the miniaturization of HSI systems while incorporating edge computing for real-time data processing. Such capabilities would facilitate on-platform decision-making, which is particularly advantageous for UAV and satellite-based applications where data bandwidth is limited. The replacement of mechanical components with solid-state solutions, such as MEMS mirrors or electro-optic modulators, could enhance system reliability and durability, making HSI more feasible in harsh environmental conditions. Furthermore, integrating AI and machine learning for data compression, anomaly detection, and adaptive scanning will improve the efficiency of data acquisition and analysis. Such advancements are crucial for high-throughput applications like environmental monitoring and precision agriculture. Besides, enhanced tunable filter technologies, such as liquid crystal structures, MEMS-based FPIs, or metasurface-integrated designs, could pave the way for rapid, lightweight solutions capable of handling the challenges associated with real-time spectral imaging in dynamic environments.

VI. CONCLUSION AND FUTURE DIRECTIONS

This review has comprehensively examined the evolution of spectral imaging system, focusing on their fundamental acquisition methodologies, technological advancements, and future directions. The discussion has underscored the strengths and limitations of staring, snapshot, and spatial scanning techniques, providing their comparative analysis. The work further provides a structured comparison of MSI and HSI acquisition techniques, highlighting MSI as a practical and cost-effective solution for large-scale applications that require spectral information at specific bands. It explores the role of LED arrays, filter wheels, and single-pixel detection techniques in improving MSI's efficiency and affordability, making it well-suited for applications in agriculture, environmental monitoring, and industrial inspection. Moreover, the

review provides an in-depth analysis of HSI advancements, which have enabled superior material identification and quantitative spectral analysis. HSI systems are categorized into staring, snapshot, and spatial scanning approaches, this work clarifies how each technique optimizes spectral fidelity, spatial resolution, and acquisition speed. The discussion on tunable filters—including AOTFs, LCTFs, FPIs, and LVFs—offers valuable insights into how different filter types balance spectral accuracy and mechanical complexity in staring HSI systems. Furthermore, by emphasizing the significance of snapshot HSI innovations, such as advanced filter arrays, illustrating how real-time spectral imaging has revolutionized dynamic event monitoring, including biomedical and remote sensing. Another key contribution is the examination of spatial scanning-based designs, particularly pushbroom and whiskbroom techniques, which continue to be pivotal for airborne and spaceborne spectral imaging. By bridging hardware advancements with computational modeling, this review sheds light on new paradigms for achieving highly detailed reconstructions with reduced data redundancy.

The future of spectral imaging will be shaped by advancements in miniaturization, computational imaging, and machine learning integration, expanding its applications across various fields. Emerging technologies are expected to enhance system portability, processing efficiency, and analytical capabilities, leading to more widespread adoption and novel uses. Miniaturization and portability will be key drivers of future spectral imaging development. Advancements in sensor fabrication, optical component design, and metasurface integration will enable the creation of ultra-compact spectral imaging systems. MEMS-based tunable filters and metasurface-integrated optics will allow for lightweight, field-deployable systems suited for applications in remote sensing, biomedical diagnostics, and consumer electronics. These compact designs will facilitate spectral imaging adoption in handheld and wearable devices, enhancing accessibility and usability.

The integration of real-time processing will further transform spectral imaging capabilities. Processing spectral data directly on imaging devices will minimize latency and reduce data transmission requirements, making spectral imaging more viable for applications such as industrial inspection and UAV-based remote sensing. Artificial intelligence and machine learning will play an increasingly critical role in spectral imaging. AI-driven algorithms will enhance spectral image reconstruction, feature extraction, and anomaly detection, improving material classification and medical diagnostics. Deep learning models will refine spectral image analysis by optimizing noise reduction, data compression, and computational efficiency. These capabilities will be particularly beneficial for spectral imaging in medical applications, enabling real-time, non-invasive disease detection and surgical guidance.

Spectral imaging is poised to make significant contributions to personalized medicine. The ability to capture molecular-level spectral signatures will allow for more

precise diagnostics, particularly in cancer detection, wound assessment, and tissue analysis. AI-enhanced spectral imaging will provide detailed biochemical information, offering a non-invasive alternative to traditional diagnostic methods. By integrating it with clinical workflows, spectral imaging will improve early detection, treatment planning, and real-time monitoring of medical conditions. Computational spectral imaging techniques will also redefine data acquisition and processing methods. The development of computational approaches such as compressive sensing and diffractive neural networks will improve spectral resolution while reducing data acquisition times. These advancements will lead to more efficient spectral imaging systems that require fewer measurements while maintaining high levels of accuracy. Open-source development and collaborative innovations will continue to drive progress in spectral imaging [338], [339], [340].

Open-source spectral imaging involves the creation and dissemination of spectral imaging tools, algorithms, and systems that are freely accessible for modification, improvement, and distribution. This collaborative approach will significantly reduce the barriers to entry for researchers and developers, fostering a more inclusive and dynamic field. The availability of open-source resources allows for rapid prototyping and testing of new ideas, enabling researchers to build on existing work without the constraints of proprietary technologies. Additionally, the shared nature of open-source development encourages collaboration across institutions and disciplines, leading to a more exchange of knowledge and expertise [323]. In addition, open-source platforms will continue to provide educational opportunities for students and newcomers to the field, offering hands-on experience with state-of-the-art technologies without the need for equipment. This democratization of technology has the potential to accelerate advancements in HSI, as a broader and more diverse group of contributors can participate in the development process. The availability of open-source spectral imaging datasets and tools will further stimulate innovation by enabling researchers to experiment with new imaging algorithms and system designs.

Looking ahead, the convergence of hardware advancements, AI-powered processing, and novel computational methods will continue to shape spectral imaging. Future research will focus on overcoming current limitations, including the quest in miniaturization of optical components, increasing acquisition speed, enhancing real-time imaging capabilities, and improving data processing efficiency. By addressing these challenges, spectral imaging will become an even more powerful tool across diverse domains, unlocking new possibilities for scientific research, industrial applications, and medical advancements.

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SANI MUKHTAR received the B.Sc. degree in electrical and control engineering from the Arab Academy for Science, Technology, and Maritime Transport, Alexandria, Egypt, in 2016, and the M.Sc. degree in electrical engineering from the King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia, in 2020. He is currently pursuing the Ph.D. degree in electrical engineering with the Khalifa University of Science and Technology, Abu Dhabi, United Arab Emirates. He was a Visiting Scholar with the King Abdullah University of Science and Technology (KAUST), Saudi Arabia, in 2019, and the University of Massachusetts Amherst, USA, in 2023. His research focuses on optoelectronics, and the design and fabrication of optical metasurfaces for applications in imaging, sensing, and communication. He has received awards, such as the Best Poster Paper Award from the UAE-GSRC Conference, in 2023, and is supported by prestigious scholarships, including Jigawa State Government Fellowship for undergraduate studies abroad, Saudi Arabian Government Fellowship for graduate studies, and the Khalifa University Postgraduate Scholarship Award.



AMIR ARBABI received the B.Sc. degree in electrical engineering from the University of Tehran, in 2006, the M.Sc. degree in electrical engineering from the University of Waterloo, in 2009, and the Ph.D. degree in electrical engineering from the University of Illinois at Urbana–Champaign, in 2013. He is currently an Associate Professor of electrical and computer engineering with the University of Massachusetts Amherst (UMass Amherst). Before joining UMass Amherst, in 2017, he was a Postdoctoral Scholar and later a Senior Research Scientist with California Institute of Technology's T. J. Watson Laboratories of Applied Physics, from 2013 to 2016. He holds over 35 patents in areas, such as metasurface spectrometers and imaging systems. His research interests include the experimental and theoretical aspects of nanophotonic, flat optics, and photonic integrated circuits, with applications in optical data processing, sensors, consumer electronics, optical communications, and imaging. He has received multiple awards, including the Best ECE Junior Faculty Award, in 2021. He serves as an Associate Editor for *Journal of Nanophotonics*, *Sensors*, *IEEE PHOTONICS JOURNAL*, and *Frontiers in Electronic Materials*.



JAIME VIEGAS (Member, IEEE) received the M.Sc. degree in optoelectronics and lasers and the Ph.D. degree in physics from the University of Porto, Portugal. In 2010, he joined the Masdar Institute, Khalifa University of Science and Technology, as a Faculty Member at the Microsystems Engineering Program. He is currently an Associate Professor with the Department of Electrical Engineering and Computer Science, Khalifa University of Science and Technology. He has co-authored seven U.S. patents. His research interests include silicon photonics, optical sensors, fiber optics sensors, digital holography, metamaterials design and fabrication, MEMS for photonics, and advanced lithography.

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